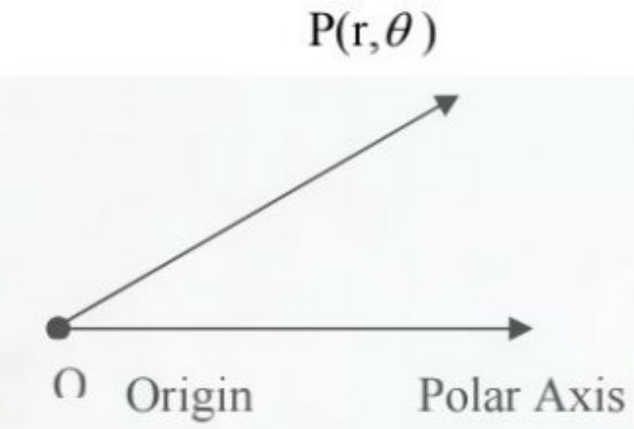
Mth301 Final Term
Highlighted 23-45
By:Sir Afzal

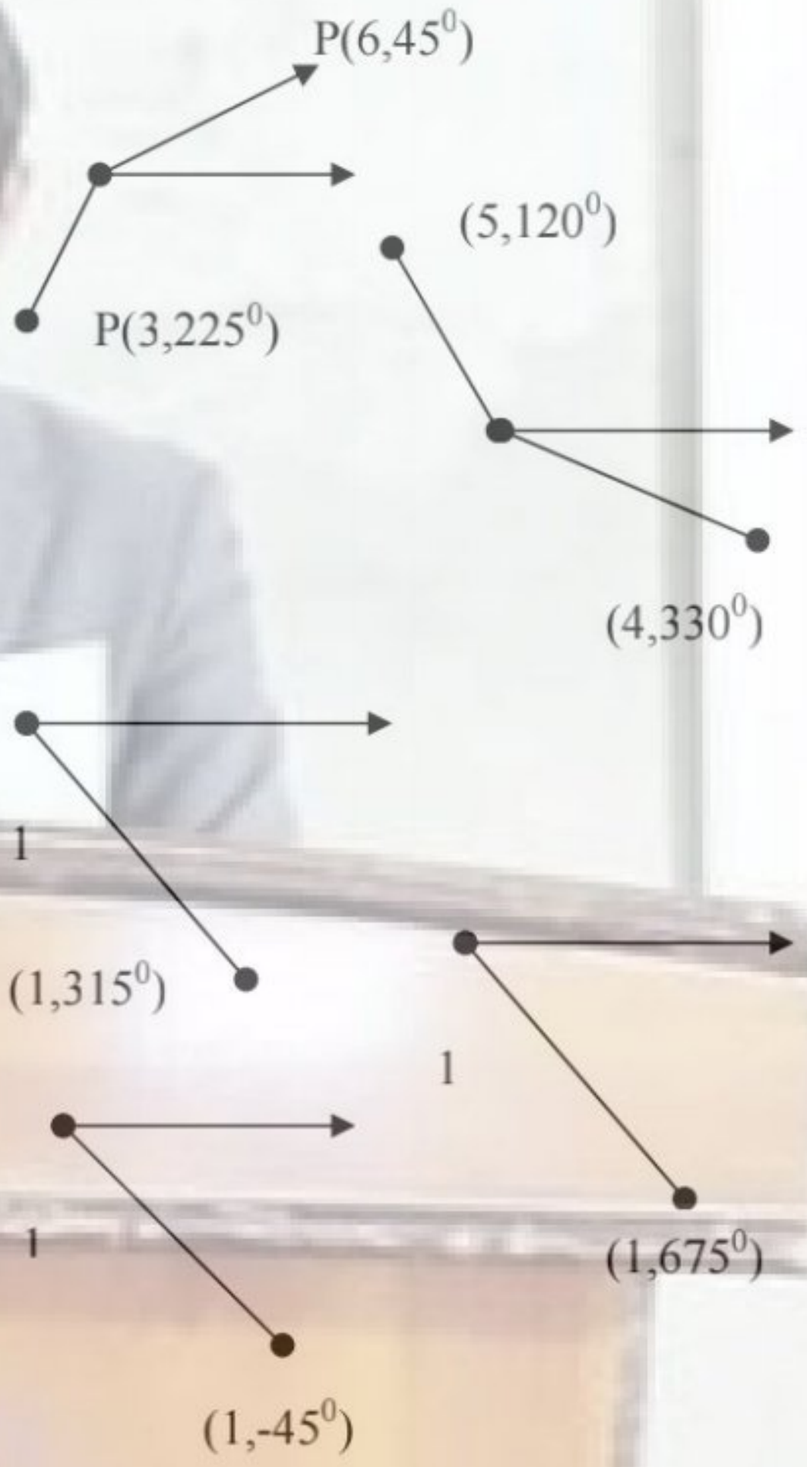
Lecture No - 23 Polar Coordinate Systems

POLAR COORDIANTE SYSTEMS

To form a polar coordinate system in a plane, we pick a fixed point O, called the origin or pole, and using the origin as an endpoint we construct a ray, called the polar axis. After selecting a unit of measurement, we may associate with any point P in the plane a pair of polar coordinates (r, θ), where r is the distance from P to the origin and θ measures the angle from the polar axis to the line segment OP.



The number r is called the radial distance of P and θ is called a polar angle of P. In the points (6, 45°), (3, 225°), (5, 120°), and (4, 330°) are plotted in polar coordinate systems.



THE POLAR COORDINATES OF A POINT ARE NOT UNIQUE.

For example, the polar coordinates
(1, 315°), (1, – 45°), and (1, 675°)
all represent the same point
In general, if a point P has polar co-ordinate
(r, θ), then for any integer n=0,1,2,3,……
(r,θ+n.360⁰) and (r,θ+n.360⁰)
are also polar co-ordinates of p
In the case where P is the origin, the line line seg
Because there is no clearly defined polar angle ir
Polar angle θ may be used. Thus, for every θ ma
(0, θ) is the origin.

NEGATIVE VALUES OF R

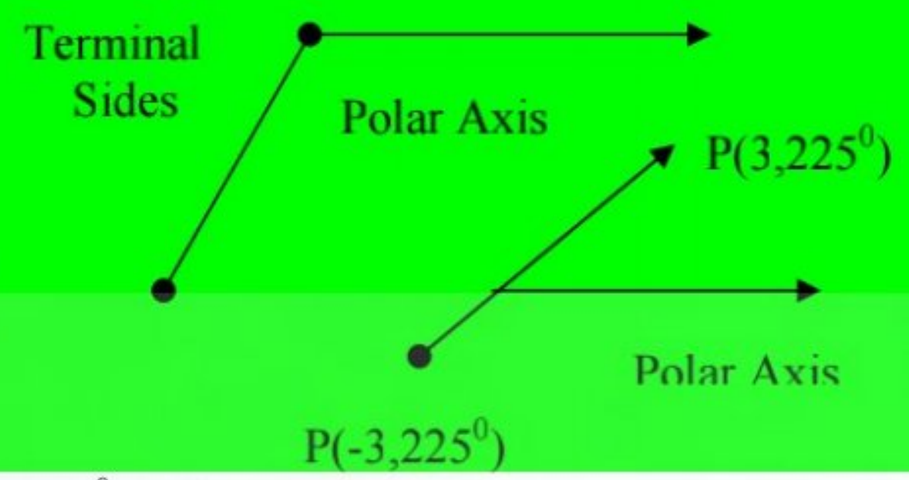
When we start graphing curves in polar coordinates, it will be desirable to allow negative values for r. This will require a special definition. For motivation, consider the point P with polar coordinates (3, 225⁰) We can reach this point by rotating the polar axis 225⁰ and then moving forward from the origin 3 units along the terminal side of the angle. On the other hand, we can also reach the point P by rotating the polar axis 45⁰ and then moving backward 3 units from the origin along the extension of the terminal side of the angle

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

This suggests that the point (3, 225 might also be denoted by (− 3, 45 with the minus sign serving to indicate that the point is on the extension of the angle's terminal side rather than on the terminal side itself.

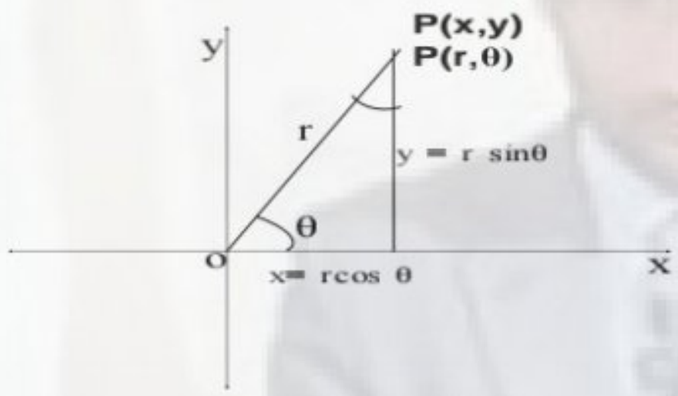


Since the terminal side of the angle $\theta+180^0$ is the extension of the terminal side other angle θ, We shall define.

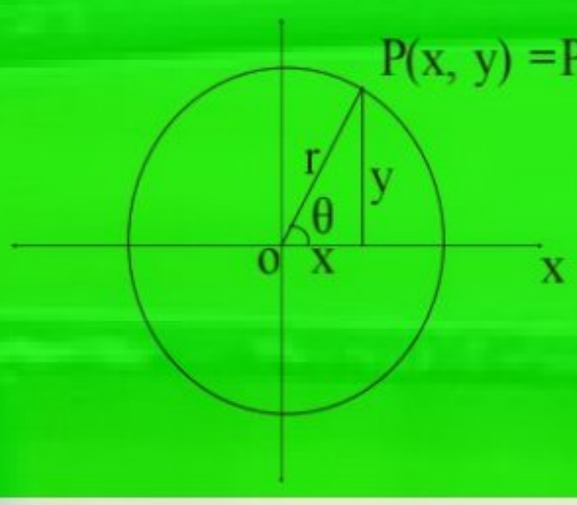
(-r, θ) and (r, $\theta+180^0$) to be polar coordinates for the same point.

With r=3 and $\theta=45^0$ in (2) if follows that (-3, 45^0) and (-3, 225^0) represent the same point.

RELATION BETWEEN POLAR AND RECTANGULAR COORDINATES



CONVERSION FORMULA FROM POLAR TO CARTESIAN COORDINATES AND VICE VERSA



Example

Find the rectangular coordinates of the point P whose polar co-ordinates are $(6,135^0)$

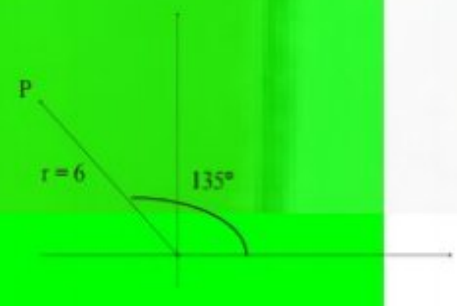
Solution:

Substituting the polar coordinates

r = 6 and $\theta=135^0$ in $x=\cos\theta$ and $y=\sin\theta$ yeilds

$x = 6\cos 135° = 6(-\sqrt{2}/2) = -3\sqrt{2}$

$y = 6\sin 135° = 6(\sqrt{2}/2) = 3\sqrt{2}$



Thus, the rectangular coordinates of the point P are $(-3\sqrt{2}, 3\sqrt{2})$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example:

Find polar coordinates of the point P whose rectangular coordinates are $(-2, 2\sqrt{3})$

Solution:

We will find polar coordinates (r, θ) of P such that r >0 and $0 \le \theta \le 2\pi$.

$$r = \sqrt{x^2 + y^2} = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4+12} = \sqrt{16} = 4$$

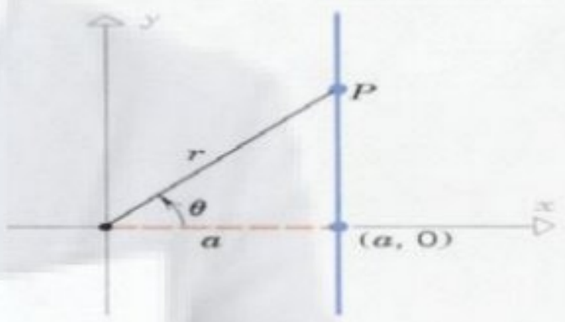
$$\tan\theta = \frac{y}{x} = \frac{2\sqrt{3}}{-2} = -\sqrt{3} \Rightarrow \theta = \tan^{-1}(-\sqrt{3}) = \frac{2\pi}{3}$$

From this we have (-2,2 $\sqrt{3}$) lies in the second quadrant of P. All other Polar co-ordinates of P have the form

$$(4, \frac{2\pi}{3} + 2n\pi) \quad or \quad (-4, \frac{5\pi}{3} + 2n\pi) \quad , \quad Where\ n\ is\ \text{int}eger$$

LINES IN POLAR COORDIANTES

A line perpendicular to the x-axis and passing through the point with xy co-ordinates with (a,0) has the equation x=a . To express this equation in polar co-ordinates we substitute $x = r\cos\theta \Rightarrow a = r\cos\theta$ --------(1)



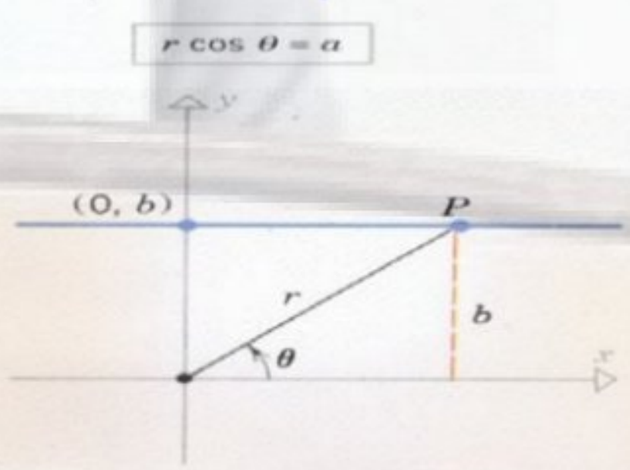
This result makes sense geometrically since each point P (r, θ) on this line will yield the value a for r cos θ.

A line parallel to the x-axis that meets the y-axis it the point with xy-coordinates (0, b) has the equation y = b.

Substituting $y = r\sin\theta$ yields.

$r\sin\theta = b$ (2)

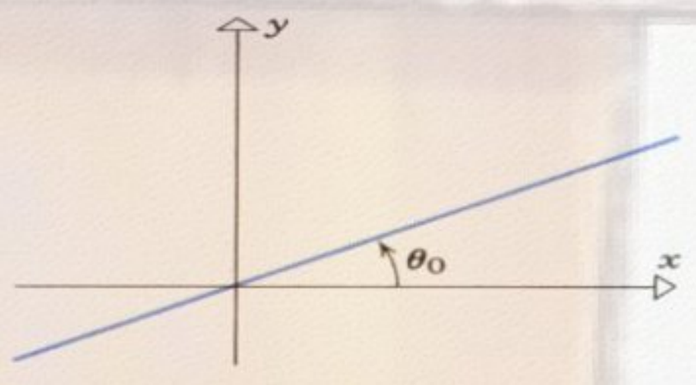
as the polar equation of this line. This makes sense geometrically since each point P (r, θ) on this line will yield the value b for r sin θ

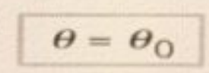


For Any constant θ_0, the equation

$\theta = \theta_0$ (3)

is satisfied by the coordinates of all points of the form P (r, θ_0), regardless of the value of r. Thus, the equation represents the line through the origin making an angle of θ_0 (radians) with the polar axis.





By substitution $x = r\cos\theta$ and $y = r\sin\theta$ in the equation $Ax + By + C = 0$. We obtain the general polar form of the line,

$$r(A\cos\theta + B\sin\theta) + C = 0$$

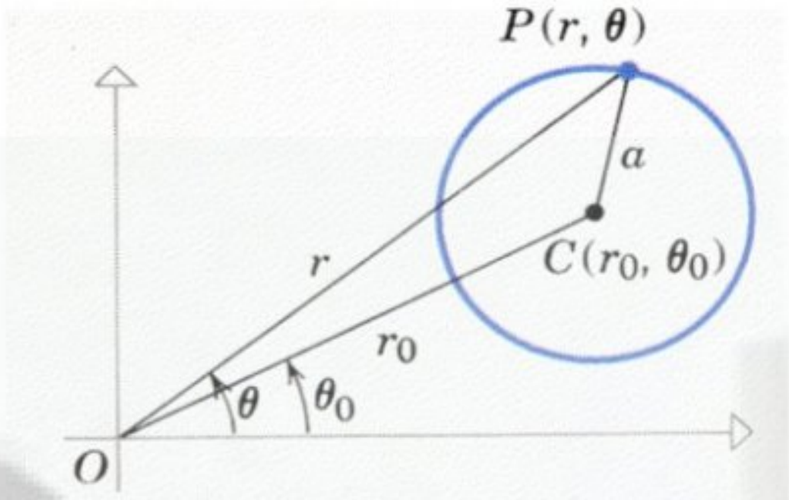
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

CIRCLES IN POLAR COORDINATES

Let us try to find the polar equation of a circle whose radius is a and whose center has polar coordinates (r_0, θ_0). If we let $P(r, \theta)$ be an arbitrary point on the circle, and if we apply the law of cosines to the triangle OCP we obtain

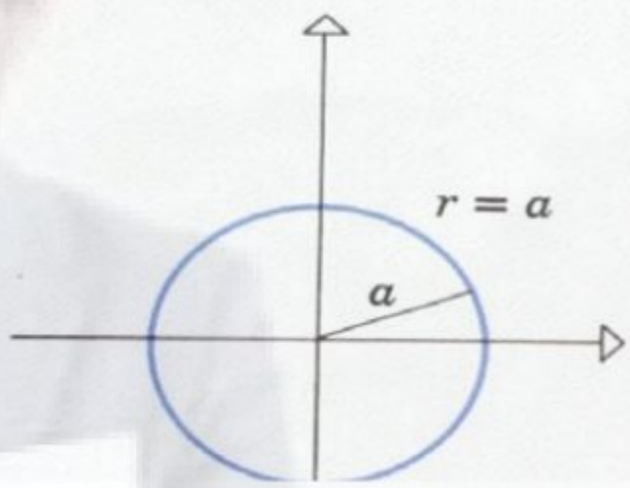


$$r^2 - 2rr_0 \cos(\theta - \theta_0) + r_0^2 = a^2 \qquad (1)$$

SOME SPECIAL CASES OF EQUATION OF CIRCLE IN POLAR COORDINATES

A circle of radius a, centered at the origin, has an especially simple polar equation. If we let $r_0 = 0$ in (1), we obtain $r^2 = a^2$ or, since $a \geq 0$, $r = a$

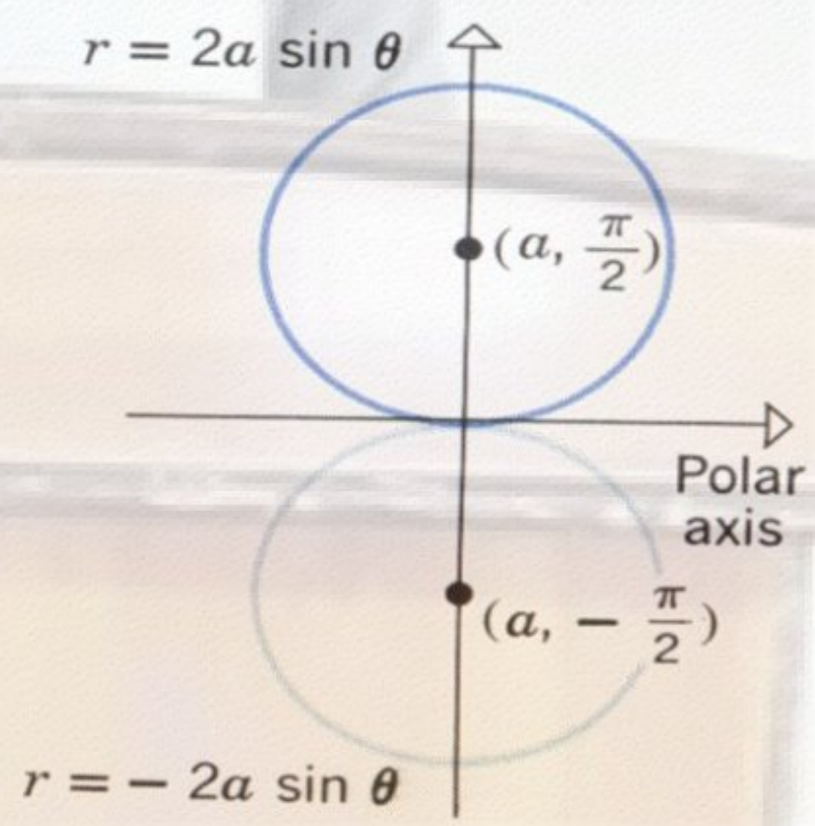
This equation makes sense geometrically since the circle of radius a, centered at the origin, consists of all points P (r, θ) for which r = a, regardless of the value of θ



If a circle of radius a has its center on the x-axis and passes through the origin, then the polar coordinates of the center are either

(a, 0) or (a, π)

depending on whether the center is to the right or left of the origin



Lecture No -24 Sketching

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Draw graph of the curve having the equation r = sin θ

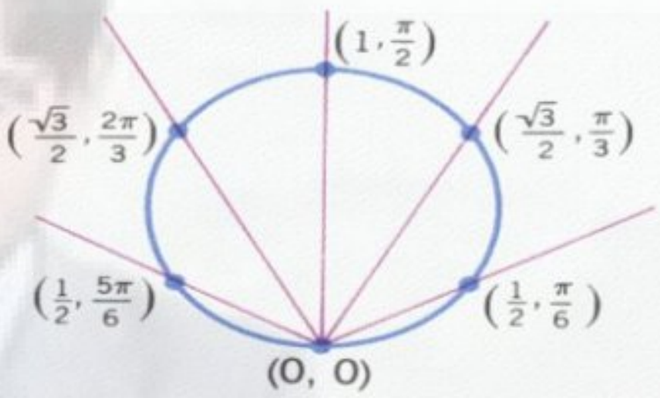
By substituting values for θ at increments of $\frac{\pi}{6}(30^0)$ and calculating r , we can construct

The following table:

θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
r = sin θ	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$

θ (radians)	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
r = sin θ	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	−1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0

Note that there are 13 pairs listed in Table, but only 6 points plotted in This is because the pairs from θ = π on yield duplicates of the preceding points. For example, (– ½, 7π/6) and (1/2, π/6) represent the same point.
The points appear to lie on a circle.



hat this is indeed the case may be seen by expressing the given equation in terms of x and y. We first multiply the given equation through by r to obtain $r^2 = r\sin\theta$ which can be rewritten as

$$x^2 + y^2 = y \quad \text{or} \quad x^2 + y^2 - y = 0$$

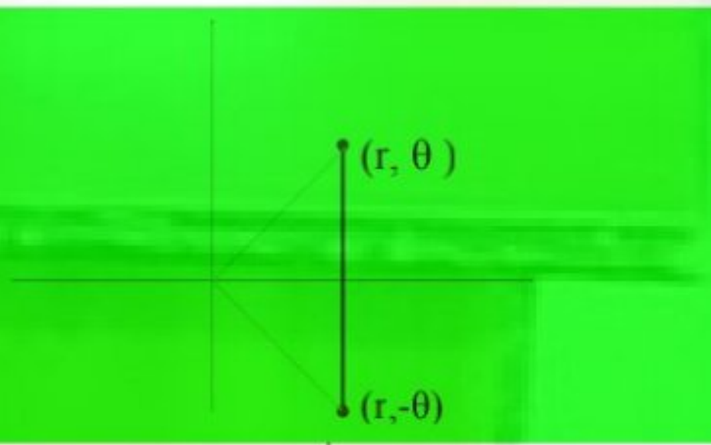
or on completing the square $x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}$ .This is a circle of radius 1\2 centered at the point (0,1/2) in the xy-plane.

Sketching of Curves in Polar Coordinates

1.SYMMETRY

(i) Symmetry about the Initial Line

If the equation of a curve remains unchanged when (r,θ) is replaced by either $(r,-\theta)$ in its equation ,then the curve is symmetric about initial line.

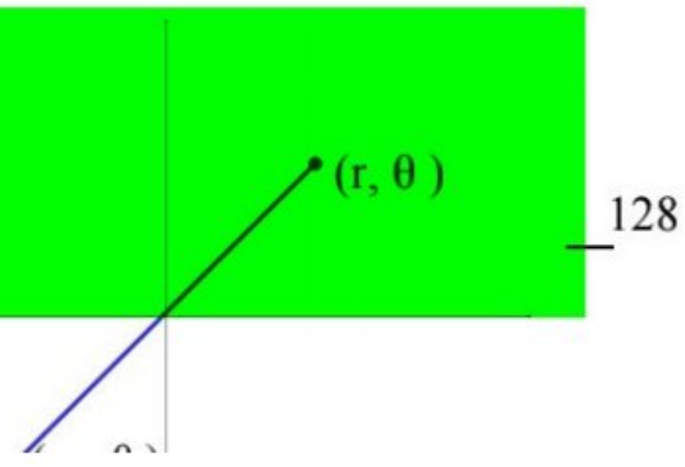


(ii) Symmetry about the y-axis

If when (r, θ) is replaced by either $(r, \pi-\theta)$ in The equation of a curve and an equivalent equation is obtained ,then the curve is symmetric about the line perpendicular to the initial i.e, the y-axis

(ii) Symmetry about the Pole

If the equation of a curve remains unchanged



© Copyright Virtual Unive

Scanned with CamScanner

Scanned with CamScanner

when either (-r, θ) or is substituted for (r, θ) in its equation ,then the curve is symmetric about the pole. In such a case ,the center of the curve.

2. Position Of The Pole Relative To The Curve

See whether the pole on the curve by putting r=0 in the equation of the curve and solving for θ.

3. Table Of Values

Construct a sufficiently complete table of values. This can be of great help in sketching the graph of a curve.

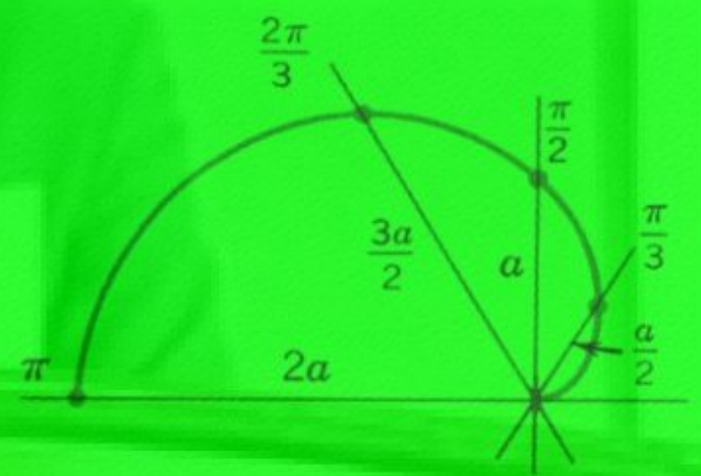
II Position Of The Pole Relative To The Curve.

When $r = 0$, $\theta = 0$ .Hence the curve passes through the pole.

III. Table of Values

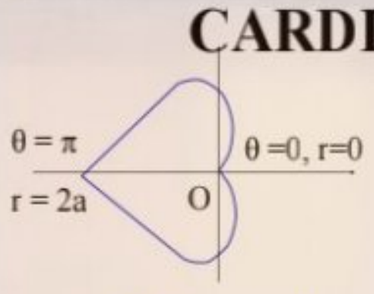
θ	0	π/3	π/2	2π/3	π
r=a (1−cosθ)	0	a/2	a	3a/2	2a

As θ varies from 0 to π, cos θ decreases steadily from 1 to − 1, and 1 − cos θ increases steadily from 0 to 2. Thus, as θ varies from 0 to π, the value of r = a (1 − cos θ) will increase steadily from an initial value of r = 0 to a final value of r = 2a.

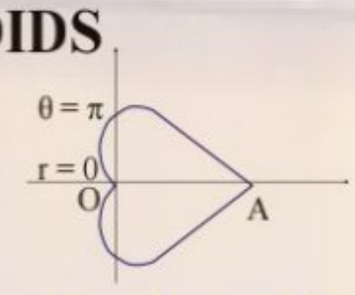


On reflecting the curve in about the x-axis, we obtain the curve.

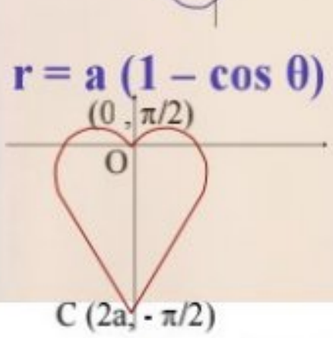
CARDIOIDS



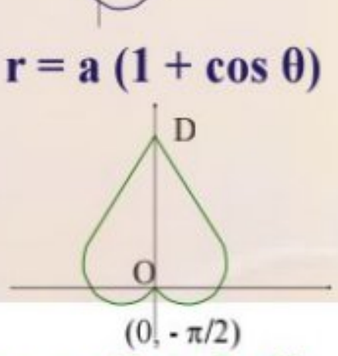
r = a (1 – cos θ)



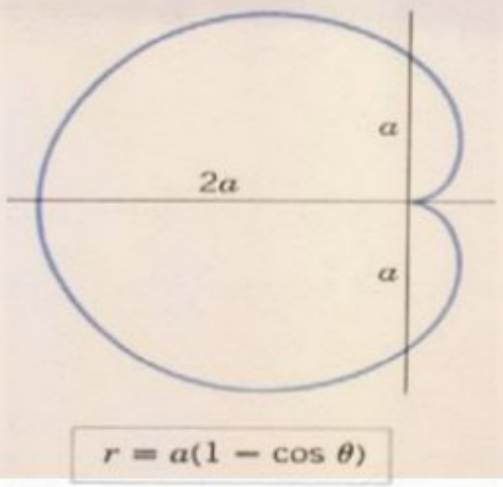
r = a (1 + cos θ)



r = a (1 – sin θ)



r = a (1 + sin θ)



CARDIDOIDS AND LIMACONS

© Copyright Virtual University of Pakistan

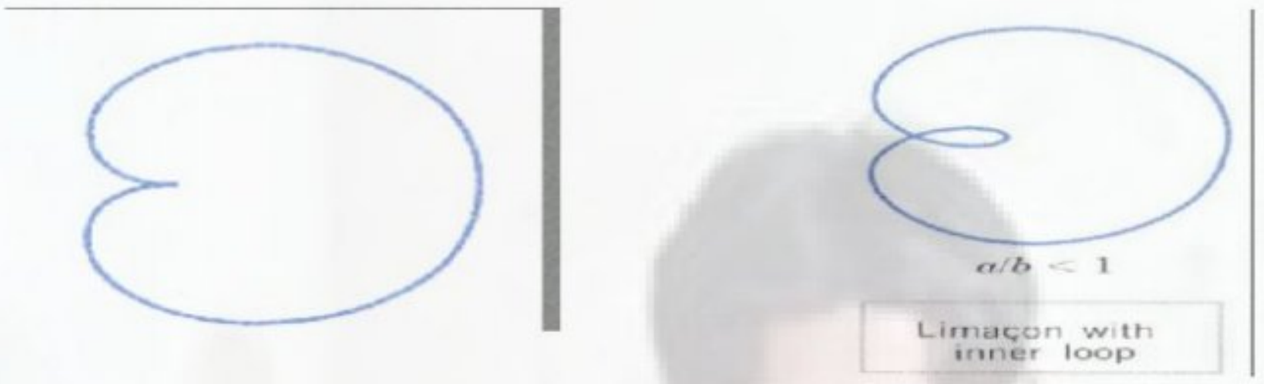
Scanned with CamScanner

Scanned with CamScanner

$r=a+b\sin\theta$, $r=a-b\sin\theta$

$r=a+b\cos\theta$, $r=a-b\cos\theta$

The equations of above form produce polar curves called limacons. Because of the heart-shaped appearance of the curve in the case a = b, limacons of this type are called cardioids. The position of the limacon relative to the polar axis depends on whether sin θ or cos θ appears in the equation and whether the + or – occurs.



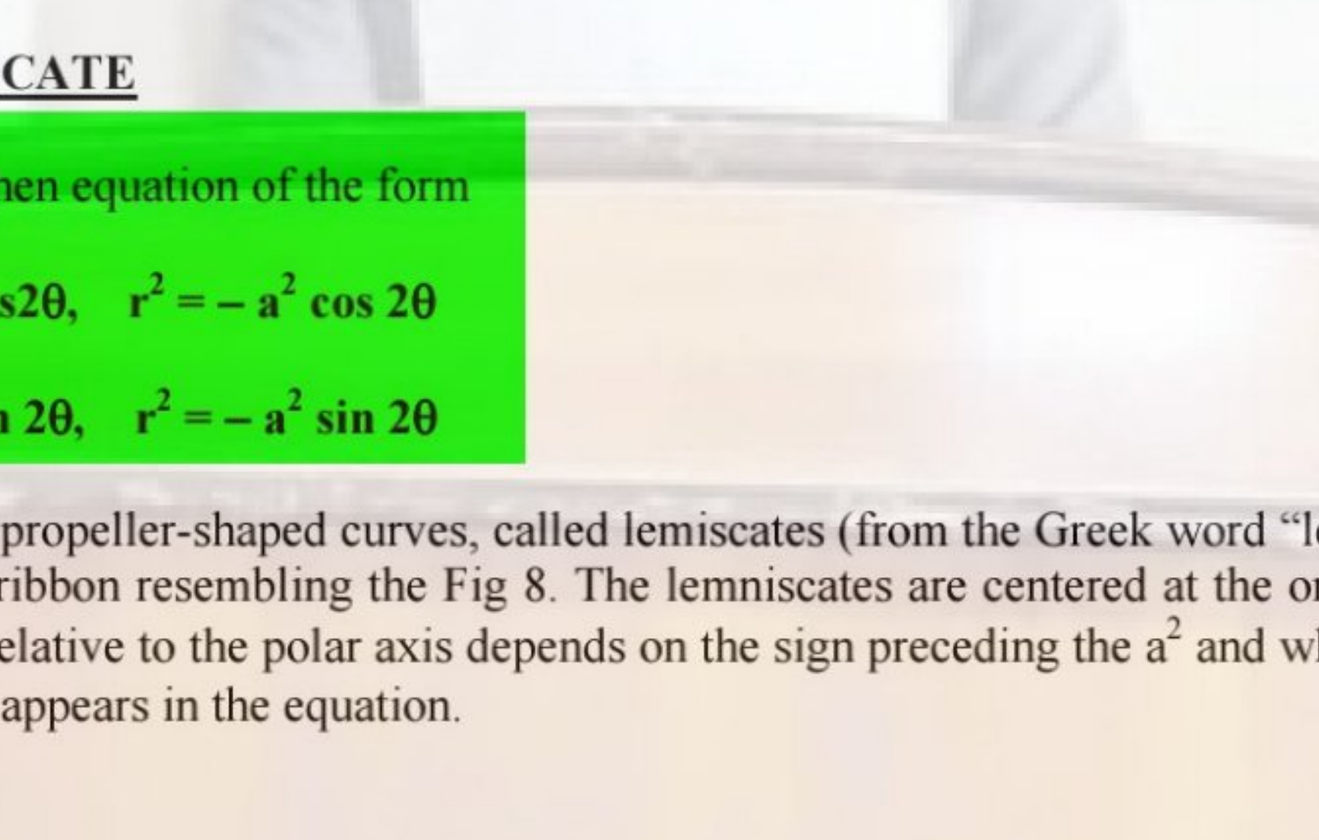
LEMINSCATE

If $a > 0$, then equation of the form

$r^2 = a^2\cos 2\theta$, $r^2 = -a^2\cos 2\theta$

$r^2 = a^2\sin 2\theta$, $r^2 = -a^2\sin 2\theta$

represent propeller-shaped curves, called lemiscates (from the Greek word "lemnicos" for a looped ribbon resembling the Fig 8. The lemniscates are centered at the origin, but the position relative to the polar axis depends on the sign preceding the a^2 and whether sin 2θ or cos 2θ appears in the equation.



Examp

le

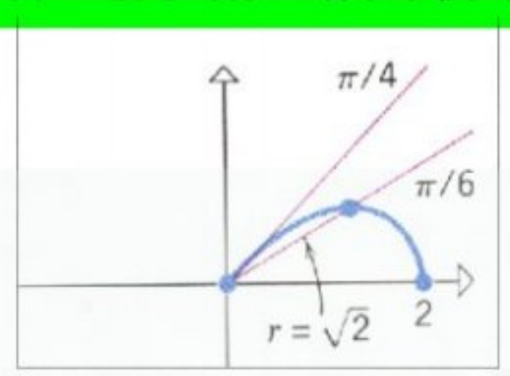
© Copyright Virtual University of Pakistan

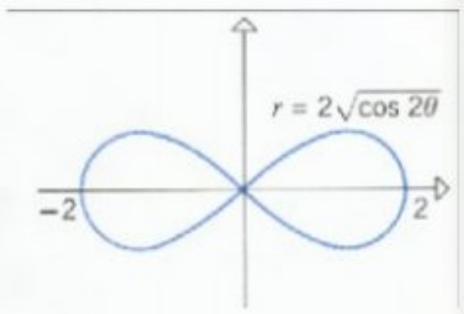
Scanned with CamScanner

Scanned with CamScanner

$$r^2 = 4 \cos 2\theta$$

The equation represents a lemniscate. The graph is **symmetric** about the **x-axis** and the **y-axis**. Therefore, we can obtain each graph by first sketching the portion of the graph in the range $0 \leq \theta < \pi/2$ and then reflecting that portion about the x- and y-axes. The curve passes through the origin when $\theta = \pi/4$, so the line $\theta = \pi/4$ is tangent to the curve at the origin. As θ varies from 0 to $\pi/4$, the value of $\cos 2\theta$ decreases steadily from 1 to 0, so that r decreases steadily from 2 to 0. For θ in the range $\pi/4 < \theta < \pi/2$, the quantity $\cos 2\theta$ is negative, so there are no real values of r satisfying first equation. Thus, there are no points on the graph for such θ. The entire graph is obtained by reflecting the curve about the x-axis and then reflecting the resulting curve about the y-axis.



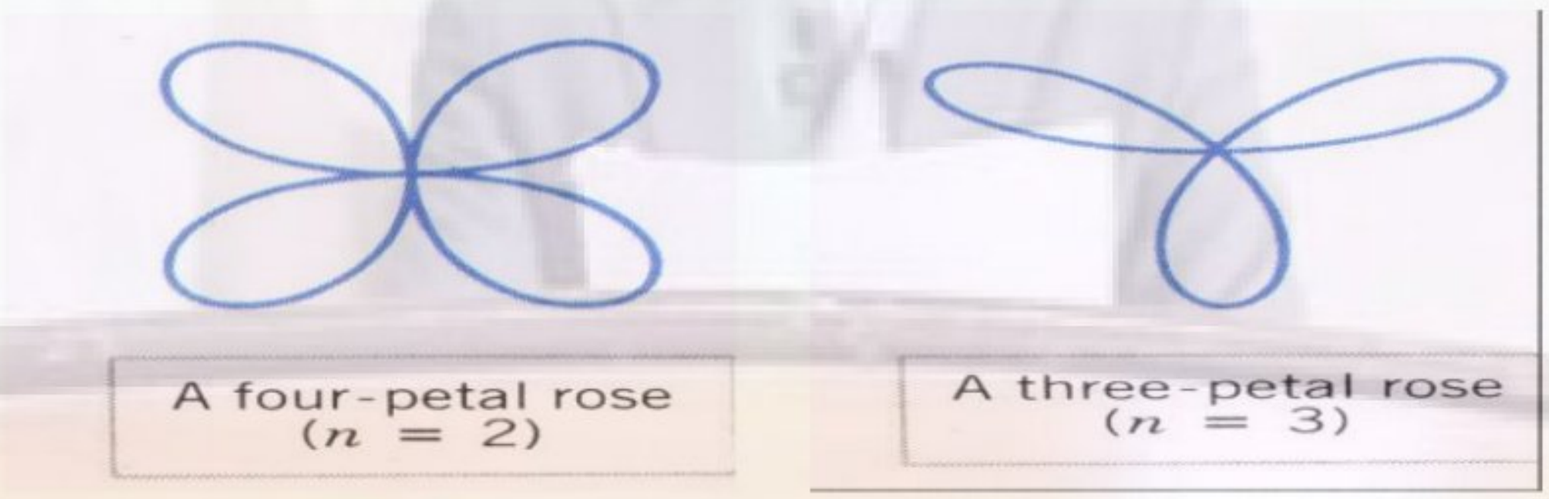


ROSE CURVES

Equations of the form

$r = a \sin n\theta$ and $r = a \cos n\theta$

represent flower-shaped curves called roses. The rose has **n equally spaced petals or loops if n is odd** and **2n equally spaced petals if n is even**



A four-petal rose ($n = 2$)

A three-petal rose ($n = 3$)

The orientation of the rose relative to the polar axis depends on the sign of the constant a and whether $\sin\theta$ or $\cos\theta$ appears in the equation.

SPIRAL

A curve that "winds around the origin" infinitely many times in such a way that r increases (or decreases) steadily as θ increases is called a spiral. The most common example is the spiral of Archimedes, which has an equation of the form.

$r = a\theta \quad (\theta \geq 0)$ or $r = a\theta \quad (\theta \leq 0)$

In these equations, θ is in radians and a is positive.

EXAMPLE

Sketch the curve $\mathbf{r = \theta \ (\theta \geq 0)}$ in polar coordinates.

This is an equation of spiral with a = 1; thus, it represents an Archimedean spiral.

Since r = 0 when $\theta = 0$, the origin is on the curve and the polar axis is tangent to the spiral.

A reasonably accurate sketch may be obtained by plotting the intersections of the spiral with the x and y axes and noting that r increases steadily as θ increases. The intersections with the x-axis occur when

$\theta = 0, \ \pi, \ 2\pi, \ 3\pi, \ldots\ldots$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

at which points r has the values

$r = 0,\ \pi,\ 2\pi,\ 3\pi,\ldots..$

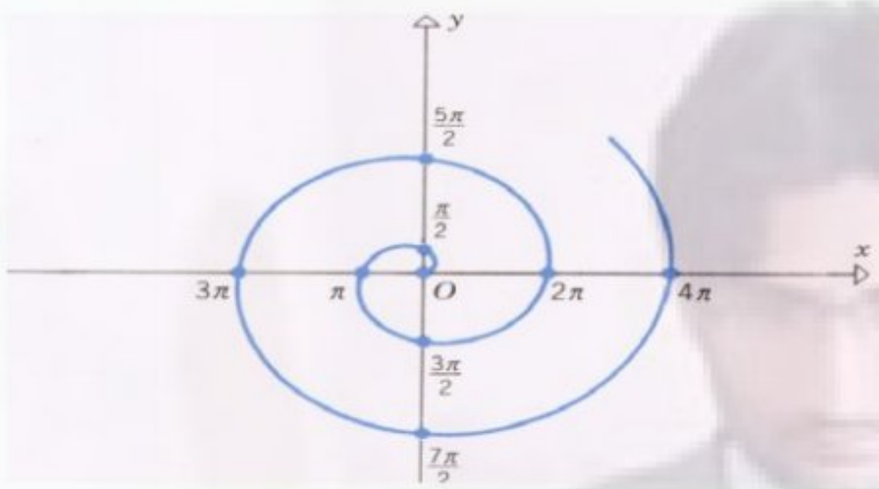
and the intersections with the y-axis occur when

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots\ldots$$

at which points r has the values

$$r = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots\ldots$$

Starting from the origin, the Archimedean spirals $r = \theta$ $(\theta \geq 0)$ loops counterclockwise around the origin.



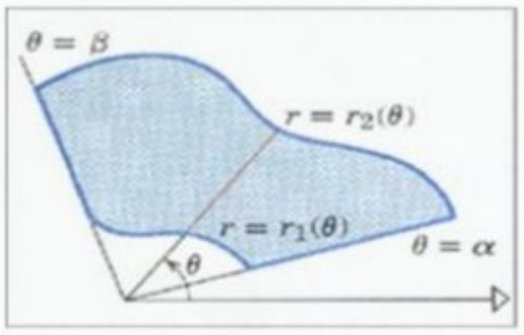
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -25 Double integrals in polar co-ordinates

Double integrals in which the integrand and the region of integration are expressed in polar coordinates are important for two reasons: First, they arise naturally in many applications, and second, many double integrals in rectangular coordinates are more easily evaluated if they are converted to polar coordinates. The function $z = f(r,\theta)$ to be integrated over the region R As shown in the Fig.



INTEGRALS IN POLAR COORDIATES

When we defined the double integral of a function over a region R in the xy-plane, we began by cutting R into rectangles whose sides were parallel to the coordinate axes. These were the natural shapes to use because their sides have either constant x-values or constant y-values. In polar coordintes, the natural shape is a "polar rectangle" whose sides have constant r and θ- values.

Suppose that a function $f(r, \theta)$ is defined over a region R that is bounded by the ray $\theta = \alpha$ and $\theta = \beta$ and by the continuous curves $r = r_1(\theta)$ and $r = r_2(\theta)$. Suppose also that $0 \le r_1(\theta) \le r_2(\theta) \le a$ for every value of θ between α and β. Then R lies in a fan-shaped region Q defined by the inequalities $0 \le r \le a$ and $\alpha \le \theta \le \beta$.

Then the double integral in polar coordinates is given as

$$\iint_R f(r,\theta)\, dA = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\, dr\, d\theta$$

How to find limits of integration from sketch

Step 1. Since θ is held fixed for the first integration, draw a radial line from the origin through the region R at a fixed angle θ. This line crosses the boundary of R at most twice. The innermost point of intersection is one the curve $r = r_1(\theta)$ and the outermost point is on the curve $r = r_2(\theta)$. These intersections determine the r-limits of integration.

Step 2. Imagine rotating a ray along the positive x-axis one revolution counterclockwise about the origin. The smallest angle at which this ray intersects the region R is $\theta = \alpha$ and the largest angle is $\theta = \beta$. This yields the θ-limits of the integration.

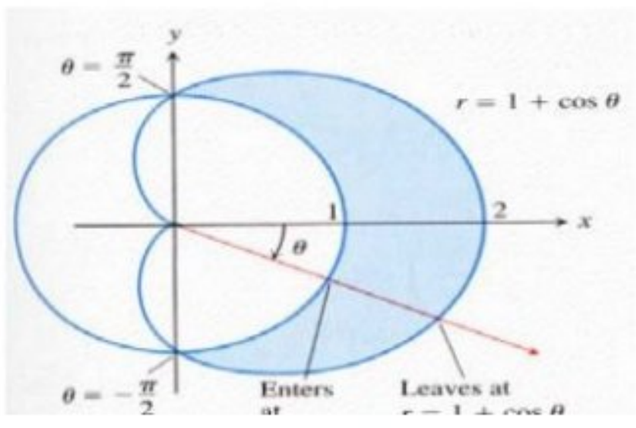
EXAMPLE

Find the limits of integration for integrating $f(r, \theta)$ over the region R that lies inside the cardioid $r = 1 + \cos\theta$ and outside the circle $r = 1$.

Solution:

Step 1. We sketch the region and label the bounding curves.

Step 2. The r-limits of integration. A typical ray from



© Copyright Virtual University

Scanned with CamScanner

Scanned with CamScanner

The origin enters R where r =1 and leaves where

r =1+cos θ.

Step 3. The θ-limits of integration. The rays from the

origin that intersect R run from $\theta = -\frac{\pi}{2}$ to $\theta = \frac{\pi}{2}$.

The integral is $\int_{-\pi/2}^{\pi/2}\int_{1}^{1+\cos\theta} f(r,\theta)\, r\, dr\, d\theta$

$= 2\int_{0}^{\pi/2}\int_{1}^{1+\cos\theta} f(r,\theta)\, r\, dr\, d\theta$

EXAMPLE

Evalaute $\iint_R \sin\theta\, dA$

Where R is the region ion the first quadrant that is outside the circle r = 2 and inside the cardioid r = 2(1+cosθ).

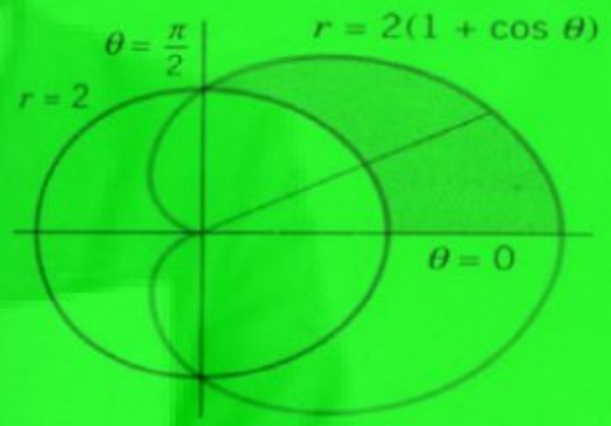
Solution:

$$\iint_R \sin\theta\, dA = \int_0^{\pi/2}\int_2^{2(1+\cos\theta)} (\sin\theta)\, r\, dr\, d\theta$$

$$= \int_0^{\pi/2} \frac{1}{2} r^2 \sin\theta \Big]_{r=2}^{2(1+\cos\theta)} d\theta$$

$$= 2\int_0^{\pi/2} [(1+\cos\theta)^2 \sin\theta - \sin\theta]\, d\theta$$

$$= 2\left[-\frac{1}{3}(1+\cos\theta)^3 + \cos\theta\right]_0^{\pi/2} = \left[-\frac{1}{3} - \left(-\frac{5}{3}\right)\right] = \frac{8}{3}$$



EXAMPLE

Use a double polar integral to find the area enclosed by the three-petaled rose r = sin 3θ.

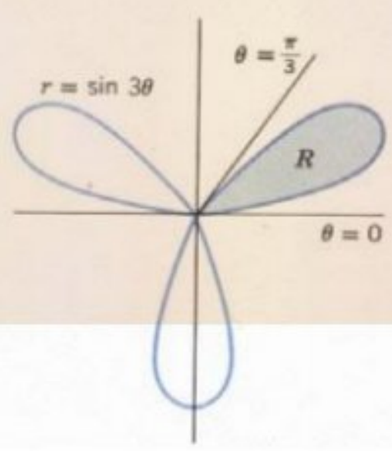
We calculate the area of the petal R in the first quadrant and multiply by three.

Solution:

$$A = 3\iint_R dA = 3\int_0^{\pi/3}\int_0^{\sin 3\theta} r\, dr\, d\theta = \int_0^{\pi/3} \frac{r^2}{2}\Big|_0^{\sin 3\theta} dr\, d\theta$$

$$= \frac{3}{2}\int_0^{\pi/3} \sin^2 3\theta\, d\theta = \frac{3}{4}\int_0^{\pi/3} (1-\cos 6\theta)\, d\theta$$

$$= \frac{3}{4}\left[\theta - \frac{\sin 6\theta}{6}\right]_0^{\pi/3} = \left[\frac{3}{4}\theta - \frac{3}{24}\sin 6\theta\right]_0^{\pi/3} = \frac{1}{4}\pi$$



EXAMPLE

© Copyright Virtual University of Pakistan

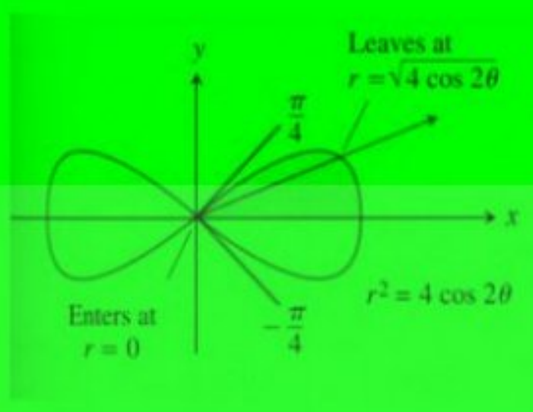
Scanned with CamScanner

Scanned with CamScanner

Find the area enclosed by the lemniscate $r^2 = 4\cos 2\theta$. The total area is four times the first-quadrant portion.

Solution:

$$A = 4\int_0^{\pi/4}\int_0^{\sqrt{4\cos 2\theta}} r\,dr\,d\theta = A\int_0^{\pi/4}\left[\frac{r2}{2}\right]_{r=0}^{r=\sqrt{4\cos 2\theta}} d\theta$$



$$= 4\int_0^{\pi/4} 2\cos 2\theta\,d\theta = 4\sin 2\theta\Big]_0^{\pi/4} = 4.$$

CHANGING CARTESIAN INTEGRALS INTO POLAR INTEGRALS

The procedure for changing a Cartesian integral $\iint_R f(x, y)\,dx\,dy$ into a polar integral has two steps.

Step 1. Substiute $x = r\cos\theta$ and $y = r\sin\theta$, and replace $dx\,dy$ by $r\,dr\,d\theta$ in the Cartesian integral.

Step 2. Supply polar limits of integration for the boundary of R. The Cartesian integral then becomes

$$\iint_R f(x,y)\,dx\,dy = \iint_G f(r\cos\theta, r\sin\theta) r\,dr\,d\theta$$

where G denotes the region of integration in polar coordinates.

Notice that dx dy is not replaced by dr dθ but by r dr dθ.

EXAMPLE

Evalaute the double integral $\int_0^1\int_0^{\sqrt{1-x^2}} (x^2 + y^2)\,dy\,dx$ by changing to polar coordinates.

The region of integration is bounded by

$0 \le y \le \sqrt{1 - x^2}$ and $0 \le x \le 1$

$y = \sqrt{1 - x^2}$ is the circle

$x^2 + y^2 = 1, \quad r = 1$

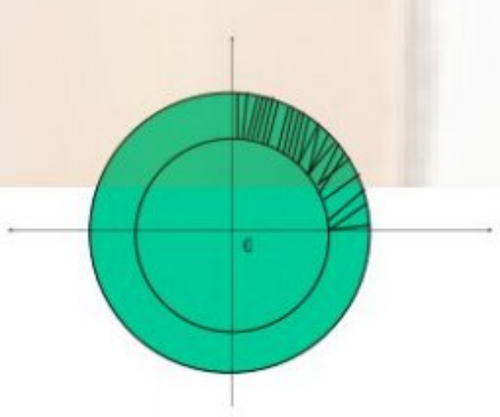
On changing into the polar coordinates, the given integral is

$$\int_0^{\pi/2}\int_0^1 r^3\,dr\,d\theta = \int_0^{\pi/2}\left|\frac{r^4}{4}\right|_0^1 d\theta = \int_0^{\pi/2}\frac{1}{4}\,d\theta = \left|\frac{\theta}{4}\right|_0^{\pi/2} = \frac{\pi}{8}$$

EXAMPLE

Evaluate $I = \int_D\int \frac{dx\,dy}{x^2 + y^2}$ by changing to polar coordinates,

where D is the region in the first quadrant between the circles $x^2+y^2=a^2$ and $x^2+y^2=b^2$, $0<a<b$



$$I = \int_0^{\pi/2}\int_a^b \frac{r\,dr\,d\theta}{r^2} = \int_0^{\pi/2} [\ln r]_a^b\,d\theta$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$= \int_0^{\pi/2} \ln\left(\frac{b}{a}\right) d\theta = \left[\theta \ln\left(\frac{b}{a}\right)\right]_0^{\pi/2} = \frac{\pi}{2} \ln\left(\frac{b}{a}\right).$$

EXAMPLE

Evaluate the double integral $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy\, dx$ by changing to polar coordinates.

The region of integration is bounded by $0 < y < \sqrt{1-x^2}$ and $0 \le x \le 1$

$y = \sqrt{1-x^2}$ is the circle $x^2 + y^2 = 1$, $r = 1$

On changing into the polar coordinates, the given integral is

$$\int_0^{\pi/2} \int_0^1 r^3 dr d\theta = \int_0^{\pi/2} \left|\frac{r^4}{4}\right|_0^1 d\theta = \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{1}{4} |\theta|_0^{\pi/2} = \frac{1}{4}(\pi/2) = \pi/8$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No - 26 EXAMPLES

EXAMPLE

Evalaute $I=\int_0^4 \int_0^{\sqrt{4y-y^2}} (x^2+y^2)dx\,dy$ by changing into polar coordinates.

The region of integration is bounded by $0\le x\le\sqrt{4y-y^2}$ and $0\le y\le 4$

Now $x=\sqrt{4y-y^2}$ is the circle $x^2+y^2-4y=0 \Rightarrow x^2+y^2=4y$.In polar coordinates this takes the form $r^2=4r\sin\theta,\quad r=4\sin\theta$

On changing the integral into polar coordinates, we have

$$I=\int_0^{\pi/2}\int_0^{4\sin\theta} r^2 . r\,dr\,d\theta=\int_0^{\pi/2} 64\sin^4\theta\,d\theta=64.\frac{3.1}{4.2}.\frac{\pi}{2}=12\pi \quad \text{(using Walli's formula)}$$

EXAMPLE

Evalaute $\iint_R e^{x^2+y^2}\,dy\,dx.$,where R is the semicircular region bounded by the x-axis and the curve $y=\sqrt{1-x^2}$

In Cartesian coordinates, the integral in question is a nonelementary integral and there is no direct way to integrate $e^{x^2+y^2}$ with respect to either x or y.

Substituting $x=r\cos\theta$, $y=r\sin\theta$, and replacing dy dx by r dr dθ enables us to evaluate the integral as

$$\iint_R e^{x^2+y^2}dy\,dx=\int_0^{\pi}\int_0^{1} e^{r^2} r\,dr\,d\theta=\int_0^{\pi}\left[\frac{1}{2}e^{r^2}\right]_0^1 d\theta=\int_0^{\pi}\frac{1}{2}(e-1)d\theta=\frac{\pi}{2}(e-1).$$

EXAMPLE

Let R_a be the region bounded by the circle $x^2+y^2=a^2$. Define

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}dx\,dy=\lim_{a\to\infty}\iint_R e^{-(x^2+y^2)}dx\,dy$$

To evaluate this improper integral.

$$I=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\exp\{-(x^2+y^2)\}\,dx\,dy=\lim_{a\to\infty}\iint_{Da}\exp\{-(x^2+y^2)\}\,dx\,dy$$

$$=\lim_{a\to\infty}\int_0^{2\pi}\int_0^{a}\exp\{-r^2\}r\,dr\,d\theta=\lim_{a\to\infty}\int_0^{2\pi}\frac{1}{2}(1-\exp\{-a^2\}d\theta=\lim_{a\to\infty}\frac{1}{2}(1-\exp\{-a^2\}\,\theta\Big|_0^{2\pi}$$

$$=\pi-\lim_{a\to\infty}\frac{\pi}{\exp\{-a^2\}}=\pi$$

EXAMPLE

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Prove that $\int_0^\infty e^{-t^2}\,dt=\frac{1}{2}\sqrt{\pi}$.

$$I=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\exp\{-(x^2+y^2)\}\,dx\,dy=\int_{-\infty}^{\infty}\exp\{-y^2\}\left[\int_{-\infty}^{\infty}\exp\{-x^2\}dx\right]dy=\left[\int_{-\infty}^{\infty}\exp\{-y^2\}dy\right]\left[\int_{-\infty}^{\infty}\exp\{-x^2\}dx\right]$$

$$=\left[\int_{-\infty}^{\infty}\exp\{-t^2\}dt\right]\left[\int_{-\infty}^{\infty}\exp\{-t^2\}\,dt\right]=\lim_{a\to\infty}\left[\int_{-a}^{a}\exp\{-t^2\}\,dt\right]^2=4\lim_{a\to\infty}\left[\int_0^a\exp\{-t^2\}\,dt\right]^2$$

Hence we have $4\lim_{a\to\infty}\left[\int_0^a\exp\{-t^2\}\,dt\right]=\lim_{a\to\infty}\int_{-a}^{a}\int_{-a}^{a}\exp\{-(x^2+y^2)\}dx\,dy\;4\lim_{a\to\infty}\left[\int_0^\infty\exp\{-t^2\}dt\right]^2=\pi$

$$\lim_{a\to\infty}\left[\int_0^\infty\exp\{-t^2\}dt\right]^2=\pi/4\;\Rightarrow\;\int_0^\infty\exp\{-t^2\}\,dt=\frac{\sqrt{\pi}}{2}$$

THEOREM

Let G be the rectangular box defined by the inequalities

$a\le x\le b,\quad c\le y\le d,\quad k\le z\le \ell$

If f is continuous on the region G, then $\iiint_G f(x,y,z)\,dV=\int_a^b\int_c^d\int_k^\ell f(x,y,z)\,dz\,dy\,dx$

Moreover, the iterated integral on the right can be replaced with any of the five other iterated integrals that result by altering the order of integration.

$$=\int_c^d\int_k^\ell\int_a^b f(x,y,z)\,dy\,dz\,dx=\int_c^d\int_a^b\int_k^\ell f(x,y,z)\,dy\,dx\,dz=\int_a^b\int_c^d\int_k^\ell f(x,y,z)\,dx\,dy\,dz$$

$$=\int_a^b\int_k^\ell\int_c^d f(x,y,z)\,dx\,dz\,dy=\int_k^\ell\int_a^b\int_c^d f(x,y,z)\,dz\,dx\,dy$$

EXAMPLE

Evalaute the triple integral $\iiint_G 12xy^2z^3\,dV$ over the rectangular box G defined by the inequalities $-1\le x\le 2,\ 0\le y\le 3,\ 0\le z\le 2$.

We first integrate with respect to z, holding x and y fixed, then with respect to y holding x fixed, and finally with respect to x.

$$\iiint_G 12xy^2z^3\,dV=\int_{-1}^{2}\int_0^3\int_0^2 12xy^2z^3\,dz\,dy\,dx=\int_{-1}^{2}\int_0^3\left[3xy^2z^4\right]_{z=0}^{2}dy\,dx=\int_{-1}^{2}\int_0^3 48xy^2\,dy\,dx$$

$$=\int_{-1}^{2}\left[16xy^3\right]_{y=0}^{3}dx=\int_{-1}^{2}432x\,dx=216x^2\Big]_{-1}^{2}=648$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

EXAMPLE

Evalaute $\iiint_R (x - 2y + z)\, dx\, dy\, dz$ Region R : $0 \le x, 1, 0 \le y \le x^2, 0 \le z \le x + y$

$$= \int_0^1 \int_0^{x^2} \int_0^{x+y} (x - 2y + z)\, dz\, dy\, dx = \int_0^1 \int_0^{x^2} \left| \frac{(x - 2y + z)^2}{2} \right|_0^{x+y} dy\, dx$$

$$= \int_0^1 \int_0^{x^2} \left[\frac{(x-2y+x+y)^2}{2} - \frac{(x-2y)^2}{2} \right] dy\, dx = \frac{1}{2} \int_0^1 \int_0^{x^2} (3x2 - 3y2)\, dy\, dx = \frac{3}{2} \int_0^1 \left| x^2 y - \frac{y^3}{3} \right|_0^{x^2} dx$$

$$= \frac{3}{2} \int_0^1 \left(x4 - \frac{x^6}{z} \right) dx = \frac{3}{2} \left| \frac{x^5}{5} - \frac{x^7}{21} \right|_0^1 = \frac{3}{2} \left[\frac{1}{5} - \frac{1}{21} \right] = \frac{8}{35}$$

Example:

Evalaute $\iiint_S xyz\, d x\, dy\, dz$ Where $S = \{(x,y,z): x^2+y^2+z^2 \le 1, x \ge 0, y \ge 0, z \ge 0\}$

S is the sphere $x^2 + y^2 + z^2 = 1$ .Since x, y, z are al +ve so we have to consider only the +ve octant of the sphere.

Now $x^2 + y^2 + z^2 = 1$. So that $z = \sqrt{1 - x^2 - y^2}$

The Projection of the sphere on xy plan is the circle $x^2 + y^2 = 1$.

This circle is covered as y-varies from 0 to $\sqrt{1 - x^2}$ and x varies from 0 to 1.

$$\iiint_R xyz\, dx\, dy\, dz = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dz\, dy\, dx = \int_0^1 \int_0^{\sqrt{1-x^2}} xy \left| \frac{z^5}{2} \right|_0^{\sqrt{1-x^2-y^2}} dy\, dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \left(\frac{1 - x^2 - y^2}{2} \right) dy\, dx = \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} x\,(y - x^2y - y^3)\, dy\, dx$$

$$= \frac{1}{2} \int_0^1 x \left(\frac{y^2}{2} - \frac{x^2y^2}{2} - \frac{y^4}{4} \right) \Big|_0^{\sqrt{1-x^2}} dx = \frac{1}{4} \int_0^1 x \left[1 - x^2 - x^2\,(1-x^2) - \frac{(1-x^2)^2}{2} \right] dx$$

$$= \frac{1}{8} \int_0^1 (x - 2x^3 + x^5)\, dx = \frac{1}{8} \left| \frac{x^2}{2} - \frac{x^4}{2} + \frac{x^6}{6} \right|_0^1 = \frac{1}{8} \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right) = \frac{1}{48}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -27 Vector Valued Functions

Recall that a function is a rule that assigns to each element in its domain one and only one element in its range. Thus far, we have considered only functions for which the domain and range are sets of real numbers; such functions are called real-valued functions of a real variable or sometimes simply real-valued functions. In this section we shall consider functions for which the domain consists of real numbers and the range consists of vectors in 2-space or 3-space; such functions are called vector-valued functions of a real variable or more simply vector-valued functions. In 2-space such functions can be expressed in the form.

$\mathbf{r}(t) = (x(t), y(t)) = x(t)\,\mathbf{i} + y(t)\,\mathbf{j}$

and in 3-space in the form

$\mathbf{r}(t) = (x(t), y(t), z(t))\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$

where x(t), y(t), and z(t) are real-valued functions of the real variable t. These real-valued functions are called the component functions or components of **r**. As a matter of notation, we shall denote vector-valued functions with boldface type [**f(t), g(t)**, and **r** (t) and real-valued functions, as usual, with lightface italic type [*f(t), g(t), and r(t)*].

EXAMPLE

$\mathbf{r}(t) = (\ln t)\,\mathbf{i} + \sqrt{t^2 + 2}\,\mathbf{j} + (\cos t\pi)\mathbf{k}$

then the component functions are $x(t) = \ln t, y(t) = \sqrt{t^2 + 2}$, and $z(t) = \cos t\pi$

The vector that r(t) associates with t = 1 is $r(1) = (\ln 1)\,\mathbf{i} + \sqrt{3}\mathbf{j} + (\cos \pi)\,\mathbf{k} = \sqrt{3}\mathbf{j} - \mathbf{k}$

The function **r** is undefined if $t \leq 0$ because ln *t* is undefined for such *t*.

If the domain of a vector-valued function is not stated explicitly, then it is understood to consist of all real numbers for which every component is defined and yields a real value. This is called the natural domain of the function. Thus the natural domain of a vector-valued function is the intersection of the natural domains of its components.

PARAMETRIC EQUATIONS IN VECTOR FORM

Vector-valued functions can be used to express parametric equations in 2-space or 3-space in a compact form.

For example, consider the parametric equations $x = x(t),\ y = y(t)$

Because two vectors are equivalent if and only if their corresponding components are equal, this pair of equations can be replaced by the single vector equation.

$x = x(t),\ y = y(t)$

$x\,\mathbf{i} + y\,\mathbf{j} = x(t)\mathbf{i} + y(t)\,\mathbf{j}$

Similarly, in 3-space the three parametric equations

$x = x(t),\ y = y(t),\ z = z(t)$

can be replaced by the single vector equation

$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$

if we let $r = x\,\mathbf{i} + y\,\mathbf{j}$ and $\mathbf{r}(t) = x(t)\,\mathbf{i} + y(t)\,\mathbf{j}$ in 2-sapce

and let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{r}(t) = x(t)\,\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$

in 3-space, then both (2) and (4) can be written as $\mathbf{r} = \mathbf{r}(t)$

which is the vector form of the parametric equations in (1) and (3). Conversely, every vector equation of form (5) can be rewritten as parametric equations by equating components on the two sides.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

EXAMPLE

Express the given parametric equations as a single vector equation.

(a) $x = t^2$, $y = 3t$

(b) $x = \cos t$, $y = \sin t$, $z = t$

(a) Using the two sides of the equations as components of a vector yields.

$$x\,\mathbf{i} + y\,\mathbf{j} = t^2\,\mathbf{i} + 3t\,\mathbf{j}$$

(b) Proceeding as in part (a) yields

$$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$$

EXAMPLE

Find parametric equations that correspond to the vector equation

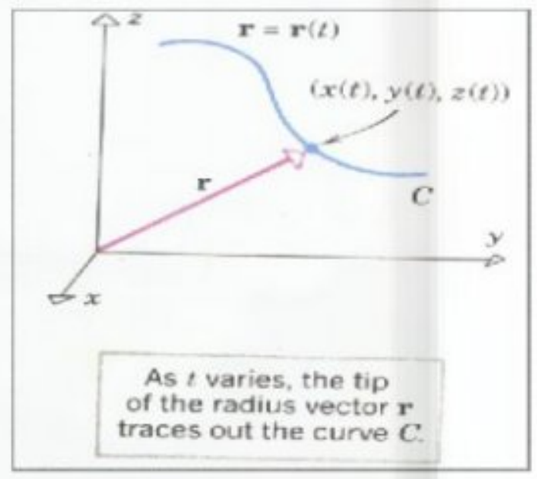
$$x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k} = (t^3 + 1)\,\mathbf{i} + 3\,\mathbf{j} + e^t\,\mathbf{k}$$

Equating corresponding components yields.

$x = t^3 + 1$, $y = 3$, $z = e^t$

GRAPHS OF VECOR-VALUED FUNCTOINS

One method for interpreting a vector-valued function r(t) in 2-space or 3-space geometrically is to position the vector $r = r(t)$ with its initial point at the origin, and let C be the curve generated by the tip of the vector r as the parameter t varies



The vector r, when positioned in this way, is called the radius vector or position vector of C, and C is called the graph of the function r (t) or, equivalently, the graph of the equation r = r (t). The vector equation r = r (t) is equivalent to a set of parametric equations, so C is also called the graph of these parametric equations.

EXAMPLE

Sketch the graph of the vector-valued function $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}$, $0 \le t \le 2\pi$

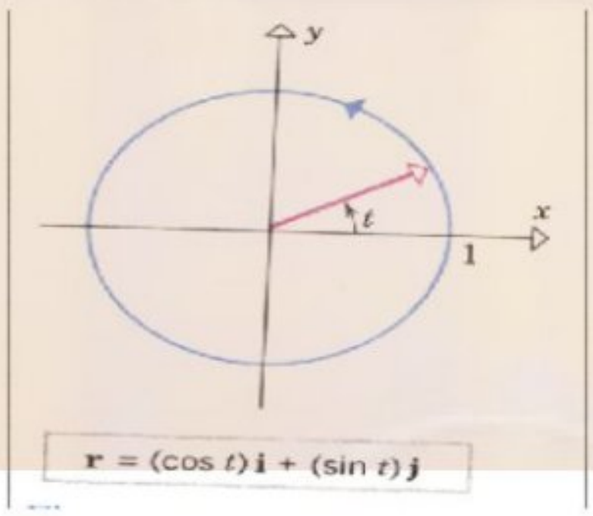
The graph of r (t) is the graph of the vector equation

$x\mathbf{i} + y\mathbf{j} = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}$, $0 \le t \le 2\pi$

or equivalently, it is the graph of the parametric equations

$x = \cos t$, $y = \sin t$ $\quad (0 \le t \le 2\pi)$

This is a circle of radius 1 that is centered at the origin with the direction of increasing t counterclockwise. The graph and a radius vector are shown in Fig.



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

EXAMPLE

Sketch the graph of the vector-valued function $\mathbf{r}(t) = (\cos t)\mathbf{i}+(\sin t)\mathbf{j}+2\mathbf{k}$, $0 \le t \le 2\pi$

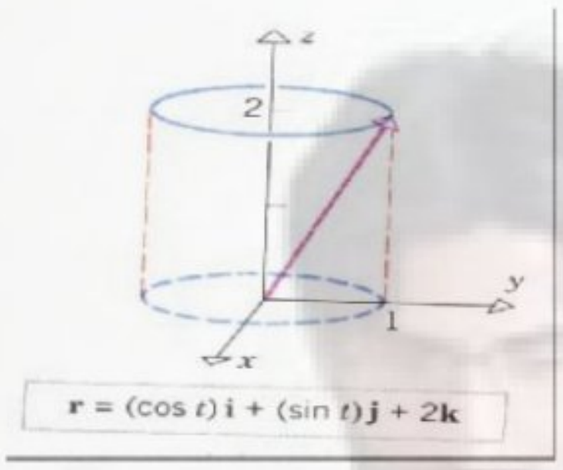
The graph of **r**(t) is the graph of the vector equation

$x\mathbf{i}+y\mathbf{j}+z\mathbf{k}=(\cos t)\mathbf{i}+(\sin t)\mathbf{j}+2\mathbf{k}$, $0 \le t \le 2\pi$

or, equivalently, it is the graph o the parametric equations

$x=\cos t, \quad y = \sin t, \quad z=2$ $(0 \le t \le 2\pi)$

From the last equation, the tip o the radius vecor traces a curve in the plane z = 2, and from the first two equations and the preceding example, the curve is a circle of radius 1 centered on the z-axis and traced counterclockwise looking down the z-axis. The graph and a radius vector are shown in Fig.



EXAMPLE

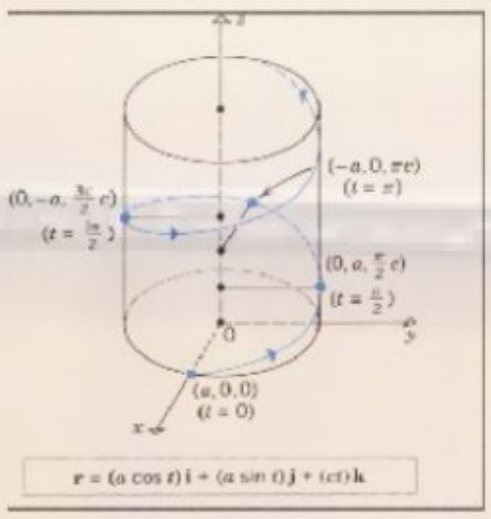
Sketch the graph of the vector-valued function $\mathbf{r}(t) = (a\cos t)\mathbf{i} + (a\sin t)\mathbf{j} + (ct)\mathbf{k}$ where a and c are positive constant.

The graph of **r**(t) is the graph of the parametric equations.

$x = a\cos t, \quad y = a \sin t, \quad z = ct$

As the parameter t increases, the value of z = ct also increases, so the point (x, y, z) moves upward. However, as t increases, the point (x, y, z) also moves in a path directly over the circle. $x = a\cos t, \quad y = a \sin t$ in the xy-plane. The combination of these upward and circular motions produces a corkscrew-shaped curve that wraps around a right-circular cylinder of radius a centered on the z-axis.

This curve is called a circular helix.



EXAMPLE

Describe the graph of the vector equation $\mathbf{r} = (-2 + t)\mathbf{i} + 3t\mathbf{j} + (5 - 4t)\mathbf{k}$

The corresponding parametric equations are $x = -2 + t, \quad y = 3t, \quad z = 5 - 4t$

The graph is the line in 3-space that passes through the point (– 2, 0, 5) and is parallel to the vector $\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$.

EXAMPLE

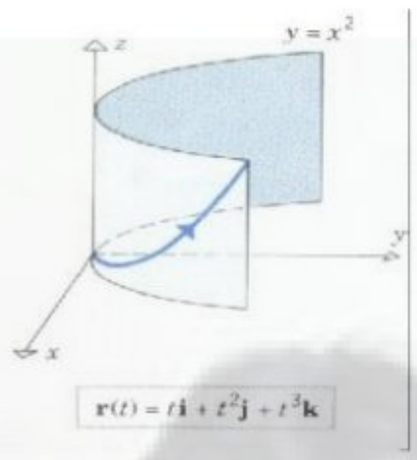
The graph of the vector-valued function $r(t) = t\,\mathbf{i} + t^2\,\mathbf{j} + t^3\,\mathbf{k}$ is called a twisted cubic. Show that this curve lies o the parabolic cylinder $y = x^2$, and sketch the graph for $t \ge 0$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

The corresponding parametric equations are $x = t, \quad y = t^2, \quad z = t^3$
Eliminating the parameter t in the equations for x and y yields $y = x^2$, so the curve lies on the parabolic cylinder with this equation. The curve starts at the origin for t = 0; as t increases, so do x, y, and z, so the curve is traced in the upward direction, moving away from the origin along the cylinder.



GRAPHS OF CONSTANT VECOR-VALUED FUNCTIONS

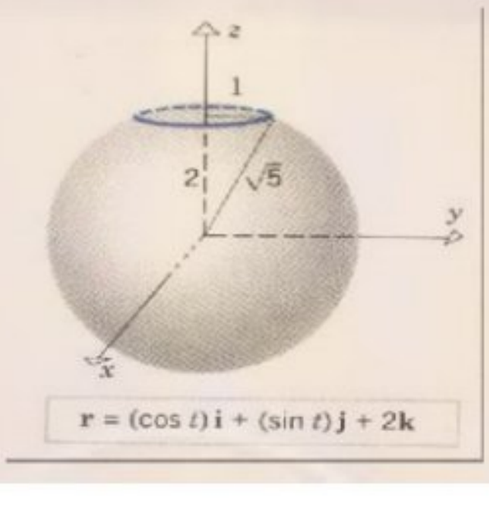
If c is a constant vector in the sense that it does not depend on a parameter, then the graph of $\mathbf{r} = \mathbf{c}$ is a single point since the radius vector remains fixed with its tip at **c**.
If $\mathbf{c} = x_0\mathbf{i} + y_0\mathbf{j}$ (in 2-space), then the graph is the point (x_0, y_0), and if $c = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ (in 3-space), then the graph is the point (x_0, y_0, z_0).

EXAMPLE

The graph of the equation $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ is the point (2, 3, – 1) in 3-space.
If r(t) is a vector-valued function, then for each value of the parameter t, the expression ||r(t)|| is a real-valued function of t because the norm (or length of r(t) is a real number.
For example,
If $\mathbf{r}(t) = t\,\mathbf{i} + (t-1)\,\mathbf{j}$
Then $\|r(t)\| = \sqrt{t^2 + (t-1)^2}$ which is a real-valued function of t.

EXAMPLE

The graph of $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + 2\mathbf{k}, \quad 0 \le t \le 2\pi$
is a circle of radius 1 centered on the z-axis and lying in the plane $z = 2$. This circle lies on the surface of a sphere of radius $\sqrt{5}$ because for each value of t
$\|r(t)\| = \sqrt{\cos^2 t + \sin^2 t + t} = \sqrt{1+4} = \sqrt{5}$
which shows that each point on the circle is a distance of $\sqrt{5}$ units from the origin.



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -28 Limits of Vector Valued Functions

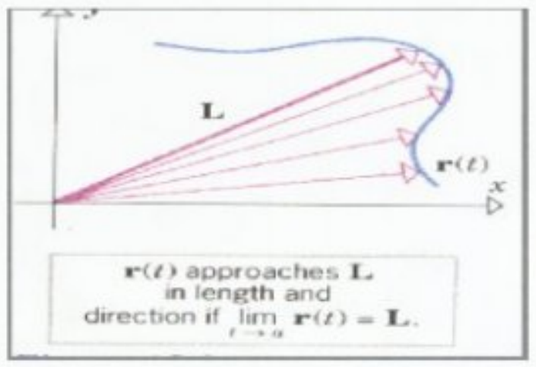
The limit of a vector-valued functions is defined to be the vector that results by taking the limit of each component. Thus, for a function $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ in 2-space we define.

$$\lim_{t\to\alpha} r(t) = (\lim_{t\to\alpha} x(t))\mathbf{i} + (\lim_{t\to\alpha} y(t))\mathbf{j}$$

and for a function $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ in 3-space we define.

$$\lim_{t\to\alpha} r(t) = (\lim_{t\to\alpha} x(t))\mathbf{i} + (\lim_{t\to\alpha} y(t))\mathbf{j} + (\lim_{t\to\alpha} z(t))k$$

If the limit of any component does not exist, then we shall agree that the limit of r (t) does not exist.



These definitions are also applicable to the one-sided and infinite limits $\lim_{t\to\alpha^+}$, $\lim_{t\to\alpha^-}$ $\lim_{t\to+\infty}$, and $\lim_{t\to-\infty}$. It follows from (1) and (2) that

$$\lim_{t\to\alpha} r(t) = L$$

if and only if the components of r(t) approach the components of L as $t \to a$. Geometrically, this is equivalent to stating that the length and direction of r (t) approach the length and direction of L as $t \to \alpha$

CONTINUITY OF VECTOR-VALUED FUNCTIONS

The definition of continuity for vector-valued functions is similar to that for real-valued functions. We shall say that r is continuous at t_0 if

1. $r(t_0)$ is defined;
2. $\lim_{t\to t_0} r(t)$ exists;
3. $\lim_{t\to t_0} r(t) = r(t_0)$.

It can be shown that r is continuous at t_0 if and only if each component of r is continuous. As with real-valued functions, we shall call r continuous everywhere or simply continuous if r is continuous at al real values of t. geometrically, the graph of a continuous vector-valued function is an unbroken curve.

DERIVATIVES OF VECOR-VALUED FUNCTIONS

The definition of a derivative for vector-valued functions is analogous to the definition for real-valued functions.

DEFINITION

The derivative $r'(t)$ of a vector-valued function r(t) is defined by

$$\mathbf{r'(t)} = \lim_{\mathbf{h}\to 0} \frac{\mathbf{r\,(t+h) - r\,(t)}}{\mathbf{h}}$$

Provided this limit exists.

For computational purposes the following theorem is extremely useful; it states that the derivative of a vector-valued function can be computed by differentiating each components.

THEOREM

(a) If $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ is a vector-valued function in 2-space, and if x(t) and y(t) are differentiable, then

$$r'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(b) If $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ is a vector-valued function in 3-space, and if x(t), y(t), and z(t) are differentiable, then

$$r'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}$$

We shall prove part (a). The proof of (b) is similar.

Proof (a):

$$\mathbf{r}'(t) = \lim_{h\to 0}\frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = \lim_{h\to 0}\frac{[x(t+h)-x(t)]}{h}\mathbf{i} + \lim_{h\to 0}\frac{[y(t+h) - y(t)]}{h}\mathbf{j}$$

$$= x'(t)\mathbf{i} + y'(t)\mathbf{j}$$

As with real-valued functions, there are various notations for the derivative of a vector-valued function. If r = r (t), then some possibilities are

$\frac{d}{dt}[r(t)]$, $\frac{dr}{dt}$, $\mathbf{r}'(t)$, and $\mathbf{r}'$

EXAMPLE

Let $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j}$. Find $\mathbf{r}'(t)$ and $\mathbf{r}'(1)$

$$\mathbf{r}'(t) = \frac{d}{dt}[t^2]\,\mathbf{i} + \frac{d}{dt}[t^3]\,\mathbf{j}$$

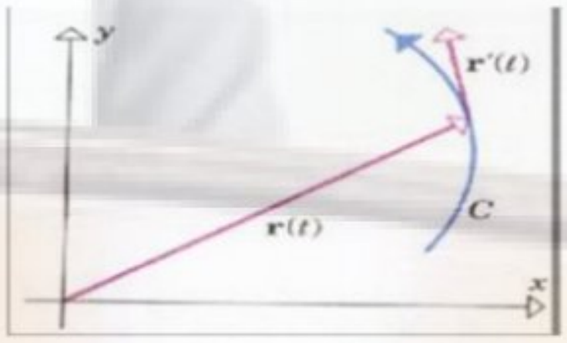
$$= 2t\,\mathbf{i}\;\; 3t^2\,\mathbf{j}$$

Substituting t=1 yields

$\mathbf{r}'(1) = 2\mathbf{i}+3\mathbf{j}$.

TAGENT VECTORS AND TANGENT LINES
GEOMETRIC INTERPRETATIONS OF THE DERIVATIVE.

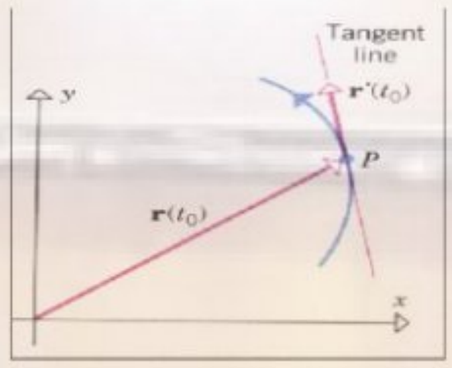
Suppose that C is the graph of a vector-valued function r(t) and that $\mathbf{r}'(t)$ exists and is nonzero for a given value of t. If the vector $\mathbf{r}'(t)$ is positioned with its initial point at the terminal point of the radius vector



DEFINITION

Let P be a point on the graph of a vector-valued function $\mathbf{r}(t)$, and let $\mathbf{r}(t_0)$ be the radius vector from the origin to P

If $\mathbf{r}'(t_0)$ exists and $\mathbf{r}'(t_0) \neq \mathbf{0}$, then we call $\mathbf{r}'(t_0)$ the tangent vector to the graph of $\mathbf{r}$ at $\mathbf{r}(t_0)$



REMARKS

Observe that the graph of a vector-valued function can fail to have a tangent vector at a point either because the derivative in (4) does not exist or because the derivative is zero at the point.If a vector-valued function $\mathbf{r}(t)$ has a tangent vector $\mathbf{r}'(t_0)$ at a point onits graph, then the line that is parallel to $\mathbf{r}'(t_0)$ and passes through the tip of the radius vector $\mathbf{r}(t_0)$ is called the tangent line of the graph of $\mathbf{r}(t_)$ at $\mathbf{r}(t_0)$

Vector equation of the tangent line is

$$\mathbf{r} = \mathbf{r}(t_0) + t\,\mathbf{r}'(t_0)$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

EXAMPLE

Find parametric equation of the tangent line to the circular helix

$x = \cos t$, $y = \sin t$, $z = 1$ at the point where $t = \pi/6$

To find a vector equation of the tangent line, then we shall equate components to obtain the parametric equations. A vector equation r=r(t) of the helix is

$$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$$

Thus, $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$

$\Rightarrow \mathbf{r}'(t) = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j} + k$

At the point where $t = \pi/6$, these vectors are

$$\mathbf{r}\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{\pi}{6}\mathbf{k} \quad \text{and}$$

$$\mathbf{r}'\left(\frac{\pi}{6}\right) = -\frac{1}{2}i + \frac{\sqrt{3}}{2}\mathbf{j} + \mathbf{k}$$

so from (5) with $t_0 = \pi/6$ a vector equation of the tangent line is

$$\mathbf{r}\left(\frac{\pi}{6}\right) + t\,\mathbf{r}'\left(\frac{\pi}{6}\right) = \left(\frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{\pi}{6}\mathbf{k}\right) + t\left(-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j} + \mathbf{k}\right)$$

Simplifying, then equating the resulting components with the corresponding components of $r = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ yields the parametric equation.

$$x = \frac{\sqrt{3}}{2} - \frac{1}{2}t, \quad y = \frac{1}{2} + \frac{\sqrt{3}}{2}t, \quad z = \frac{\pi}{6} + t$$

EXAMPLE

The graph of $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j}$ is called a semicubical parabola

Find a vector equation of the tangent line to the graph of **r**(t) at

(a) the point (0,0) (b) the point (1,1)

The derivative of **r**(t) is

$\mathbf{r}'(t) = 2t\mathbf{i} + 3t^2\mathbf{j}$

(a) The point (0, 0) on the graph of **r** corresponds to $t = 0$. As this point we have $\mathbf{r}'(0) = 0$, so there is no tangent vector at the point and consequently a tangent line does not exist at this point.

(b) The point (1, 1) on the graph of r corresponds to $t = 1$, so from (5) a vector equation of the tangent line at this point is $\mathbf{r} = \mathbf{r}(1) + t\,\mathbf{r}'(1)$

From the formulas for r (t) and $\mathbf{r}'(t)$ with $t = 1$, this equation becomes

$$\mathbf{r} = (\mathbf{i} + \mathbf{j}) + t\,(2\mathbf{i} + 3\mathbf{j})$$

If **r** is a vector-valued function in 2-space or 3-space, then we say that **r**(t) is smoothly parameterized or that **r** is a smooth function of t if the components of r have continuous derivatives with respect to t and $\mathbf{r}'(t) \neq \mathbf{0}$ for any value of t. Thus, in 3-space $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$

is a smooth function of t if $x'(t)$, $y'(t)$, and $z'(t)$ are continuous and there is no value of t at which al three derivatives are zero. A parametric curve C in 2-space or 3-space will be called smooth if it is the graph of some smooth vector-valued function.

It can be shown that a smooth vector-valued function has a tangent line at every point on its graph.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

PROPERTIES OF DERIVATIVES

(Rules of Differentiation).

In either 2-space or 3-space let $\mathbf{r}(t)$, $\mathbf{r}_1(t)$, and $\mathbf{r}_2(t)$ be vector-valued functions, f(t) a real-valued function, k a scalar, and c a fixed (constant) vector. Then the following rules of differentiation hold:

$$\frac{d}{dt}[c] = 0$$

$$\frac{d}{dt}[kr(t)] = k\frac{d}{dt}[r(t)]$$

$$\frac{d}{dt}[\mathbf{r}_1(t) + \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] + \frac{d}{dt}[\mathbf{r}_2(t)]$$

$$\frac{d}{dt}[\mathbf{r}_1(t) - \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] - \frac{d}{dt}[\mathbf{r}_2(t)]$$

$$\frac{d}{dt}[f(t)\mathbf{r}(t)] = f(t)\frac{d}{dt}[\mathbf{r}(t)] + \mathbf{r}(t)\frac{d}{dt}[f(t)]$$

In addition to the rules listed in the foregoing theorem, we have the following rules for differentiating dot products in 2-space or 3-space and cross products in 3-space:

$$\frac{d}{dt}[\mathbf{r}_1(t).\mathbf{r}_2(t)] = \mathbf{r}_1.\frac{d\mathbf{r}_2}{dt} + \frac{d\mathbf{r}_1}{dt}.\mathbf{r}_2 \quad (6)$$

$$\frac{d}{dt}[\mathbf{r}_1(t)\times\mathbf{r}_2(t)] = \mathbf{r}_1\times\frac{d\mathbf{r}_2}{dt} + \frac{d\mathbf{r}_1}{dt}\times\mathbf{r}_2 \quad (7)$$

REMARK:

In (6), the order of the factors in each term on the right does not matter, but in (7) it does.

In plane geometry one learns that a tangent line to a circle is perpendicular to the radius at the point of tangency. Consequently, if a point moves along a circular arc in 2-space, one would expect the radius vector and the tangent vector at any point on the arc to be perpendicular. This is the motivation for the following useful theorem, which is applicable in both 2-space and 3-space.

THEOREM:

If r (t) is a vector-valued function in 2-space or 3-space and ||r(t)|| is constant for all t, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$

that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t.It follows from (6) with $\mathbf{r}_1(t)=\mathbf{r}_2(t)=\mathbf{r}(t)$ that

$$\frac{d}{dt}[\mathbf{r}(t).\mathbf{r}(t)] = \mathbf{r}(t).\frac{d\mathbf{r}}{dt} + \frac{d\mathbf{r}}{dt}.\mathbf{r}(t)$$

or, equivalently, $\frac{d}{dt}[\|\mathbf{r}(t)\|]^2 = 2\mathbf{r}(t) \cdot \frac{d\mathbf{r}}{dt}$

But $\|\mathbf{r}(t)\|^2$ is constant, so its derivative is zero. Thus $2r(t).\frac{d\mathbf{r}}{dt} = 0$ that is $\mathbf{r}(t) \cdot \frac{d\mathbf{r}}{dt} = 0$

That is the $\mathbf{r}(t)$ is perpendicular $\frac{d\mathbf{r}}{dt}$

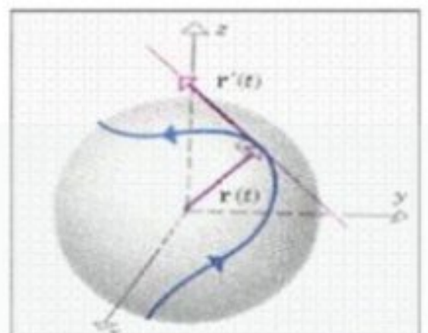
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

EXAMPLE

Just as a tangent line to a circular arc in 2-space is perpendicular to the radius at the point of tangency, so a tangent line to a curve on the surface of a sphere in 3-space is perpendicular to the radius at the point of tangency.



To see that this is so, suppose that the graph of **r**(t) lies on the surface of the sphere of radius k > 0 centered at the origin.For each value of *t* we have ||**r**(t)||=k,

$$\mathbf{r}(t).\ \mathbf{r}'(t) = 0$$

which implies that the radius vector **r**(t) and the tangent vector **r**′(t) are perpendicular. This completes the argument because the tangent line, where it exists, is parallel to the tangent vector.

INTEGRALS OF VECTOR VALUED FUNCTION

(a) If r(t)=x(t)I +y(t) j is a vector-valued function in 2-space ,the we define.

$$\int r(t)dt = (\int x(t)dt)i + (\int y(t)dt)\,j \qquad (1a)$$

$$\int_a^b r(t)dt = (\int_a^b x(t)dt)i + (\int_a^b y(t)dt)\,j \qquad (1b)$$

(b) If r(t) = x (t) **i** + y(t) **j** + z(t)k is a vector-valued function in 3-space, then we define.

$$\int \mathbf{r}(t)dt = \left(\int x(t)dt\right)\mathbf{i} + \left(\int y(t)\,dt\right)\mathbf{j} + \left(\int z(t)dt\right)\mathbf{k} \qquad \mathbf{(2a)}$$

$$\int_a^b \mathbf{r}(t)dt = \left(\int_a^b x(t)dt\right)\mathbf{i} + \left(\int_a^b y(t)dt\right)\mathbf{j} + \left(\int_a^b z(t)dt\right)\mathbf{k} \qquad \mathbf{(2b)}$$

Let $r(t) = 2ti + 3t^2 j$

$(a)\int r(t)dt \qquad (b)\int_0^2 r(t)dt$

$$\int r(t)dt = \int (2ti + 3t^2 j)dt = (\int 2tdt)i + (\int\ 3t^2 dt)\,j$$

$$(t^2 + C_1)i + (t^3 + C_2)\,j = t^2 i + C_1 i + t^3 j + C_2 j = t^2 i + t^3 j + C_1 i + C_2 j = t^2 i + t^3 j + C$$

Where $C = C_1 i + C_2 j$ *is an arbitrary vector constant of integration*

$$(b)\int_0^2 r(t)dt = \int_0^2 (2ti + 3t^2 j)dt = (\int_0^2 2tdt)i + (\int_0^2 3t^2 dt)\,j = \left[t^2\right]_0^2 i + \left[t^3\right]_0^2 j = (2^2 - 0)i + (2^3 - 0)\,j = 4i + 8j$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

PROPERTEIS OF INTEGRALS

$$\int cr(t)\,dt = c\int r(t)\,dt \qquad (3)$$

$$\int [r_1(t)+r_2(t)]\,dt = \int r1(t)dt + \int r_2(t)\,dt \qquad (4)$$

$$\int [r_1(t)-r_2(t)]dt = \int r_1(t)\,dt - \int r_2(t)\,dt \qquad (5)$$

These properties also hold for definite integrals of vector-valued functions. In addition, we leave it for the reader to show that if r is a vector-valued function in 2-space or 3-space, then $\frac{d}{dt}\left[\int r(t)\,dt\right] = r(t)$ (6)

This shows that an indefinite integrals of **r**(t) is, in fact, the set of antiderivatives of r(t), just as for real-valued functions.

If **r**(t) is any antiderivative or **r**(t) in the sense that $\mathbf{R}'(t) = \mathbf{r}(t)$, then

$$\int r(t)\,dt = \mathbf{R}(t) + \mathbf{C} \qquad (7)$$

where C is an arbitrary vector constant of integration. Moreover,

$$\int_a^b \mathbf{r}(t)\,dt = \mathbf{R}(t)\Big]_a^b = \mathbf{R}(b) - \mathbf{R}(a).$$

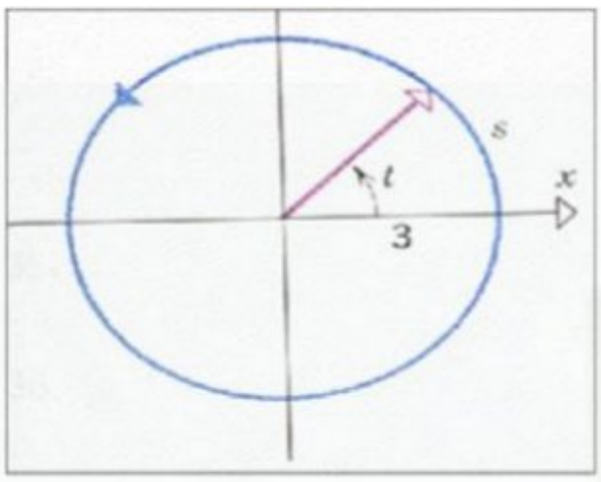
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -29 Change of parameter

It is possible for different vector-valued functions to have the same graph.
For example, the graph of the function
$r = (3 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j}$, $0 \leq t \leq 2\pi$
is the circular of radius 3 centered at the origin with counterclockwise orientation. The parameter t can be interpreted geometrically as the positive angle in radians from the x-axis to the radius vector. For each value of t, let s be the length of the arc subtended by this angle on the circle



The parameters s and t are related by
$t = s/3, \quad 0 < s < 6\pi$
if we substitute this in (10), we obtain a vector-valued function of the parameter s, namely
$r = 3 \cos (s/3)\mathbf{i} + 3 \sin (s/3)\mathbf{j}, \quad 0 \leq s \leq 6\pi$
whose graph is also the circle of radius 3 centered at the origin with counterclockwise orientation .In various problems it is helpful to change the parameter in a vector-valued function by making an appropriate substitution. For example, we changed the parameter above from t to s by substituting t=s/3 in (10).
In general, if g is a real-valued function, then substituting $t = g(u)$ in r(t) changes the parameter from t to u.

When making such a change of parameter, it is important to ensure that the new vector-valued function of u is smooth if the original vector-valued function of t is smooth. It can be proved that this will be so if g satisfies the following conditions:

1. g is differentiable.
2. g' is continuous.
3. $g'(u) \neq 0$ for any u in the domain of g.
4. The range of g is the domain of **r**.

If g satisfies these conditions, then we call $t = g(u)$ a smooth change of parameter. Henceforth, we shall assume that all changes of parameter are smooth, even if it is not stated explicitly.

ARC LENGTH

Because derivatives of vector-valued functions are calculated by differentiating components, it is natural do define integrals of vector-functions in terms of components.

EXAMPLE

If $x'(t)$ and $y'(t)$ are continuous for $a \leq t \leq b$, then the curve given by the parametric equations
$x = x(t), \quad y = y(t) \quad (a \leq t \leq b) \qquad (9)$
has arc length

$$\mathbf{L} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \qquad (10)$$

This result generalizes to curves in 3-spaces exactly as one would expect:
If $x'(t)$, $y'(t)$, and $z'(t)$ are continuous for $a \leq t \leq b$, then the curve given by the parametric equations
$x = x(t), \quad y = y(t), \quad z = z(t) \quad (a \leq t \leq b)$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

has arc length

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \qquad (12)$$

EXAMPLE

Find the arc length of that portion of the circular helix

$x = \cos t, \quad y = \sin t, \quad z = t$

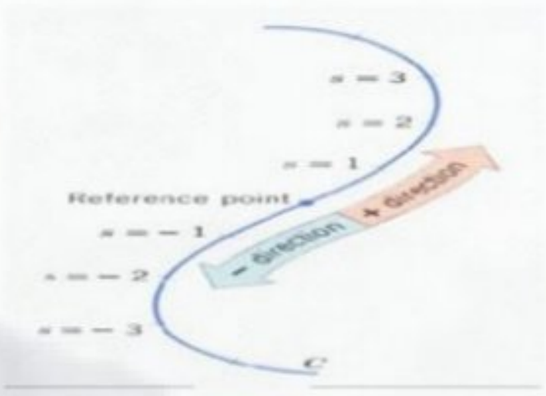
From $t = 0$ to $t = \pi$

The arc length is

$$L = \int_0^{\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt = \int_0^{\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + 1}\, dt$$

$$= \int_0^{\pi} \sqrt{2}\, dt = \sqrt{2}\, \pi$$

ARC LENTH AS A PARAMETER



For many purposes the best parameter to use for representing a curve in 2-space or 3-space parametrically is the length of arc measured along the curve from some fixed reference point. This can be done as follows:

Step 1: Select an arbitrary point on the curve C to serve as a reference point.

Step 2: Starting from the reference point, choose one direction along the curve to be the positive direction and the other to be the negative direction.

Step 3: If P is a point on the curve, let s be the "signed" arc length along C from the reference point to P, where s is positive if P is in the positive direction from the reference point, and s is negative if P is in the negative direction.

By this procedure, a unique point P on the curve is determined when a value for s is given. For example, $s = 2$ determines the point that is 2 units along the curve in the positive direction from the reference point, and $s = -\frac{3}{2}$ determines the point that is $\frac{3}{2}$ units along the curve in the negative direction from the reference point.

Let us now treat s as a variable. As the value of s changes, the corresponding point P moves along C and the coordinates of P become functions of s. Thus, in 2-space the coordinates of P are (x(x,), y(s)), and in 3-space they are (x(s), y(s), z(s)). Therefore, in 2-space the curve C is given by the parametric equations $x = x(s), \quad y = y(s)$

and in 3-space by $x = x(s), \quad y = y(s), \quad z = z(s)$

REMARKS

When defining the parameter s, the choice of positive and negative directions is arbitrary. However, it may be that the curve C is already specified in terms of some other parameter t, in which case we shall agree always to take the direction of increasing t as the positive direction for the parameter s. By so doing, s will increase as t increases and vice versa.

The following theorem gives a formula for computing an arc-length parameter s when the curve C is expressed in terms of some other parameter t. This result will be used when we want to change the parameterization for C from t to s.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

THEOREM

(a) Let C be a curve in 2-space given parametrically by

$x = x(t)$, $y = y(t)$

where $x'(t)$ and $y'(t)$ are continuous functions. If an arc-length parameter s is introduced with its reference point at $(x(t_0), y(t_0))$, then the parameters s and t are related by

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du \qquad (13a)$$

(b) Let C be a curve in 3-space given parametrically by

$x = x(t)$, $y = y(t)$, $z = z(t)$

where $x'(t)$, $y'(t)$, and $z'(t)$ are continuous functions. If an arc-length parameter s is introduced with its reference point at $(x(t_0), y(t_0), z(t_0))$, then the parameters s and t are related by

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}\, du \qquad (13b)$$

Proof

If $t > t_0$, then from (10) (with *u* as the variable of integration rather than *t*) it follows that

$$\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du \qquad (14)$$

represents the arc length of that portion of the curve C that lies between $(x(t_0), y(t_0))$ and $(x(t), y(t))$. If $t < t_0$, then (14) is the negative of this arc length. In either case, integral (14) represents the "signed" arc length *s* between these points, which proves (13a).

It follows from Formulas (13a) and (13b) and the Second Fundamental Theorem of Calculus (Theorem 5.9.3) that in 2-space.

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du\right] = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

and in 3-space

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}\, dt\right] = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

Thus, in 2-space and 3-space, respectively,

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \qquad (15a)$$

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \qquad (15b)$$

REMARKS:

Formulas (15a) and (15b) reveal two facts worth noting. First, ds/dt does not depend on t_0; that is, the value of ds/dt is independent of where the reference point for the parameter s is located. This is to be expected since changing the position of the reference point shifts each value of s by a constant (the arc length between the reference points), and this constant drops out when we differentiate. The second fact to be noted from (15a) and (15b) is that $ds/dt \geq 0$ for all t. This is also to be expected since s increases with t by the remark preceding Theorem 15.3.2. If the curve C is smooth, then it follows from (15a) and (15b) that $ds/dt \geq 0$ for all t .

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

EXAMPLE

$x = 2t + 1,\ y = 3t - 2$ (16)

using arc length s as a parameter, where the reference point for s is the point $(1, -2)$.

In formula (13a) we used u as the variable of integration because t was needed as a limit of integration. To apply (13a), we first rewrite the given parametric equations with u in place of t; this gives

from which we obtain

$x = 2u + 1, \quad y = 3u - 2$

$\frac{dx}{du} = 2, \quad \frac{dy}{du} = 3$

we see that the reference point $(1,-2)$ corresponds to $t = t_0 = 0$

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du = \int_{t_0}^{t} \sqrt{13}\, du = \sqrt{13u}\,\Big]_{u=0}^{u=t} = \sqrt{13t}$$

Therefore, $t = \frac{1}{\sqrt{13}}\, s$

Substituting this expression in the given parametric equations yields.

$$x = 2\left(\frac{1}{\sqrt{13}}s\right) + 1 = \frac{2}{\sqrt{13}}s + 1$$

$$y = 3\left(\frac{1}{\sqrt{13}}s\right) - 2 = \frac{3}{\sqrt{13}}s - 2$$

EXAMPLE

Find parametric equations for the circle $x = a\cos t,\ y = a\sin t\ \ (0 \le t \le 2\pi)$ using arc length s as a parameter, with the reference point for s being $(a, 0)$, where $a > 0$.

We first replace t by u in the given equations so that $x = a\cos u,\ \ y = a\sin u$

And $\frac{dx}{du} = -a\sin u,\ \ \frac{dy}{du} = a\cos u$

Since the reference point $(a, 0)$ corresponds to $t = 0$, we obtain

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du = \int_{t_0}^{t} \sqrt{(-a\sin u)^2 + (a\cos u)^2}\, du = \int_{0}^{t} a\, du = au\,\Big]_{u=0}^{u=t} = at$$

Solving for t in terms of s yields $t = s/a$

Substituting this in the given parametric equations and using the fact that $s = at$ ranges from 0 to $2\pi a$ as t ranges from 0 to 2π, we obtain

$x = a\cos(s/a),\ \ y = a\sin(s/a)\ \ (0 \le s \le 2\pi a)$

Example

Find Arc length of the curve $\mathbf{r}(t) = t^3\mathbf{i} + t\mathbf{j} + \frac{1}{2}\sqrt{6}\,t^2\,\mathbf{k},\ 1 \le t \le 3$

Here $x = t^3,\ y = t,\ z = \frac{1}{2}\sqrt{6}\,t^2$

$\frac{dx}{dt} = 3t^2,\ \ \frac{dy}{dt} = 1,\ \ \frac{dz}{dt} = \sqrt{6}\,t$

$$\text{Arc length} = \int_1^3 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt = \int_1^3 \sqrt{9t^4 + 1 + 6t^2}\, dt = \int_1^3 \sqrt{(3t^2 + 1)^2}\, dt$$

$$= \left|t^3 + t\right|_1^3 = (3)^3 + 3 - (1)^3 - 1 = 27 + 3 - 1 - 1 = 28$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

EXAMPLE

Calculate $\frac{dr}{du}$ by chain Rule.

$\mathbf{r} = e^{t}\mathbf{i} + 4e^{-t}\mathbf{j}$

$\frac{dr}{dt} = e^{t}\mathbf{i} - 4e^{-t}\mathbf{j}$

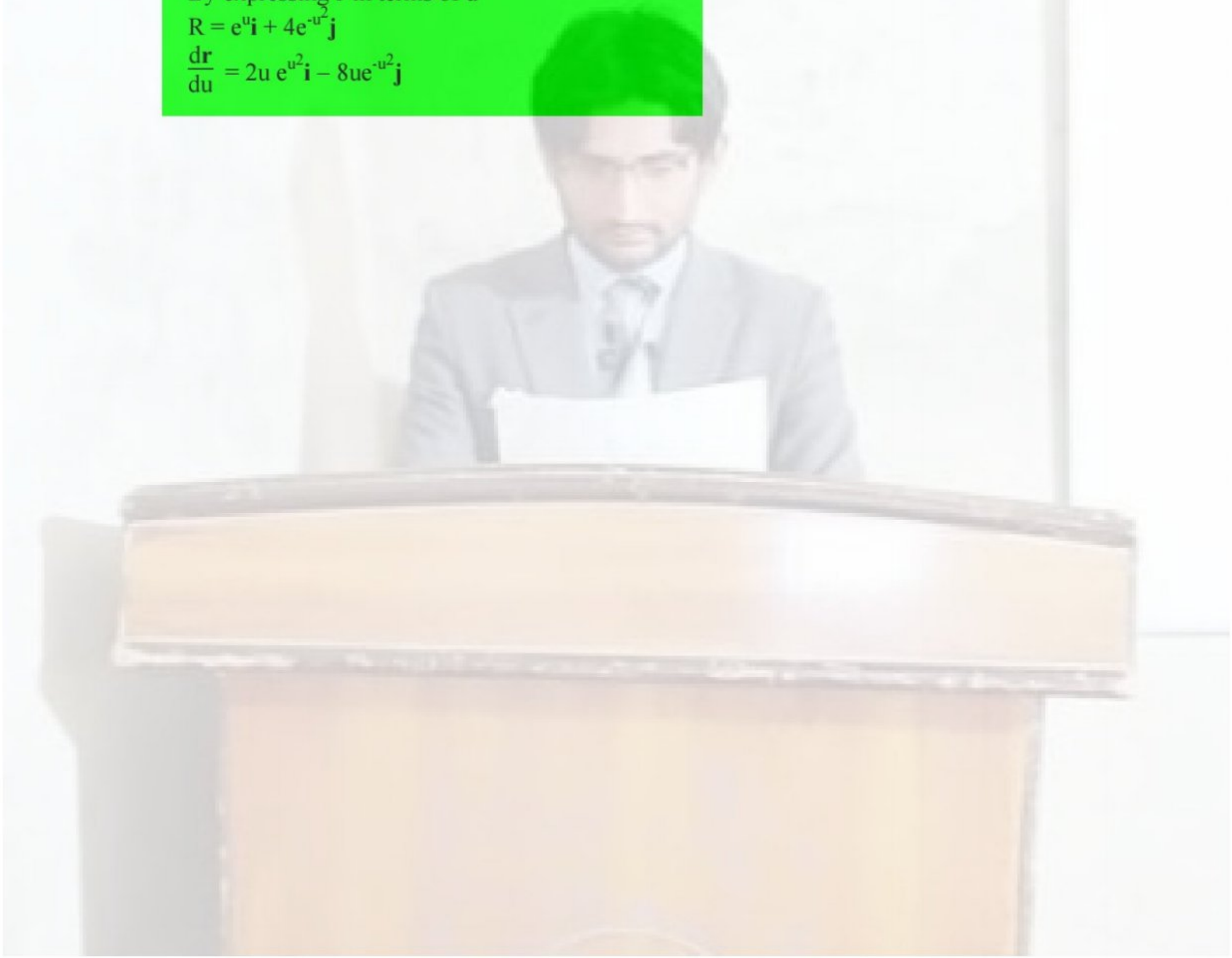
$\frac{dt}{du} = 2u$

$\frac{dr}{du} = \frac{dr}{dt} \cdot \frac{dt}{du} = (e^{t}\mathbf{i} - 4e^{t}\mathbf{j}).(2u) = 2u\, e^{u2}\mathbf{i} - 8ue^{-u^2}\mathbf{j}$

By expressing **r** in terms of u

$R = e^{u}\mathbf{i} + 4e^{-u^2}\mathbf{j}$

$\frac{dr}{du} = 2u\, e^{u^2}\mathbf{i} - 8ue^{-u^2}\mathbf{j}$



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -30 Exact Differential

If $z = f(x, y)$, then $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

The result can be extended to functions of more than two independent variables.

If $z = f(x, y, w)$, $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy + \frac{\partial z}{\partial w} dw$

Make a note of these results in differential form as shown.

Exercise

Determine the differential dz for each of the following functions.

1. $z = x^2 + y^2$
2. $z = x^3 \sin 2y$
3. $z = (2x - 1)\, e^{3y}$
4. $z = x^2 + 2y^2 + 3w^2$
5. $z = x^3 y^2 w$.

Finish all five and then check the result.

1. $dz = 2 (x\, dx + y\, dy)$
2. $dz = x^2 (3 \sin 2y\, dx + 2x \cos 2y\, dy)$
3. $dz = e^{3y} \{2dx + (6x - 3)\, dy\}$
4. $dz = 2 (xdx + 2ydy + 3wdw)$
5. $dz = x^2 y (3ywdx + 2xwdy + xydw)$

Exact Differential

We have just established that if $z=f(x, y)$

$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

We now work in reverse.

Any expression $dz = Pdx + Qdy$, where P and Q are functions of x and y, is an exact differential if it can be integrated to determine z.

$\therefore\ P = \frac{\partial z}{\partial x}$ and $Q = \frac{\partial z}{\partial y}$

Now $\frac{\partial P}{\partial y} = \frac{\partial^2 z}{\partial y \partial x}$ and $\frac{\partial Q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y}$ and we know that $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

Therefore, for dz to be an exact differential $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ and this is the test we apply.

Example

$dz = (3x^2 + 4y^2)\, dx + 8xy\, dy$.

If we compare the right-hand side with $Pdx + Qdy$, then

$P = 3x^2 + 4y^2 \ \therefore\ \frac{\partial P}{\partial y} = 8y$

$Q = 8xy \qquad \therefore\ \frac{\partial Q}{\partial x} = 8y$

$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ $\therefore$ dz is an exact differential

Similarly, we can test this one.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example

$dz = (1 + 8xy)\,dx + 5x^2\,dy$.

From this we find dz is not an exact differential

for $dz = (1 + 8xy)\,dx + 5x^2\,dy$

$\therefore\ P = 1 + 8xy \quad \therefore\ \frac{\partial P}{\partial y} = 8x$

$Q = 5x^2 \qquad \therefore\ \frac{\partial Q}{\partial x} = 10x$

$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$ $\therefore$ dz is not an exact differential

Exercise

Determine whether each of the following is an exact differential.

1. $dz = 4x^3y^3dx + 3x^4y^2\,dy$
2. $dz = (4x^3y+2xy^3)\,dx+(x^4+3x_2y^2)\,dy$
3. $dz=(15y^2e^{3x}+2xy^2)dx+(10ye^{3x}+x^2y)dy$
4. $dz=(3x^2e^{2y}-2y^2e^{3x})dx+(2x^3e^{2y}-2ye^{3x})dy$
5. $dz=(4y^3\cos4x+3x^2\cos2y)dx+(3y^2\sin4x-2x^3\sin 2y)\,dy$.

1. Yes 2. Yes 3. No 4. No 5. Yes

We have just tested whether certain expressions are, in fact, exact differentials–and we said previously that, by definition, an exact differential can be integrated. But how exactly do we go about it? The following examples will show.

Integration Of Exact Differentials

$dz = Pdx+Qdy$ where $P=\frac{\partial z}{\partial x}$ and $Q=\frac{\partial z}{\partial y}$

$\therefore\ z = \int P dx$ and also $z = \int Q dy$

Example

$dz = (2xy + 6x)\,dx + (x2 + 2y3)\,dy$.

$P = \frac{\partial z}{\partial x} = 2xy + 6x \quad \therefore\ z=\int (2xy+6x)dx$

$\therefore\ z = x^2y + 3x^2 + f(y)$ where f(y) is an arbitrary function of y only, and is akin to the constant of integration in a normal integral.

Also

$Q = \frac{\partial z}{\partial y} = x^2 + 2y^3 \quad \therefore\ z = \int (x^2+2y^3)\,dy$

$\therefore\ z = x2y + \frac{y^4}{2} + F(x)$ where F(x) is an arbitrary function of x only.

$z = x^2y + 3x^2 + f(y) \quad$ (i)

and $z = x^2y + \frac{y^4}{2} + F(x) \quad$ (ii)

For these two expressions to represent the same function, then

f(y) in (i) must be $\frac{y^4}{2}$ already in (i)

and F(x) in (ii) must be $3x^2$ already in (i)

$\therefore \quad z = x^2y + 3x^2 + \frac{y^4}{2}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example

Integrate $dz = (8e^{4x} + 2xy^2)\,dx + (4\cos 4y + 2x^2y)\,dy$.

Argue through the working in just the same way, from which we obtain

$z = 2e^{4x} + x^2y^2 + \sin 4y$

Here it is.

$dz = (8e^{4x} + 2xy^2)\,dx + (4\cos 4y + 2x^2y)\,dy$

$$P = \frac{\partial x}{\partial x} = 8e^{4x} + 2xy^2$$

$$\therefore\ z = \int (8e^{4x} + 2xy^2)\,dx$$

$$\therefore\ z = 2e^{4x} + x^2y^2 + f(y) \qquad \text{(i)}$$

$$Q = \frac{\partial z}{\partial y} = 4\cos 4y + 2x^2y$$

$$\therefore\ z = \int (4\cos 4y + 2x^2y)\,dy$$

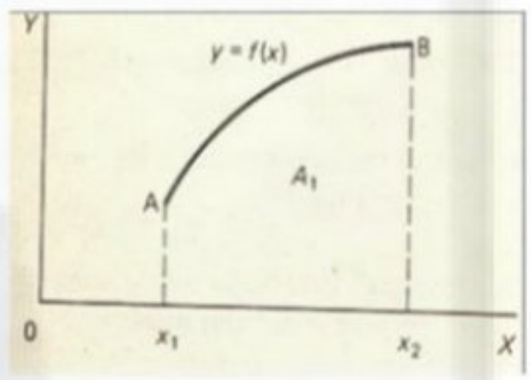
$$\therefore\ z = \sin 4y + x^2y^2 + F(x) \qquad \text{(ii)}$$

For (i) and (ii) to agree, $f(y) = \sin 4y$ and $F(x) = 2e^{4x}$

$$\therefore\ z = 2e^{4x} + x^2y^2 + \sin 4y$$

Area enclosed by the closed curve

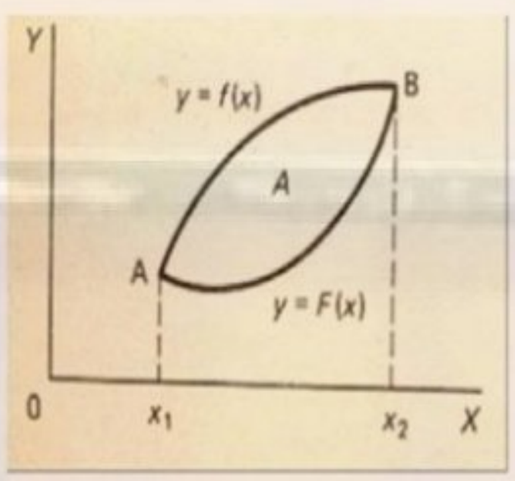
One of the earliest application of integration is finding the area of a plane figure bounded by the x-axis, the curve $y = f(x)$ and ordinates at $x=x_1$ and $x=x_2$.

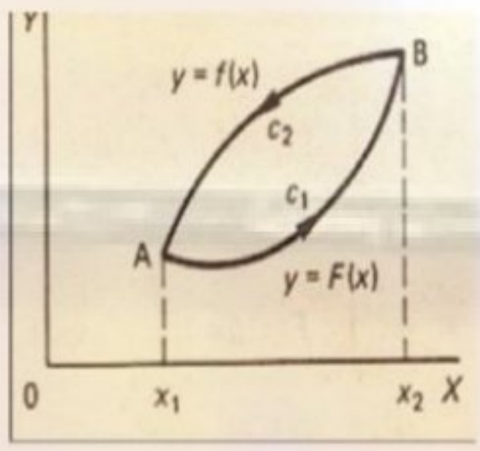


$$A_1 = \int_{x_1}^{x_2} y\,dx = \int_{x_1}^{x_2} f(x)\,dx$$

If points A and B are joined by another curve, $y = F(x)$

$$A_2 = \int_{x_1}^{x_2} f(x)\,dx$$





Combining the two figures, we have

$$A = A_1 - A_2 \qquad \therefore\ A = \int_{x_1}^{x_2} f(x)\,dx - \int_{x_1}^{x_2} f(x)\,dx$$

The final result above can be written in the form

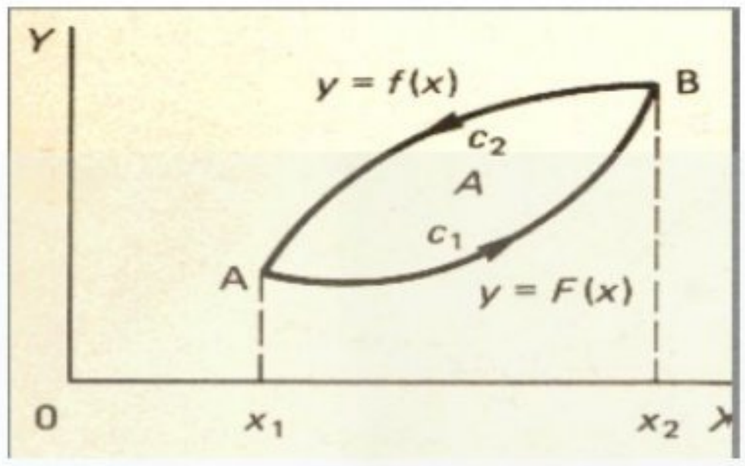
$$A = -\oint y\,dx$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Where the symbol $\oint$ indicates that the integral is to be evaluated round the closed boundary in the positive



Example

Determine the area enclosed by the graph of $y = x^3$ and $y = 4x$ for $x \geq 0$.

First we need to know the points of intersection. These are $x = 0$ and $x = 2$

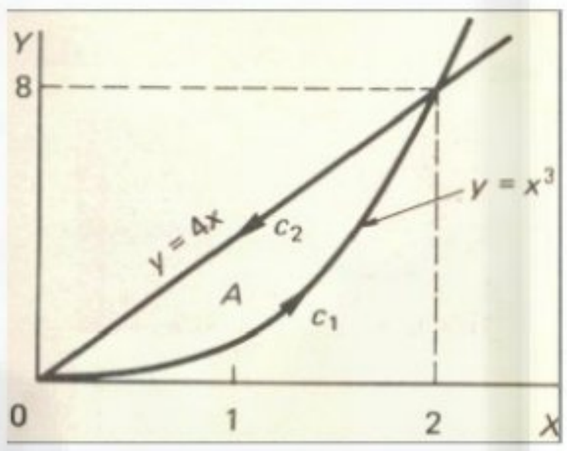
We integrate in a an anticlockwise manner

c_1: $y = x^3$, limits $x = 0$ to $x = 2$

c_2: $y = 4x$, limits $x = 2$ to $x = 0$.

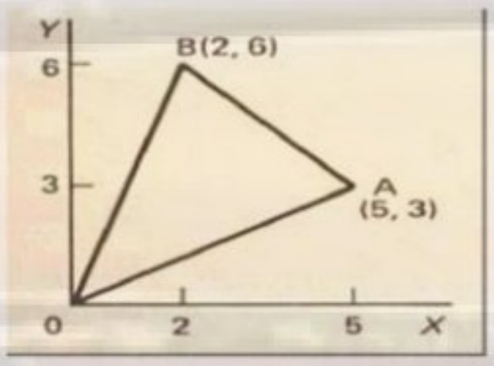
$A = -\oint y\,dx = A = 4$ square units

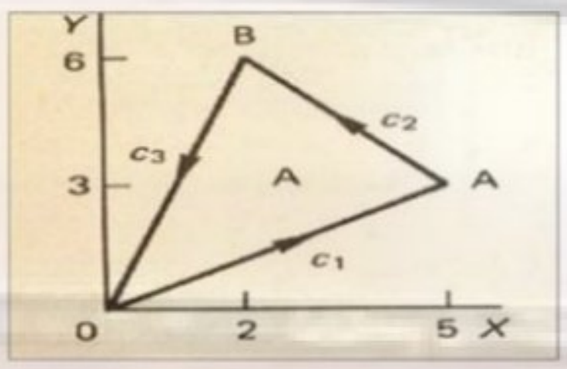
$$\text{For } A = -\oint y dx = -\left\{\int_1^2 x^3 dx + \int_2^0 4x dx\right\} = -\left\{\left(\frac{x^4}{4}\right)_0^2 + \left[2x^2\right]_0^2\right\} = 4$$



Example

Find the area of the triangle with vertices (0, 0), (5, 3) and (2, 6).





The equation of OA is $y = \frac{3}{5}x$, BA is $y = 8 - x$, OB is $y = 3x$

Then $A = -\oint y\,dx$

Write down the component integrals with appropriate limits.

$$A = -\oint y dx = -\left\{\int_0^5 \frac{3}{5}x dx + \int_5^2 (8-x) dx + \int_2^0 3x dx\right\}$$

The limits chosen must progress the integration round the boundary of the figure in an anticlockwise manner. Finishing off the integration, we have

$A = 12$ square units

© Copyright Virtual University of Pakistan

Scanned with CamScanner

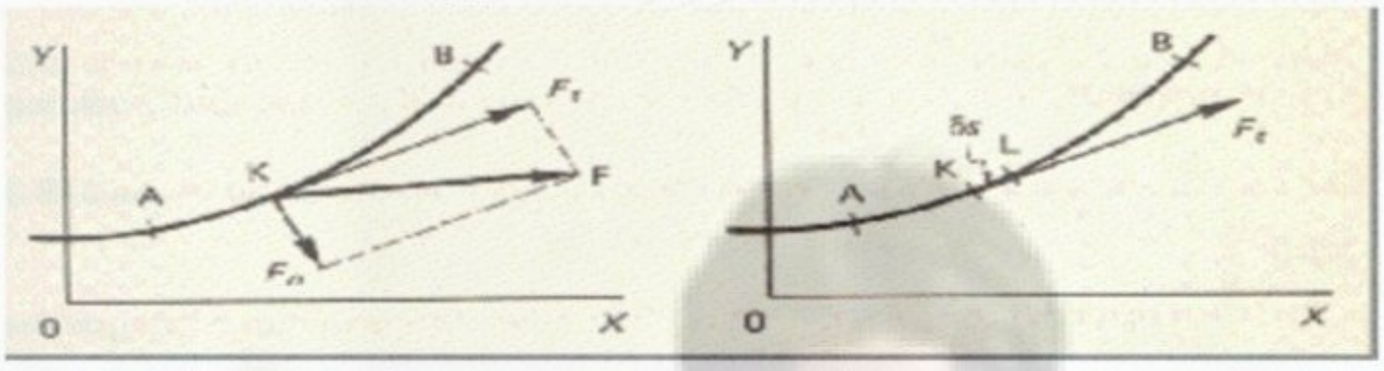
Scanned with CamScanner

The actual integration is easy enough. The work we have just done leads us on to consider line integrals, so let us make a fresh start in the next frame.

Line Integrals

If a field exists in the xy-plane, producing a force F on a particle at K, then F can be resolved into two components.F_1 along the tangent to the curve AB at K. Fa along the normal to the curve AB at K.

Line Integrals



The work done in moving the particle through a small distance δs from K to L along the curve is then approximately F_1 δs. So the total work done in moving a particle along the curve from A to B is given by

$$\lim_{\delta \to 0} \sum F_t \, \delta s = \int F_t \, ds \text{ from A to B}$$

This is normally written $\int_{AB} F_t \, ds$ where A and B are the end points of the curve,

or as $\int_C F_t \, ds$ where the curve c connecting A and B is defined.

Such an integral thus formed, is called a line integral since integration is carried out along the path of the particular curve c joining A and B.

$$\therefore \; I = \int_{AB} F_t \, dx = \int_C F_t \, ds$$

where c is the curve y = f(x) between A(x_1, y_2) and B (x_2, y_2).

There is in fact an alternative form of the integral which is often useful, so let us also consider that.

Alternative form of a line integral

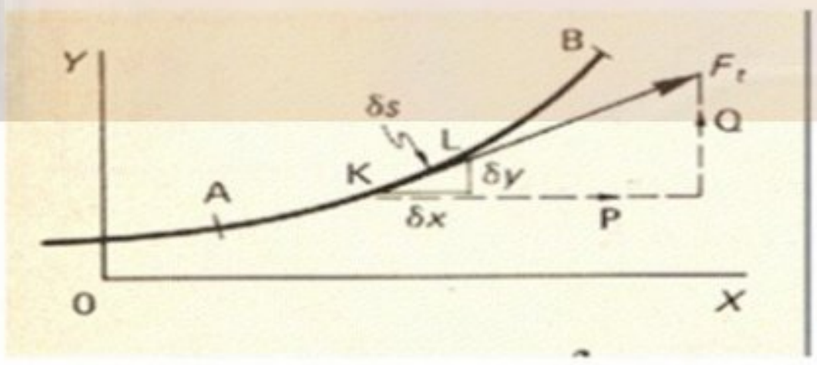
It is often more convenient to integrate with respect to x or y than to take arc length as the variable.

If F_t has a component

P in the x-direction

Q in the y-direction

then the work done from K to L can be stated as Pδx + Qδy



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$\therefore \int_{AB} F_t \, ds = \int_{AB} (P\,dx + Q dy)$

where P and Q are functions of x and y.

In general then, the line integral can be expressed as

$$I = \int_C F_t \, ds = \int_C (P\,dx + Q dy)$$

where c is the prescribed curve and F, or P and Q, are functions of x and y.

Make a note of these results –then we will apply them to one or two examples.

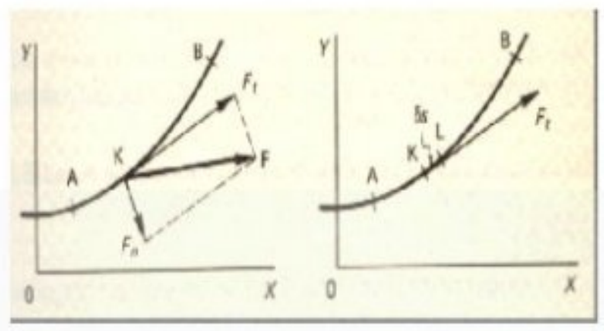
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -31 Line Integral

The work done in moving the particle through a small distance δs from K to L along the curve is then approximately F_1 δs. So the total work done in moving a particle along the curve from A to B is given by



$$\lim_{\delta \to 0} \sum F_t \, \delta s = \int F_t \, ds \text{ from A to B}$$

This is normally written $\int_{AB} F_t \, ds$ where A and B are the end points of the curve, or as $\int_C F_t \, ds$ where the curve c connecting A and B is defined.Such an integral thus formed, is called a line integral since integration is carried out along the path of the particular curve c joining A and B.

$$\therefore \; I = \int_{AB} F_t \, dx = \int_C F_t \, ds$$

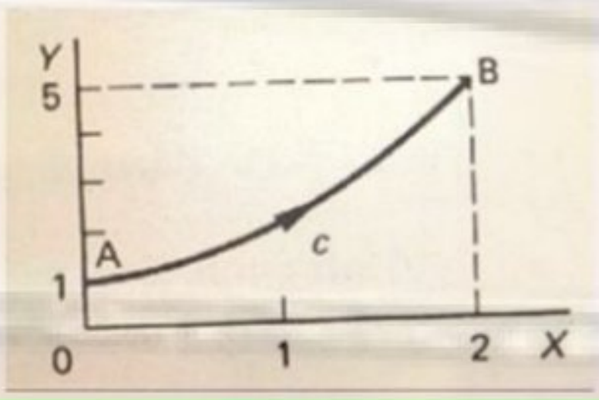
where c is the curve $y = f(x)$ between $A(x_1, y_2)$ and $B(x_2, y_2)$.
There is in fact an alternative form of the integral which is often useful, so let us also consider that.

Alternative form of a line integral

It is often more convenient to integrate with respect to x or y than to take arc length as the variable.
If F_t has a component ,P in the x-direction ,Q in the y-direction
then the work done from K to L can be stated as $P\delta x + Q\delta y$

Example 1

Evaluate $\int_C (x + 3y)\, dx$ from A (0, 1) to B (2, 5)
along the curve $y = 1 + x^2$.
The line integral is of the form
$\int_C (P\, dx + Q dy)$ where, in this case, $Q = 0$ and c is the curve $y = 1 + x^2$.

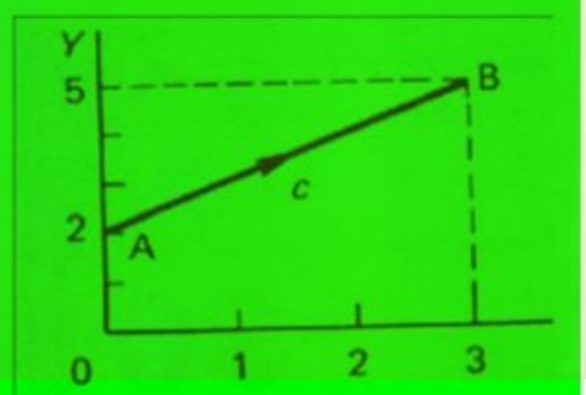


It can be converted at once into an ordinary integral by substituting for y and applying the appropriate limits of x.

$$I = \int_C (Pdx+Qdy) = \int_C (x+3y)dx = \int_C^2 (x+3+3x^2)dx$$

$$= \left[\frac{x^2}{2}+3x+x^3\right]_0^2 = 16$$

Now for another, so turn on.



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example 2

Evaluate $I = \int_C (x^2 + y)\,dx + (x - y^2)dy$ from A (0, 2) to B (2, 5) along the curve $y = 2 + x$.

$I = \int_C (Pdx + Qdy)$

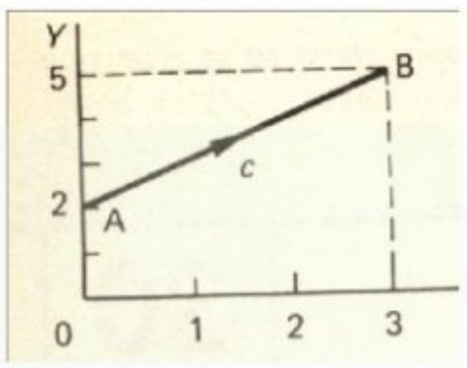
$P = x^2 + y = x^2 + 2 + x = x^2 + x + 2$

$Q = x - y^2 = x - (4 + 4x + x^2) = -(x^2 + 3x + 4)$

Also $y = 2 + x$

$\therefore$ $dy = dx$ and the limits are $x=0$ to $x=3$

$\therefore$ $I = -15$



for $I = \int_0^2 \{(x^2+x+2)\,dx - (x^2+3x+4)\,dx\}$ or $\int_0^2 -(2x+2)\,dx = \left[-x^2 - 2x\right]_0^2 = -4 - 4 = -8$

Example 3

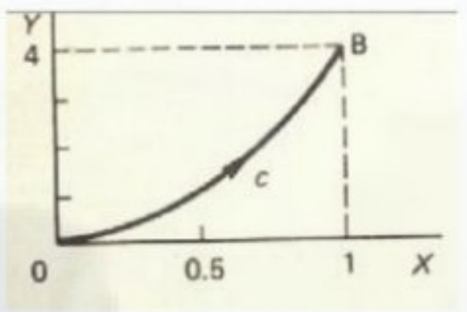
Evaluate $I = \int_C \{(x^2+2y)dx + xydy\}$ from O(0, 0) to B(1, 4) along the curve $y=4x^2$.

In this case, c is the curve $y = 4x2$.

$\therefore$ $dy = 8x\,dx$

Substitute for y in the integral and apply the limits.

Then $I = 9.4$



for $I = \int_C \{(x^2+2y)\,dx + xydy\}$

$y = 4x^2$ $\quad \therefore$ $dy = 8xdx$

also $x^2 + 2y = x^2 + 8x^2 = 9x^2$; $xy = 4x^3$

$\therefore$ $I = \int_0^1 \{9x^2dx + 32x^4dx\} = \int_0^1 (9x^2 + 32x^4)\,dx = 9.4$

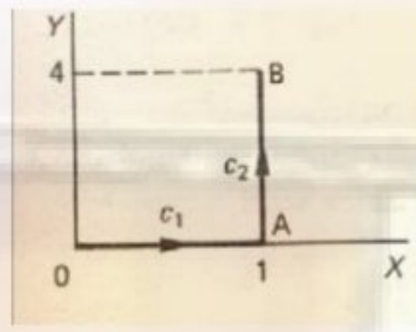
They are all done in very much the same way.

Example 4

Evaluate $I = \int_C \{(x^2 + 2y)\,dx + xydy\}$ from O(0, 0) to A (1, 0) along the line $y = 0$ and then from A (1, 0) to B (1, 4) along the line $x = 1$.

(i) OA : c1 is the line $y = 0$ $\quad \therefore$ $dy = 0$.

Substituting $y = 0$ and $dy = 0$ in the given integral gives.



$$I_{OA} = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$$

(ii) AB: Here c_2 is the line $x = 1$ $\therefore$ $dx=0$

$\therefore$ $I_{AB} = 8$

$$\text{For } I_{AB} = \int_0^4 \{(1 + 2y)\,(0) + ydy\} = \int_0^4 ydy = \left[\frac{y^2}{2}\right]_0^4 = 8$$

$$\text{Then } I = I_{OA} + I_{AB} = \frac{1}{3} + 8 = 8\frac{1}{3} \quad \therefore I = 8\frac{1}{3}$$

If we now look back to Example 3 and 4 just completed, we find that we have evaluated the same integral between the same two end points, but along different paths of integration.If we combine the two diagrams, we have

where c is the curve $y = 4x^2$ and $c1 + c^2$ are the lines

$y = 0$ and $x = 1$. The result obtained were

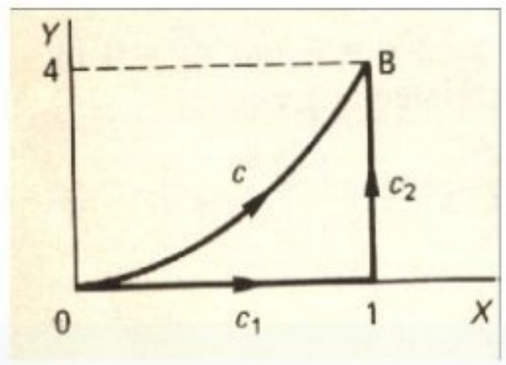
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$I_c = 9\frac{2}{3}$ and $I_{c_1+c_2} = 8\frac{1}{3}$

Notice therefore that integration along two distinct paths joining the same two end points does not necessarily give the same results.



Properties of line integrals

1. $\int_C Fds = \int_C \{Pdx + Qdy\}$

2. $\int_{AB} Fds = -\int_{BA} Fds$ and $\int_{AB} \{Pdx+Qdy\} = \int_{BA} \{Pdx+Qdy\}$

i.e. the sign of a line integral is reversed when the direction of the integration along the path is reversed.

3. (a) For a path of integration parallel to the y-axis, i.e. $x = k$, $dx = 0$

$\therefore \int_C Pdx = 0 \quad \therefore I_C = \int_C Qdy.$

(b) For a path of integration parallel to the x-axis, i.e. $y = k$, $dy = 0$.

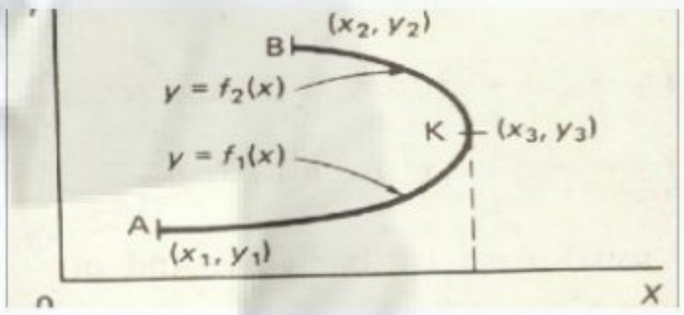
$\therefore \int_C Qdy = 0 \ \therefore I_C = \int_C Pdx.$

4. If the path of integration c joining A to B is divided into two parts AK and KB, then

$I_c = I_{AB} = I_{AK} + I_{KB}$.

5 .If the path of integration c is not single valued for part of its extent, the path is divided into two sections.

$y = f_1(x)$ from A to K, $y = f_2(x)$ from K to B.



6. In all cases, the actual path of integration involved must be continuous and single-valued.

Example

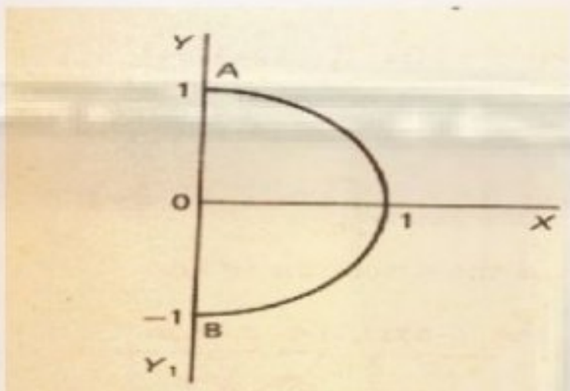
Evaluate $I = \int_C (x + y)\,dx$ from A(0, 1) to B (0, – 1) along the semi-circle $x^2+y^2=1$

for $x \geq 0$. The first thing we notice is that the path of integration c is not single-valued

For any value of x, $y = \pm\sqrt{1 - x^2}$. Therefore, we divided c into two parts

(i) $y = \sqrt{1 - x^2}$ from A to K

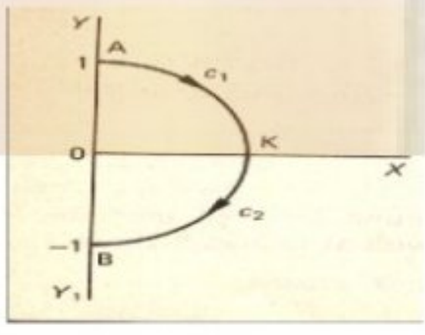
(ii) $y = -\sqrt{1 - x^2}$ from K to B



As usual, $I = \int_C (Pdx + Qdy)$ and in this particular case, $Q = 0$

$$\therefore I = \int_C Pdx = \int_0^1 (x+\sqrt{1 - x^2})\,dx + \int_1^0 (x\sqrt{1 - x^2})\,dx$$

$$= \int_0^1 (x + \sqrt{1 - x^2} - x+\sqrt{1 - x^2}\,dx = 2\int_0^1 \sqrt{1 - x^2}\,dx$$



Now substitute $x = \sin\theta$ and finish it off. $I = \frac{\pi}{2}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

for $I = 2\int_0^1 \sqrt{1-x^2}dx$ $\quad x = \sin\theta$

$\therefore \quad dx = \cos\theta\, d\theta \quad \sqrt{1-x^2} = \cos\theta$

Limits : $x = 0, \quad \theta = 0; \; x = 1, \; \theta = \frac{\pi}{2}$

$$\therefore \; I = 2\int_0^{\pi/2} \cos 2\theta d\theta = \int_0^{\pi/2} (1+\cos 2\theta)d\theta = \left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\pi/2} = \frac{\pi}{2}$$

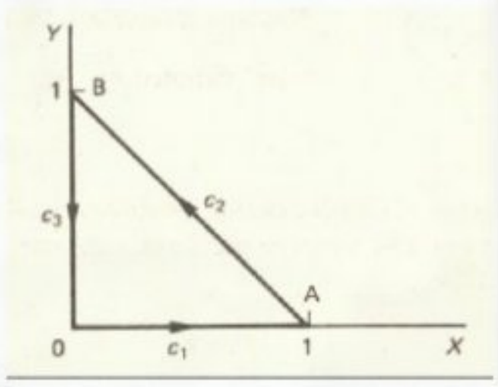
Now let us extend this line of development a stage further.

Example

Evaluate the line integral

$I = \oint (x^2dx - 2xy\, dy)$ where c comparises the three sides of the triangle joining O(0, 0), A (1, 0) and B (0, 1).

First draw the diagram and mark in c_1, c_2 and c_3, the proposed directions of integration. Do just that. The three sections of the path of integration must be arranged in an anticlockwise manner round the figure.



Now we deal with each pat separately.

(a) OA : c_1 is the line $y = 0$

Therefore, $dy = 0$.

Then $I = \oint (x^2dx - 2xy\, dy)$ for this part becomes

$$I_1 = \int_0^1 x^2dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3} \text{ therefore } I_1 = \frac{1}{3}$$

(b) AB : for c_2 is the line $y = 1 - x$

$\therefore \; dy = -dx.$

$$I_2 = \int_0^1 \{x^2dx + 2x(1-x)dx\} = \int_0^1 (x^2+2x-2x^2)dx = \int_0^1 (2x-x^2)\, dx = \left[x^2 - \frac{x^3}{3}\right]_0^1 = -\frac{2}{3}$$

$$\therefore \qquad I_2 = -\frac{2}{3}$$

Note that anticlockwise progression is obtained by arranging the limits in the appropriate order.

Now we have to determine I_3 for BO.

(c) BO: c_3 is the line $x = 0$

$\therefore \; dx = 0 \qquad \therefore \; I_3 = \int 0dy = 0 \; \therefore \quad I_3 = 0$

Finally, $I = I_1 + I_2 + I_3 = \frac{1}{3} - \frac{2}{3} + 0 = -\frac{1}{3} \qquad \therefore \quad I = -\frac{1}{3}$

Example

Evaluate $\oint_c y\, dx$ when c is the circle $x^2+y^2 = 4$.

$x^2 + y^2 = 4 \qquad \therefore \qquad y = \pm\sqrt{4-x^2}$

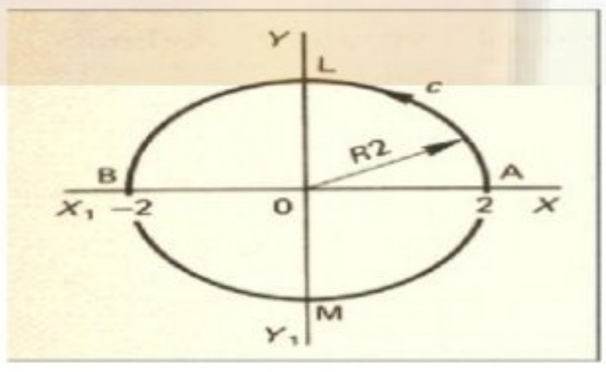
y is thus not single-valued. Therefore use

$y = \sqrt{4-x^2}$ for ALB between

$x = 2$ and $x = -2$ and

$y = -\sqrt{4-x^2}$ for BMA between

$x = -2$ and $x = 2$.



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$\therefore\ I = \int_{2}^{-2} \sqrt{4 - x^2}\, dx + \int_{-2}^{2} \{-\sqrt{4 - x^2}\}\, dx$$

$$= 2\int_{2}^{-2} \sqrt{4 - x^2}\, dx = -2\int_{-2}^{2} \sqrt{4 - x^2}\, dx \qquad = -4\int_{0}^{2} \sqrt{4 - x^2}\, dx.$$

To evaluate this integral, substitute $x = 2 \sin \theta$ and finish it off. $I = -4\pi$

© Copyright Virtual University of Pakistan

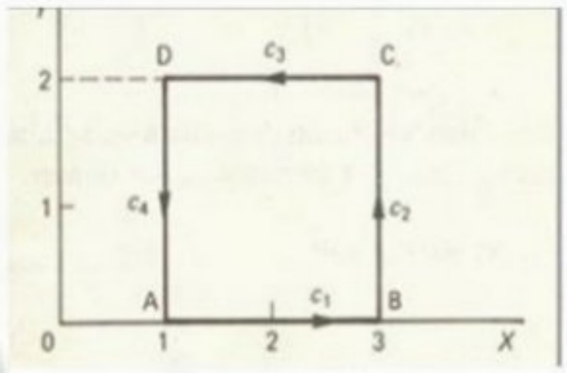
Scanned with CamScanner

Scanned with CamScanner

Lecture No -32 Examples

Example

Evaluate $I = \oint \{xydx+(1+y^2)dy\}$ where c is the boundary of the rectangle joining A(1,0), B (3, 0), C(3, 2), D (1, 2).



First draw the diagram and insert c_1, c_2, c_3, c_4.

That give

Now evaluate I_1 for AB; I_2 for BC; I_3 for CD; I_4 for DA; and finally I.

$I_1 = 0;\ I_2 = 4\frac{2}{3};\ I_3 = -8\ \ I_4 = -4\frac{2}{3};\ I = -8$

Here is the complete working.

$I = \oint \{xydx + (1 + y^2)\, dy\}$

(a) AB: c_1 is $y = 0 \ \therefore\ dy = 0 \ \therefore\ I_1 = 0$

(b) BC: c_2 is $x = 3 \ \therefore\ dx = 0$

$$\therefore I_2 = \int_0^2 (1+y^2)dy = \left[y + \frac{y^3}{3}\right]_0^2 = 4\frac{2}{3} \quad \therefore I_2 = 4\frac{2}{3}$$

(c) CD : c_3 is $y = 2 \qquad \therefore\ dy = 0$

$$\therefore I_3 = \int_3^1 2xdx = \left[x^2\right]_3^1 = -8 \quad \therefore I_3 = -8$$

(d) DA : c_4 is $x = 1 \qquad \therefore\ dx = 0$

$$\therefore\ I_4 = \int_2^0 (1+y^2)\, dy = \left[y + \frac{y^3}{3}\right]_2^0 = -4\frac{2}{3}$$

Finally $I = I_1 + I_2 + I_3 + I_4 = 0 + 4\frac{2}{3} - 8 - 4\frac{2}{3} = -8 \ \therefore I = -8$

Remember that, unless we are directed otherwise, we always proceed round the closed boundary in an anticlockwise manner.

Line integral with respect to arc length

We have already established that

$$I = \int_{AB} F_t ds = \int_{AB} \{Pdx+Qdy\}$$

where F_t denoted the tangential force along the curve c at the sample point K(x,y).

The same kind of integral can, of course, relate to any function f(x,y) which is a function of the position of a point on the stated curve, so that

$$I = \int_C f(x, y)\, ds.$$

This can readily be converted into an integral in terms of x:

$$I = \int_C f(x,y)dx = \int_C f(x,y)\, \frac{ds}{dx}\, dx$$

where $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$$\therefore \int_C f(x,y)\, dx = \int_{x_1}^{x_2} f(x,y) \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx \text{----------} \quad (1)$$

© Copyright Virtual University of Pakistan

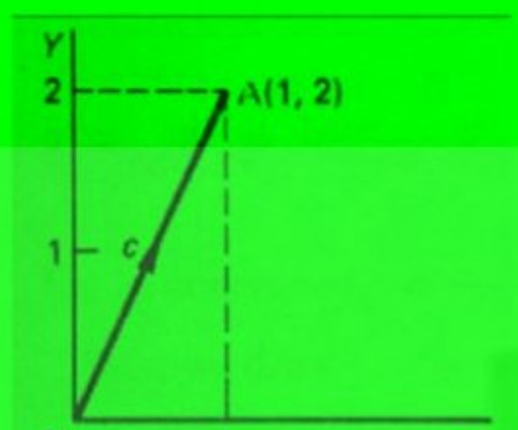
Scanned with CamScanner

Scanned with CamScanner

Example

Evalaute $I = \int_C (4x+3xy)ds$ where c is the straight line joining O(0,0) to A (1,2).

c is the line $y = 2x \;\; \therefore \;\; \frac{dy}{dx} = 2$



$$\therefore \; \frac{ds}{dx} = \sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{5}$$

$$\therefore \; I = \int_{x=0} (4x+3xy)ds = \int_0^1 (4x+3xy)(\sqrt{5})\,dx. \text{ But } y = 2x$$

for $$I = \int_0^1 (4x+6x^2)(\sqrt{5})\,dx = 2\sqrt{5}\int_0^1 (2x+3x^2)\,dx = 4\sqrt{5}$$

Parametric Equations

When x and y are expressed in parametric form, e.g. x = y (t), y = g(t), then

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} \qquad \therefore \; ds = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

$$I = \int_C f(x,y)ds = \int_{t_1}^{t_2} f(x,y)\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} \text{ ----------(2)}$$

Example

Evaluate $I = \oint 4xyds$ where c is defined as the curve x = sin t, y = cos t between t=0 and $t=\frac{\pi}{4}$.

We have $x = \sin t \qquad \therefore \; \frac{dx}{dt} = \cos t, \; y = \cos t \qquad \therefore \; \frac{dy}{dt} = -\sin t$

$$\therefore \qquad \frac{ds}{dt} = I$$

$$\text{for } \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} = \sqrt{\cos^2 t+\sin^2 t} = 1$$

$$\therefore \; I = \int_{t_1}^{t_2} f(x,y)\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

$$= \int_0^{\pi/4} 4\sin t\cos t\,dt = 2\int_0^{\pi/4} \sin 2t\,dt$$

$$= -2\left[\frac{\cos 2t}{2}\right]_0^{\pi/4} = 1$$

Dependence of the line integral on the path of integration

We know that integration along two separate paths joining the same two end points does not necessarily give identical results.With this in mind, let us investigate the following problem.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

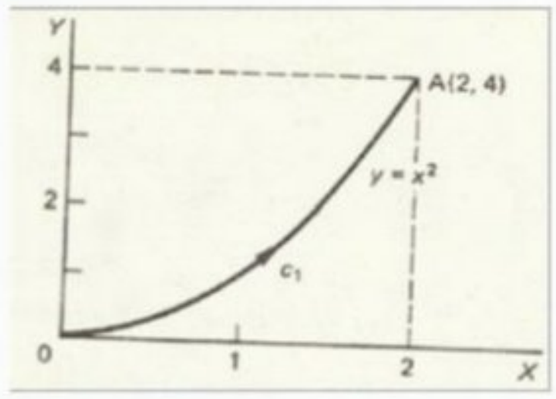
Scanned with CamScanner

EXAMPLE

Evaluate $I = \oint_C \{3x^2 y^2\, dx + 2x^3 y\, dy\}$ between O (0, 0) and A (2, 4)

(a) along c_1 i.e. $y = x^2$
(b) along c_2 i.e. $y = 2x$
(c) along c_3 i.e. $x = 0$ from (0,0) to (0,4) and $y = 4$ from (0,4) to (2,4).

(a).First we draw the figure and insert relevant information.



$$I = \int_C \{3x^2y^2dx + 2x^3ydy\}$$

The path c_1 is $y = x^2 \quad \therefore dy = 2x\,dx$

$$\therefore \quad I_1 = \int_0^2 \{3x^2x^4dx + 2x^3x^2 2xdx\} = \int_0^2 (3x^6 + 4x^6)\,dx$$

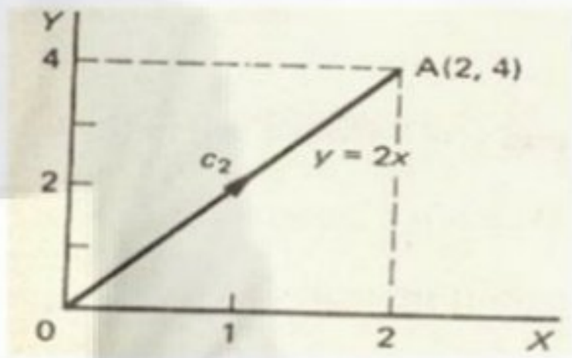
$$\therefore \quad = \left[x^7\right]_0^2 = 128 \quad \therefore \quad I_1 = 128$$

(b) In (b), the path of integration changes to c_2, i.e. $y = 2x$

So, in this case,for with c_2, $y = 2x \quad \therefore dy = 2dx$

$$\therefore I_2 = \int_0^2 (3x^2\, 4x^2\, dx + 2x^3\, 2x^2\, dx\}$$

$$= \int_0^2 20\, x^4\, dx = 4\left[x^5\right]_0^2 = 128 \qquad \therefore \quad I_2 = 128$$



(c) In the third case, the path c_3 is split

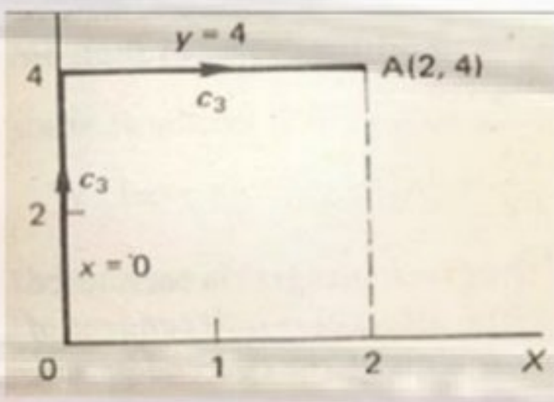
$x = 0$ from (0,0) to (0, 4), $y = 4$ from (0, 4) to (2, 4)

Sketch the diagram and determine I_3.

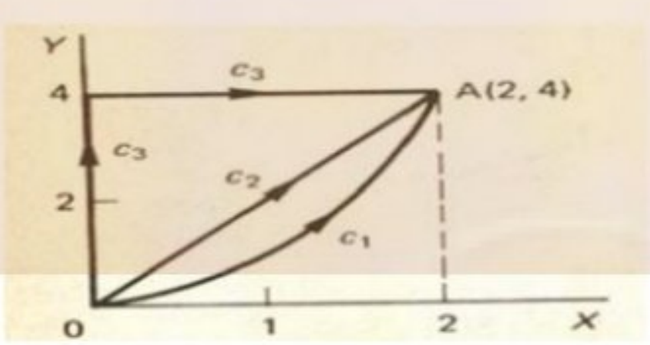
from (0,0) to (0,4) $x=0 \quad \therefore \quad dx=0 \ \therefore \ I_{3a}=0$

from (0,4) to (2,4) $y=4 \ \therefore \ dy=0 \ \therefore \ I_{3b}=48$

$$\int_0^2 48x^2\, dx = 128 \qquad \therefore \quad I_3 = 128$$



In the example we have just worked through, we took three different paths and in each case, the line integral produced the same result. It appears, therefore, that in this case, the value of the integral is independent of the path of integration taken.



We have been dealing with $I = \int_C \{3x^2y^2dx + 2x^3ydy\}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

On reflection, we see that the integrand $3x^2 y^2 dx + 2x^3 y dy$ is of the form Pdx+Qdy which we have met before and that it is, in fact, an exact differential of the function

$z = x^3y^2$, for $\frac{\partial z}{\partial x} = 3x^2 y^2$ and $\frac{\partial z}{\partial y} = 2x^3 y$

This always happens. If the integrand of the given integral is seen to be an exact differential, then the value o the line integral is independent of the path taken and depends only on the coordinates of the two end points.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -33 Examples

Example

Evaluate $I = \int_C \{3ydx + (3x+2y)dy\}$ from A(1, 2) to B (3, 5).

No path is given, so the integrand is doubtless an exact differential of some function z = f (x,y). In fact $\frac{\partial P}{\partial y} = 3 = \frac{\partial Q}{\partial x}$. We have already dealt with the integration of exact differentials, so there is no difficulty. Compare with

$I = \int_C \{P\,dx + Q\,dy\}$.

$P = \frac{\partial z}{\partial x} = 3y \qquad \therefore\ z = \int 3ydz = 3xy + f(y)$ -------- (i)

$Q = \frac{\partial z}{\partial y} = 3x + 2y \qquad \therefore\ z = \int (3x+2y)\,dy = 3xy + y^2 + F(x)$ (ii)

For (i) and (ii) to agree $f(y) = y^2$; $F(x) = 0$

Hence z = 3xy + y2

$\therefore I = \int_C \{3ydx + (3x+2y)dy\} = \int_{(1,2)}^{(3,5)} d(3xy+y^2) = \left[3xy+y^2\right]_{(1,2)}^{(3,5)} = (45+25) - (6+4) = 60$

Example

Evaluate $I = \int_C \{(x^2+ye^x)dx+(e^x+y)dy\}$ between A (0, 1) and B (1, 2).

As before, compare with $\int_C \{Pdx+Q\,dy\}$.

$P = \frac{\partial z}{\partial x} = x^2+ye^x \qquad \therefore\ z = \frac{x^3}{3} + ye^x + f(y)$

$Q = \frac{\partial z}{\partial y} = e^x+y \qquad \therefore\ z = ye^x + \frac{y^2}{2} + F(x)$

For these expressions to agree,

$f(y) = \frac{y^2}{2}$; $F(x) = \frac{x^3}{3}$ Then $I = \left[\frac{x^3}{3} + ye^x + \frac{y^2}{2}\right]_{(0,1)}^{(1,2)} = \frac{5}{6} + 2e$

So the main points are that, if (Pdx+Qdy) is an exact differential

(a) $I = \int_C (Pdx + Qdy)$ is independent of the path of integration

(b) $I = \oint_C (P\,dx + Q\,dy)$ is zero.

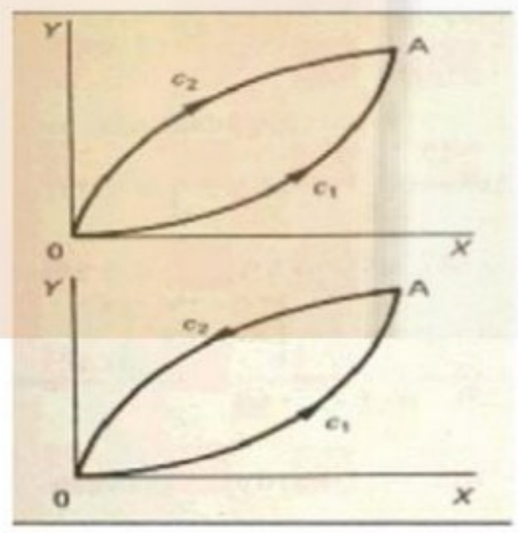
If $I = \int_C \{P\,dx + Q\,dy\}$ and (Pdx + Qdy) is an exact differential,

Then $I_{c_1} = I_{c_2}$

$I_{c_1} + I_{c_2} = 0$

Hence, the integration taken round a closed curve is zero, provided (Pdx+Q dy) is an exact differential.

$\therefore$ If (P dx + Q dy) is an exact differential, $\oint (P\,dx + Q\,dy) = 0$



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Exact differentials in three independent variables

A line integral in space naturally involves three independent variables, but the method is very much like that for two independent variables.

$dz = Pdx + Q\,dy + R\,dw$ is an exact differential of $z = f(x, y, w)$

if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}; \frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}; \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

If the test is successful, then

(a) $\int_C (P\,dx + Q\,dy + R\,dw)$ is independent of the path of integration.

(b) $\oint_C (P\,dx + Q\,dy + R\,dw)$ is zero.

Example

Verify that $dz=(3x^2yw+6x)dx+(x^3w-8y)dy+(x^3y+1)\,dw$ is inexact differential and hence evaluate $\int_C dz$ from A (1, 2, 4) to B (2, 1 3).

First check that dz is an exact differential by finding the partial derivatives above, when

$P = 3x^2yw + 6x$; $Q = x^3w - 8y$; and $R = x^3y + 1$

$\frac{\partial P}{\partial y} = 3x^2w$; $\frac{\partial Q}{\partial x} = 3x^2w$ $\therefore \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

$\frac{\partial P}{\partial w} = 3x^2y$; $\frac{\partial R}{\partial x} = 3x^2y$ $\therefore \frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$

$\frac{\partial R}{\partial y} = x^3$; $\frac{\partial Q}{\partial w} = x^3$ $\therefore \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

$\therefore$ dz is an exact differential

Now to find z. $P = \frac{\partial z}{\partial x}$; $Q = \frac{\partial z}{\partial y}$; $R = \frac{\partial z}{\partial w}$

$\therefore \frac{\partial z}{\partial x} = 3x^2yw+6x$ $\therefore$ $z = \int (3x^2yw+6x)dx = x^3yw+3x^2+f(y)+F(w)$

$\therefore \frac{\partial z}{\partial y} = x^3w-8x$ $\therefore$ $z = \int (x^3w-8y)dy = x^3yw-4y^2+g(x)+F(w)$

$\frac{\partial z}{\partial w} = x^3y+1$ $\therefore$ $z = \int (x^3y+1)dw = y^3yw+w+f(y)+g(x)$

For these three expressions for z to agree

$f(y) = -4y^2$; $F(w) = w$; $g(x) = 3x^2$

$\therefore \quad z = x^3yw + 3x^2 - 4y^2 + w$

$\therefore \quad I = \left[x^3yw + 3x^2-4y^2+w\right]_{(1,2,4)}^{(2,1,3)}$

for $I = \left[x^3yw + 3x^2-4y^2+w\right]_{(1,2,4)}^{(2,1,3)} = (24+12-4+3)-(8+3-16+4)=36$

The extension to line integrals in space is thus quite straightforward.

Finally, we have a theorem that can be very helpful on occasions and which links up with the work we have been doing. It is important, so let us start a new section.

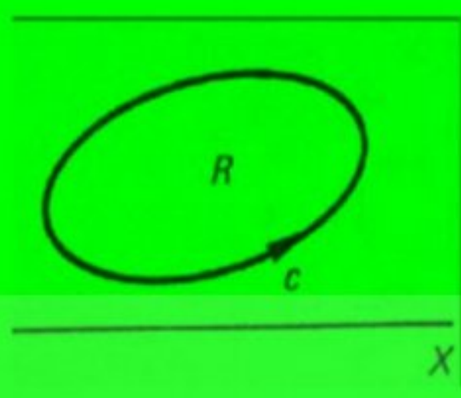
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Green's Thorem

Let P and Q be two function of x and y that are finite and continuous inside and the boundary c of a region R in the xy-plane.If the first partial derivatives are continuous within the region and on the boundary, then Green's theorem states that.



$$\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\, dy = -\oint_C (P\, dx + Q\, dy)$$

That is, a double integral over the plane region R can be transformed into a line integral over the boundary c of the region – and the action is reversible.

Let us see how it works.

EXAMPLE

Evaluate $I = \oint_C \{(2x - y)dx + (2y+x)dy\}$ around the boundary c . the ellipse $x^2 + 9y^2 = 16$.

The integral is of the form

$I = \oint_C \{P\, dx + Q\, dy)$ where $P = 2x - y \;\therefore\; \frac{\partial P}{\partial y} = -1$ and $Q = 2y + x \;\therefore\; \frac{\partial Q}{\partial x} = 1.$

$$\therefore I = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx dy = -\iint_R (-1-1) dx\, dy = 2\iint_R dx\, dy$$

But $\iint_R dx\, dy$ over any closed region give the area of the figure.

In this case, then, $I = 24$ where A is the area of the ellipse

$x^2+9y^2 = 16$ i.e. $\frac{x^2}{16} + \frac{9y^2}{16} = 1$

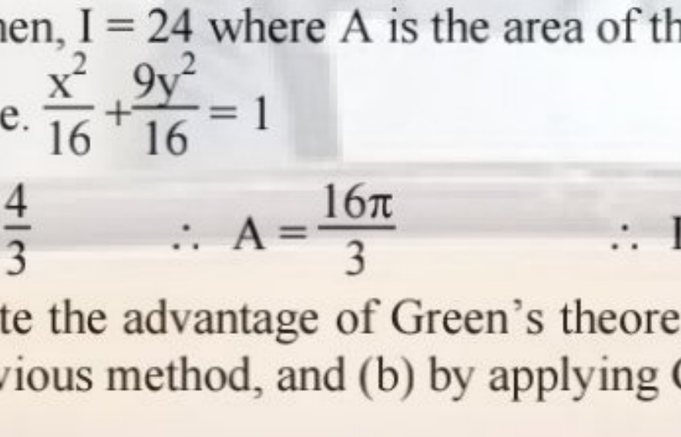
$$\therefore\; a = 4;\; b = \frac{4}{3} \qquad \therefore\; A = \frac{16\pi}{3} \qquad \therefore\; I = 2A = \frac{32\pi}{3}$$

To demonstrate the advantage of Green's theorem, let us work through the next example (a) by the previous method, and (b) by applying Green's theorem.

Example

Evaluate $I = \oint_C \{(2x+y)\, dx + (3x-2y)\, dy\}$ taken in anticlockwise manner round the triangle with vertices at O (0,0) A (1, 0) B (1, 2).

$I = \oint_C \{(2x + y)\, dx + (3x - 2y)\, dy\}$



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(a) By the previous method

There are clearly three stages with c_1, c_2, c_3. Work through the complete evaluation to determine the value of I. It will be good revision. When you have finished, check the result with the solution in the next frame. I = 2

(a) (i) c_1 is $y = 0 \quad \therefore \; dy = 0$

$$\therefore \; I_1 = \int_0^1 2x\,dx = \left[x^2\right]_0^1 = 1 \quad \therefore \; I_1 = 1$$

(ii) c_2 is $x = 1 \quad \therefore \; dx = 0$

$$\therefore \; I_2 = \int_0^2 (3-2y)\,dy = \left[3y - y^2\right]_1^0 = 2 \quad \therefore \; I_2 = 2$$

(iii) c_3 is $y = 2x \quad \therefore \; dy = 2\,dx$

$$\therefore \; I_3 = \int_1^0 \{4x\,dx + (3x - 4x)\,2\,dx\}$$

$$= \int_1^0 2x\,dx = \left[x^2\right]_1^0 = -1 \quad \therefore \; I_3 = -1$$

$$I = I_1 + I_2 + I_3 = 1 + 2 + (-1) = 2 \quad \therefore \; I = 2$$

Now we will do the same problem by applying Green's theorem, so more

(b) By Green's theorem

$$I = \oint_C \{(2x + y)\,dx + (3x - 2y)\,dy\}$$

$P = 2x + y \quad \therefore \quad \frac{\partial P}{\partial y} = 1;$

$Q = 3x - 2y \quad \therefore \quad \frac{\partial Q}{\partial x} = 3$

$$I = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

Finish it off. I = 2

For $I = -\iint_R (1-3)\,dx\,dy = 2\iint_R dx\,dy = 2A$

$= 2 \times$ the area of the triangle $= 2 \times 1 = 2$

$\therefore \; I = 2$

Application of Green's theorem is not always the quickest method. It is useful, however, to have both methods available.

If you have not already done so, make a note of Green's theorem.

$$\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy = -\oint_C (P\,dx + Q\,dy)$$

Note: Green's theorem can, in fact, be applied to a region that is not simply connected by arranging a link between outer and inner boundaries, provided the path of integration is such that the region is kept on the left-hand side.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

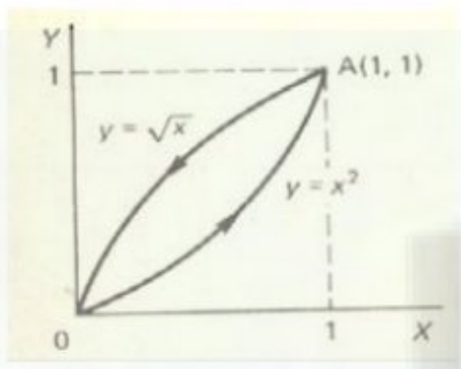
Scanned with CamScanner

Lecture -34.....Examples

Example

Evaluate the line integral $I = \oint_C \{xy\,dx + (2x - y)\,dy\}$ round the region bounded by the curves $y = x^2$ and $x = y^2$ by the use of Green's theorem.

Points of intersection are O(0, 0) and A(1, 1).

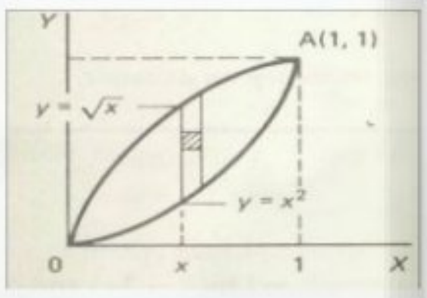


$$I = \oint_C \{xy\,dx + (2x - y)\,dy\}$$

$$\oint_C \{P dx + Q dy\} = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$P = xy \;\therefore\; \frac{\partial P}{\partial y} = x;\; Q = 2x - y \;\therefore\; \frac{\partial Q}{\partial x} = 2$$

$$I = -\iint_R (x - 2)\,dx\,dy = -\int_0^1 \int_{y=x^2}^{y=\sqrt{x}} (x - 2)\,dy\,dx$$



$$= -\int_0^1 (x - 2)\left[\,y\,\right]_{x^2}^{\sqrt{x}} dx$$

$$\therefore\; I = -\int_0^1 (x - 2)\left(\sqrt{x} - x^2\right)\,dx = -\int_0^1 (x^{3/2} - x^3 - 2x^{1/2} + 2x^2)\,dx$$

$$= -\left[\frac{2}{5}x^{5/2} - \frac{1}{4}x^4 - \frac{4}{3}x^{3/2} + \frac{2}{3}x^3\right]_0^1 = \frac{31}{60}$$

In this special case when P=y and Q= – x so $\frac{\partial P}{\partial y} = 1$ and $\frac{\partial Q}{\partial x} = -1$

Green's theorem then states $\iint_R \{1 - (-1)\}\,dx\,dy = -\oint_C (P\,dx + Q\,dy)$

i.e. $2\iint_R dx\,dy = -\oint_C (y\,dx - x\,dy) = \oint_C (x\,dy - y\,dx)$

Therefore, the area of the closed region $A = \iint_R dx\,dy = \frac{1}{2}\oint_C (x\,dy - y\,dx)$

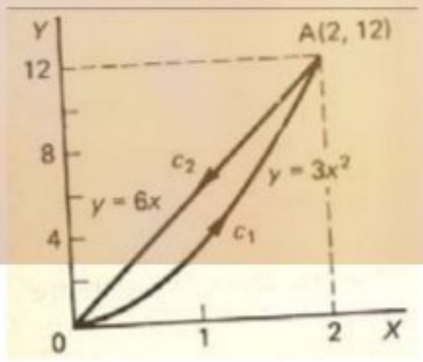
Example

Determine the area of the figure enclosed by $y = 3x^2$ and $y = 6x$.

Points of intersection : $3x^2 = 6x$ $\quad\therefore\; x = 0$ or 2

Area $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$

We evaluate the integral in two parts, i.e.
OA along c_1 and AO along c_2



$$2A = \int_{c_1\ (\text{along OA})} (x\,dy - y\,dx) + \int_{c_2\ (\text{along OA})} (x\,dy - y\,dx) = I_1 + I_2$$

I_1: c_1 is $y = 3x^2$ $\quad\therefore\; dy = 6x\,dx$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$\therefore \quad I_1 = \int_0^2 (6x^2dx - 3x^2dx) = \int_0^2 3x^2\,dx = \left[x^3 \right]_0^2 = 8 \;\therefore\; I_1 = 8$$

Similarly, for c_2 is $y = 6x \quad \therefore \; dy = 6\,dx$

$$\therefore \quad I_2 = \int_2^0 (6x\,dx - 6x\,dx) = 0$$

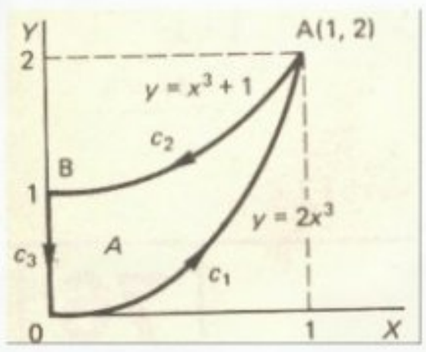
$$\therefore \quad I_2 = 0$$

$$\therefore \quad I = I_1 + I_2 = 8 + 0 = 8$$

$\therefore$ A = 4 square units

Example

Determine the area bounded by the curves $y = 2x^3$, $y = x^3 + 1$ and the axis $x = 0$ for $x \geq 0$.



Here it is $y = 2x^3$; $y = x^3 + 1$; $x = 0$

Point of intersection $2x^3 = x^3 + 1 \quad \therefore \; x^3 = 1 \quad \therefore \; x = 1$

Area $A = \frac{1}{2}\oint_C (x\,dy - y\,dx) \quad \therefore \; 2A = \oint_C (x\,dy - y\,dx)$

(a) OA : c_1 is $y = 2x^3 \quad \therefore \; dy = 6x^2\,dx$

$$\therefore I_1 = \int_{c_1} (xdy - ydx) = \int_0^1 (6x^3dx - 2x^3\,dx) \quad = \int_0^1 4x3\,dx = \left[x^4 \right]_0^1 = 1$$

$\therefore \; I_1 = 1$

(b) AB: c_2 is $y = x^3 + 1 \quad \therefore \; dy = 3x^2\,dx$

$$\therefore \quad I_2 = \int_1^0 \{3x^3\,dx - (x^3 + 1)\,dx\} = \int_1^0 (2x^3 - 1)\,dx = \left[\frac{x^4}{2} - x\right]_1^0 = -\left(\frac{1}{2} - 1\right) = \frac{1}{2}$$

$$\therefore \quad I_2 = \frac{1}{2}$$

(c) BO: c_3 is $x = 0 \quad \therefore \; dx = 0$

$$I_3 = \int_{y=1}^{y=0} (xdy - ydx) = 0 \quad \therefore \; I_3 = 0$$

$$\therefore \quad 2A = I = I_1 + I_2 + I_3 = 1 + \frac{1}{2} + 0 = 1\frac{1}{2} \quad \therefore \; A = \frac{3}{4} \text{ square units}$$

Revision Summary

Properties of line integrals

- Sign of line integral is reversed when the direction of integration along the path is reversed.
- Path of integration parallel to y-axis, $dx = 0 \quad \therefore \; I_c = \int_c^{} Q\,dy$.
- Path of integration parallel to x-axis, $dy = 0 \quad \therefore \; I_c = \int_c^{} P\,dx$.
- Path of integration must be continuous and single-valued.
- Dependence of line integral on path of integration.
- In general, the value of the line integral depends on the particular path of integration.

- Exact differential
  If $P\,dx + Q\,dy$ is an exact differential

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

(b) $I = \int_c (P\,dx + Q\,dy)$ is independent of the path of integration

(c) $I = \oint_C (P\,dx + Q\,dy)$ is zero.

- Exact differential in three variables.
  If $P\,dx + Q\,dy + R\,dw$ is an exact differential

  (a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}\ ;\ \frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}\ ;\ \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

  (b) $\int_c (P\,dx + Q\,dy + R\,dw)$ is independent of the path of integration.

  (c) $\oint_C (P\,dx + Q\,dy + R\,dw)$ is zero.

- Green's theorem

  $\oint_C (P\,dx+Q\,dy) = -\iint_R \left\{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right\} dx\,dy$ and, for a simple closed curve,

  $\oint_C (x\,dy - y\,dx) = 2\iint_R dx\,dy = 2A$

  where A is the area of the enclosed figure.

Gradient of a scalar function

Del operator is given by $\nabla = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right)$

$\nabla\phi = \text{grad}\,\phi = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right)\phi = \mathbf{i}\frac{\partial\phi}{\partial x} + \mathbf{j}\frac{\partial\phi}{\partial y} + \mathbf{k}\frac{\partial\phi}{\partial z}$

$\text{grad}\phi = \nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$

Div (Divergence of a vector function)

If $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

then $\text{div}\,A = \nabla.A = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right).\left(a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}\right)$

$\therefore \text{div}\,\mathbf{A} = \nabla.\mathbf{A} = \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$

Note that

(a) the grad operator ∇ acts on a scalar and gives a vector

(b) the div operator $\nabla.$ acts on a vector and gives a scalar.

Example

If $A = x^2y\mathbf{i} - xyz\mathbf{j} + yz^2\mathbf{k}$ then

$\text{Div}\,\mathbf{A} = \nabla.\mathbf{A} = \frac{\partial}{\partial x}(x^2y) - \frac{\partial}{\partial y}(xyz) + \frac{\partial}{\partial z}(yz^4) = 2xy - xz + 2yz$

Example

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

If $A = 2x^2y\mathbf{i} - 2(xy^2+y^3)\mathbf{j}+3y^2z^2\mathbf{k}$ determine $\nabla.\mathbf{A}$ i.e. div **A**.

$\mathbf{A}= 2x^2y\mathbf{i} - 2(xy^2 + y^3z)\mathbf{j} + 3y^2z^2\mathbf{k}$

$$\nabla.\mathbf{A}=\frac{\partial a_x}{\partial x}+\frac{\partial a_y}{\partial y}+\frac{\partial a_z}{\partial z}=4xy-2(2xy+3y^2z)+6y^2z= 4xy - 4xy - 6y^2z+6y^2z = 0$$

Such a vector A for which $\nabla.\mathbf{A} = 0$ at all points, i.e. for all values of x, y, z, is called a solenoid vector. It is rather a special case.

Curl (curl of a vector function)

The curl operator denoted by $\nabla\times$, acts on a vector and gives another vector as a result.

If $\mathbf{A}= a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ then curl $\mathbf{A}=\nabla\times\mathbf{A}$.

$$\text{i.e. curl } \mathbf{A}=\nabla\times\mathbf{A}=\left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right)\times\left(\mathbf{a_1i}+\mathbf{a_2j}+a_3\mathbf{k}\right)$$

$$=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ a_1 & a_2 & a_3\end{vmatrix}$$

$$\therefore\ \nabla\times\mathbf{A}=\mathbf{i}\left(\frac{\partial a_3}{\partial y}-\frac{\partial a_2}{\partial z}\right)+\mathbf{j}\left(\frac{\partial a_1}{\partial z}-\frac{\partial a_3}{\partial x}\right)+\mathbf{k}\left(\frac{\partial a_2}{\partial x}-\frac{\partial a_1}{\partial y}\right)$$

Curl A is thus a vector function.

Example

If $\mathbf{A}=(y^4-x^2z^2)\mathbf{i}+(x^2+y^2)\mathbf{j}-x^2yzk$,determine curl A at the point (1,3, −2).

$$\text{Curl } \mathbf{A} = \nabla\times\mathbf{A} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ y4-x^2z^2 & x^2+y^2 & -x^2yz\end{vmatrix}$$

Now we expand the determinant

$$\nabla\times\mathbf{A}=\mathbf{i}\left\{\frac{\partial}{\partial y}(-x^2yz)-\frac{\partial}{\partial z}(x^2+y^2)\right\}-\mathbf{j}\left\{\frac{\partial}{\partial x}(-x^2yz)-\frac{\partial}{\partial z}(y^4-x^2z^2)\right\}$$

$$+\mathbf{k}\left\{\frac{\partial}{\partial x}(x^2+y^2)-\frac{\partial}{\partial y}(y^4-x^2z^2)\right\}$$

$\nabla\times\mathbf{A}=\mathbf{i}\{-x^2z\}-\mathbf{j}\{-2xyz+2x^2z\}+\mathbf{k}(2x-4y^3\}$. $\therefore$ At (1, 3, − 2),

$\nabla\times\mathbf{A} = \mathbf{i}(2)-\mathbf{j}(12-4)+\mathbf{k}(2-108) = 2\mathbf{i}-8\mathbf{j}-106\mathbf{k}$

Example

Determine curl F at the point (2,0,3) given that $\mathbf{F}=ze^{2xy}\mathbf{i}+2xycosy\mathbf{j}+(x+2y)\mathbf{k}$.

$$\text{In determine form, curl } \mathbf{F} = \nabla\times\mathbf{F} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ ze^{2xy} & 2xzcosy & x+2y\end{vmatrix}$$

Now expand the determinant and substitute the values for x, y and z, finally obtaining curl

$\nabla\times\mathbf{F} = \mathbf{i}\{2-2x\cos y\}-\mathbf{j}\{1-e^{2xy}\}+\mathbf{k}(\{2z\cos y-2xze^{2xy}\}$

$\therefore$ At(2,0,3) $\nabla\times\mathbf{F}=\mathbf{i}(2-4)-\mathbf{j}(1-1)+\mathbf{k}(6-12) = -2\mathbf{i}-6\mathbf{k} = -2(\mathbf{i}+3k)$

Summary of grad, div and curl

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(a) Grad operator ∇ acts on a scalar field to give a vector field.
(b) Div operator $\nabla.$ Acts o a vector field to give a scalar field.
(c) Curl operator $\nabla \times$ acts on a vector field to give a vector field.
(d) With a scalar function ϕ (x,y,z)

$$\text{Grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

(e) With a vector function $A = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$

(i) $$\text{div } \mathbf{A} = \nabla.\mathbf{A} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z}$$

(ii) $$\text{Curl } \mathbf{A} = \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix}$$

Multiple Operations

We can combine the operators grad, div and curl in multiple operations, as in the examples that follow.

EXAMPLE

If $A = x^2y\mathbf{i} + yz^3\mathbf{j} - zx^3\mathbf{k}$

Then div $\mathbf{A}=\nabla.\mathbf{A} = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right).(x^2y\mathbf{i} + yz^3\mathbf{j} - zx^3\mathbf{k})$

$= 2xy+z^3 + x^3 = \phi$ (say)

Then grad (div $\mathbf{A}$) $= \nabla(\nabla.\mathbf{A}) = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k} = (2y+3x^2)\mathbf{i}+(2x)\mathbf{j}+(3z^2)\mathbf{k}$

i.e., grad div $A = \nabla(\nabla.\mathbf{A}) = (2y+3x^2)\mathbf{i}+2x\mathbf{j}+3z^2\mathbf{k}$

Example

If $\phi = xyz - 2y^2z + x^2z^2$ determine div grad ϕ at the point (2, 4, 1).

First find grad ϕ and then the div of the result.

div grad $\phi = \nabla.(\nabla\phi)$

We have $\phi = xyz - 2y^2z + x^2z^2$

$$\text{grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k} = (yz+2xz^2)\mathbf{i}+(xz-4yz)\mathbf{j}+(xy-2y^2+2x^2z)\mathbf{k}$$

$\therefore$ div grad $\phi=\nabla.(\nabla\phi) = 2z^2 - 4z+2x^2$

$\therefore$ At (2,4,1), div grad $\phi = \nabla.(\nabla\phi) = 2 - 4 + 8 = 6$

$$\text{grad } \phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

$$\text{Then div grad } \phi = \nabla.(\nabla\phi) = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right).\left(\frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}\right) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial z^2} + \frac{\partial^2\phi}{\partial z^2}$$

$$\therefore \text{div grad } \phi = \nabla.(\nabla\phi) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example

If $F = x^2yz\mathbf{i} + xyz^2\mathbf{j} + y^2z\mathbf{k}$ determine curl **F** at the point (2, 1, 1).Determine an expression for curl F in the usual way, which will be a vector, and then the curl of the result. Finally substitute values.

Curl curl $\mathbf{F}=\nabla\times(\nabla\times\mathbf{F}) = \mathbf{i}+2\mathbf{j}+6\mathbf{k}$

$$\text{For curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & xyz^2 & y^2z \end{vmatrix} = (2yz-2xyz)\mathbf{i}+x^2y\mathbf{j}+(yz^2-x^2z)\mathbf{k}$$

$$\text{Then Curl Curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2yz-2xyz & x^2y & yz^2-x^2z \end{vmatrix} = z^2\mathbf{i}-(-2xz-2y+2xy)\mathbf{j}+(2xy-2z+2xz)\mathbf{k}$$

$\therefore$ At (2, 1, 1), curl cul $\mathbf{F}=\nabla\times(\nabla\times\mathbf{F}) = \mathbf{i}+2\mathbf{j}+6\mathbf{k}$

Two interesting general results

(a) Curl grad ϕ where ϕ is a scalar

$$\text{grad } \phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

$$\therefore \text{ curl grad } \phi = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix}$$

$$= \mathbf{i}\left\{\frac{\partial^2\phi}{\partial y\partial z}-\frac{\partial^2\phi}{\partial z\partial x}\right\} - \mathbf{j}\left\{\frac{\partial^2\phi}{\partial z\partial x}-\frac{\partial^2\phi}{\partial x\partial z}\right\} + \mathbf{k}\left\{\frac{\partial^2\phi}{\partial x\partial y}-\frac{\partial^2\phi}{\partial y\partial x}\right\} = 0$$

$\therefore$ curl grad $\phi = \nabla\times(\nabla\phi) = 0$

(b) Div curl A where A is a vector.

$$\mathbf{A} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

$$\text{curl } \mathbf{A} = \nabla\times\mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix} = \mathbf{i}\left(\frac{\partial a_z}{\partial y}-\frac{\partial a_y}{\partial z}\right) - \mathbf{j}\left(\frac{\partial a_z}{\partial x}-\frac{\partial a_x}{\partial z}\right) + \mathbf{k}\left(\frac{\partial a_y}{\partial x}-\frac{\partial a_x}{\partial y}\right)$$

$$\text{Then div curl } \mathbf{A} = \nabla.(\nabla\times\mathbf{A}) = \left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right).(\nabla\times\mathbf{A})$$

$$= \frac{\partial^2 a_z}{\partial x\partial y}-\frac{\partial^2 a_y}{\partial z\partial x}-\frac{\partial^2 a_z}{\partial x\partial y}+\frac{\partial^2 a_x}{\partial y\partial z}-\frac{\partial^2 a_y}{\partial z\partial x}-\frac{\partial^2 a_x}{\partial y\partial z} = 0$$

$\therefore$ div curl $\mathbf{A} = \nabla.(\nabla\times\mathbf{A}) = 0$

(c) Div grad ϕ where ϕ is a scalar.

$$\text{grad } \phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

$$\text{Then div grad } \phi = \nabla.(\nabla\phi) = \left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right).\left(\frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k}\right) = \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial z^2}+\frac{\partial^2\phi}{\partial z^2}$$

$$\therefore \text{div grad } \phi = \nabla.(\nabla\phi) = \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}$$

This result is sometimes denoted by $\nabla^2\phi$.

So these general results are

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(a) curl grad $\phi = \nabla\times(\nabla\phi) = 0$

(b) div curl $\mathbf{A} = \nabla.(\nabla\times\mathbf{A}) = 0$

(c) div grad $\phi=\nabla.(\nabla\phi)=\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

v.i.p

Lecture No -35 Definite Integrals

Definite integral for $\sin^n x$ and $\cos^n x$, $0 \le x \le \pi/2$

$$\int_0^{\frac{\pi}{2}} \sin^2 x dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)dx = \frac{1}{2}\left|x - \frac{\sin 2x}{2}\right|_0^{\frac{\pi}{2}} = \frac{1}{2}\left|\frac{\pi}{2} - \frac{\sin \pi}{2}\right| = \frac{\pi}{4}$$

$$\int_0^{\frac{\pi}{2}} \sin^2 x dx = \frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} (1+\cos 2x)dx = \frac{1}{2}\left|x + \frac{\sin 2x}{2}\right|_0^{\frac{\pi}{2}} = \frac{1}{2}\left|\frac{\pi}{2} + \frac{\sin \pi}{2}\right| = \frac{\pi}{4}$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x dx = \frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^3 x dx = \int_0^{\frac{\pi}{2}} \sin^2 x \sin x dx = \int_0^{\frac{\pi}{2}} (1-\cos^2 x)\sin x dx = \int_0^{\frac{\pi}{2}} \sin x dx + \int_0^{\frac{\pi}{2}} \cos^2 x(-\sin x)dx$$

$$= \left|-\cos x\right|_0^{\frac{\pi}{2}} + \left|\frac{\cos^3 x}{3}\right|_0^{\frac{\pi}{2}} = -\cos\frac{\pi}{2} + \cos 0 + \frac{1}{3}\left[\cos^3\frac{\pi}{2} - \cos^3 0\right] = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \cos^3 x dx = \int_0^{\frac{\pi}{2}} \cos^2 x \cos x dx = \int_0^{\frac{\pi}{2}} (1-\sin^2 x)\cos x dx = \int_0^{\frac{\pi}{2}} \cos x dx - \int_0^{\frac{\pi}{2}} \sin^2 x(\cos x)dx$$

$$= \left|\sin x\right|_0^{\frac{\pi}{2}} - \left|\frac{\sin^3 x}{3}\right|_0^{\frac{\pi}{2}} = \sin\frac{\pi}{2} - \sin 0 - \frac{1}{3}\left[\sin^3\frac{\pi}{2} - \sin^3 0\right] = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^4 x dx = \int_0^{\frac{\pi}{2}} (\sin^2 x)^2 dx = \int_0^{\frac{\pi}{2}} \left[\frac{1-\cos 2x}{2}\right]^2 dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (1-2\cos 2x + \cos^2 2x)dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} (1-2\cos 2x + \frac{1+\cos 4x}{2})dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (\frac{3}{2} - 2\cos 2x + \frac{c\cos 4x}{2})dx$$

$$= \frac{1}{4}\left|\frac{3}{2}x - \sin 2x + \frac{\sin 4x}{8}\right|_0^{\frac{\pi}{2}} = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2} - \sin\pi + \frac{\sin 2\pi}{8}\right]$$

$$\int_0^{\frac{\pi}{2}} \sin^4 x dx = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2}\right] \qquad so \qquad \int_0^{\frac{\pi}{2}} \sin^4 x dx = \frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$\int_0^{\frac{\pi}{2}} \cos^4 x dx = \int_0^{\frac{\pi}{2}} (\cos^2 x)^2 dx = \int_0^{\frac{\pi}{2}} \left[\frac{1+\cos 2x}{2}\right]^2 dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (1+2\cos 2x+\cos^2 2x)dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} (1+2\cos 2x+\frac{1+\cos 4x}{2})dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (\frac{3}{2}+2\cos 2x+\frac{c\,os\,4x}{2})dx$$

$$= \frac{1}{4}\left|\frac{3}{2}x+\sin 2x+\frac{\sin 4x}{8}\right|_0^{\frac{\pi}{2}} = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2}+\sin\pi+\frac{\sin 2\pi}{8}\right]$$

$$\int_0^{\frac{\pi}{2}} \cos^4 x dx = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2}\right] \qquad \text{So} \quad \int_0^{\frac{\pi}{2}} \cos^4 x dx = \frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^5 x dx = \frac{4}{5}\frac{2}{3} \qquad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^5 x dx = \frac{4}{5}\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^6 x dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2} \qquad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^6 x dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^7 x dx = \frac{6}{7}\frac{4}{5}\frac{2}{3} \qquad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^7 x dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^8 x dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2} \qquad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^8 x dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^9 x dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3} \qquad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^9 x dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^{10} x dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2} \qquad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^8 x dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

Wallis Sine Formula

When n is even $\int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{n-1}{n}.\frac{n-3}{n-2}.\frac{n-5}{n-4}.\frac{n-7}{n-6}.\text{- - - - - - - - - -}\frac{5}{6}.\frac{3}{4}.\frac{1}{2}.\frac{\pi}{2}$

When n is odd $\int_0^{\frac{\pi}{2}} \cos^n x dx = \frac{n-1}{n}.\frac{n-3}{n-2}.\frac{n-5}{n-4}.\frac{n-7}{n-6}.\text{- - - - - - - - - -}\frac{6}{7}.\frac{4}{5}.\frac{2}{3}.$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$\int_0^{\frac{\pi}{2}} \sin^{11} x dx = \frac{10.8.6.4.2}{11.9.7.5.3} \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^{11} x dx = \frac{10.8.6.4.2}{11.9.7.5.3}$$

$$\int_0^{\frac{\pi}{2}} \sin^{12} x dx = \frac{11.9.7.5.3.1}{10.8.6.4.2}\frac{\pi}{2} \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^{12} x dx = \frac{11.9.7.5.3.1}{10.8.6.4.2}\frac{\pi}{2}$$

Integration By Parts

$$\int UVdx = U\int V\,dx - \int\left[\int Vdx.\frac{dU}{dx}\right]dx$$

Example *Evaluate* $\int$ x lnx dx

$\int$ x lnx dx= lnx $\int$ xdx – $\int$ [$\int$ x dx. $\frac{d}{dx}$(ln x)] dx (We are integrating by parts)

$= \ln x(\frac{x^2}{2}) - \int (\frac{x^2}{2})(\frac{1}{x})dx = (\frac{x^2}{2}) \ln x - \int (\frac{x}{2})d\,x = (\frac{x^2}{2}) \ln x - \frac{1}{2}(\frac{x^2}{2})$

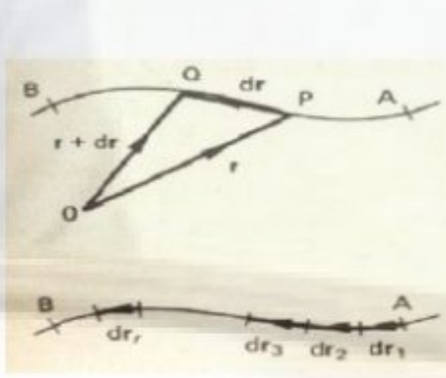
Example *Evaluate* $\int$ x sinx dx

$\int$ x sinx dx =x $\int$ sin xdx – $\int$ [$\int$ sinx dx. $\frac{d}{dx}$(x)] dx (We are integrating by parts)

$= x(-\cos x) - \int (-\cos x)(1)dx = -x(\cos x) + \int \cos x\, d\,x = -x(\cos x) + \sin x$

Line Integrals

Let a point p on the curve c joining A and B be denoted by the position vector **r** with respect to origin O. If q is a neighboring point on the curve with position vector **r+dr** , then $\overline{PQ} = \mathbf{r}$

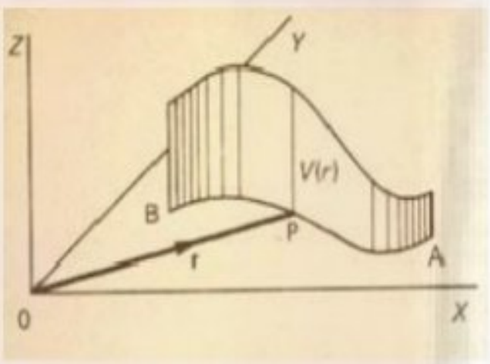


The curve c can be divided up into many n such small arcs , approximating to $d\mathbf{r}_1$, $d\mathbf{r}_2$, $d\mathbf{r}_3$,..... $d\mathbf{r}_p$,.....

so that $\overline{AB}\sum_{p=1}^{n} dr_p$ where $d\mathbf{r}_p$ is a vector representing the element of the arc in both magnitude and direction. If dr $\to 0$, then the length of the curve AB= $\int_c dr$.

Scalar Field

If a scalar field V(r) exists for all points on the curve , the $\sum_{p=1}^{n} V(r)dr_p$ with dr $\to 0$, defines the line integral of V i.e line integral = $\int_c V(r)dr$.



We can illustrate this integral by erecting a continuous Ordinate to V(r) at each point of the curve $\int_c V(r)dr$ is then represented by the area of the curved surface between the ends A and B the curve c. To evaluate a line integral , the integrand is expressed in terms of x , y ,z with dr =dx + dy j + dz k

In practice , x , y and are often expressed in terms of parametric equation of a fourth variable (say u), i.e. x = x(u) ; y = y(u) ; z = z(u) . From these , dx, dy and dz can be written in terms of u and the integral evaluate in terms of this parameter u.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

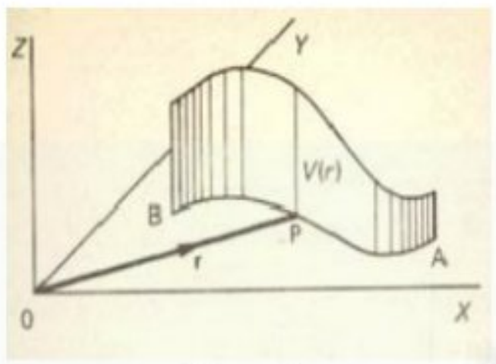
Lecture No -36 Scalar Field

Scalar Field

If a scalar field V(r) exists for all points on the curve ,

the $\sum_{p=1}^{n} V(r)dr_p$ with dr $\to 0$, defines the line integral

of V i.e line integral = $\int_c V(r)dr.$



We can illustrate this integral by erecting a continuous Ordinate to V(r) at each point of the curve $\int_c V(r)dr$ is then represented by the area of the curved surface between the ends A and B the curve c. To evaluate a line integral , the integrand is expressed in terms of x , y ,z with dr =dx + dy j + dz k
In practice , x , y and are often expressed in terms of parametric equation of a fourth variable (say u), i.e. x = x(u) ; y = y(u) ; z = z(u) . From these , dx, dy and dz can be written in terms of u and the integral evaluate in terms of this parameter u.

Example

If V=xy^{2}z , evaluate $\int_c V(r)dr$ along the curve c having parametric equations

x = 3u; y=2u^2 ;z=u^3 between A(0,0,0) and B(3,2,1)
V= xy^{2}z = (3u)(4u^4)(u^3)=12u^8
d**r**= dx**i**+ dy **j**+ dz **k** $\Rightarrow$ **dr= 3**du i +4udu j +**3u^2**du k
for x = 3u ; $\therefore$ dx = 3du ; y = 2u^2 $\therefore$ dy = 4u du ; z = u^3 $\therefore$ dz =**3u^{2}dz**
Limiting : A(0,0,0) corresponds to B(3,2,1) corresponds to u
A(0,0,0) $\equiv$ u=0 ; B(3,2,1) $\equiv$ u = 1

$$\int_c V(r)dr = \int_0^1 12u^8(3\,i + 4u\,j + 3u^2\,k)du = \left|36\frac{u^9}{9}i + 48\frac{u^{10}}{10}j + 36\frac{u^{11}}{11}\right|_0^1 = 4i + \frac{24}{5}j + \frac{36}{11}k$$

Example

If V = xy + y^{2}z Evaluate $\int_c V(r)dr$ along the curve c defined by x= t^2; y = 2t ; z= t+5

between A(0,0,5) and B(4,4,7) . As before , expressing V and d**r** in term of the parameter t .

since V=xy+y^{2}z
= (t^2)(2t)+(4t^2)(t+5)
= 6t^3 + 20t^2.
x = t^2 dx = 2t dt
y = 2t dy = 2 dt
z = t+5 dz = dt
$\therefore$ dr = dx**i** + dy**j** + dz**k**
= 2t dt **i** + 2 dt **j** + dt**k**
$\therefore \int_C$ Vdr = $\int_C$ (6t^3+20t^2)(2t **i** +2**j**+**k**) dt

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Limits: A (0, 0, 5) $\equiv t = 0$;

B (4, 4, 7) $\equiv t = 2$

$$\therefore \int_C \mathbf{V}\,d\mathbf{r} = \int_0^2 (6t^3+20t^2)(2t\,\mathbf{i}+2\,\mathbf{j}+\mathbf{k})\,dt$$

$$\int_C \mathbf{V}\,d\mathbf{r} = 2\int_0^2 \{6t^4+20t^3)\mathbf{i}+(6t^3+20t^2)\mathbf{j} +(3t^3+10t^2)\mathbf{k}\}dt.$$

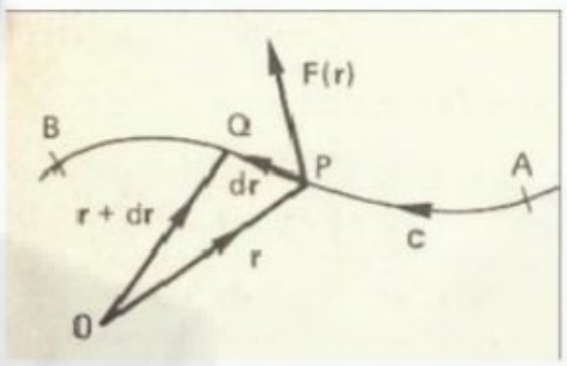
$$= \frac{8}{15}(444\mathbf{i}+290\mathbf{j}+145\mathbf{k})$$

Vector Field

If a vector field F(r) exist for all points of the curve c, then for each element of arc we can form the scalar product F.dr. Summing these products for all elements of arc, we have

$$\sum_{p=1}^{n} F.dr_p$$

The line integral of F(r) fr om A to B along the stated curve = $\int_C F.dr$.



In this case, since **F**.d**r** is a scalar product, then the line integral is a scalar. To evaluate the line integral, F and d **r** are expressed in terms of x,y,z, and the curve in parametric form. We have

$\mathbf{F} = F_1\,\mathbf{i} + F_2\,\mathbf{j} + F_3\,\mathbf{k}$

And $d\mathbf{r} = dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}$

Then $\mathbf{F}.d\mathbf{r} = (F_1\,\mathbf{i} + F_2\,\mathbf{j} + F_3\,\mathbf{k}).(dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}) = \int_c (F_1dx + F_2dy + F_3dz)$

Now for an example to show it in operation.

Example

If $F(r) = x^2y\,\mathbf{i} + xz\,\mathbf{j} + 2yz\,\mathbf{k}$, Evaluate $\int_c F.dr$ between A(0,0,0) and B(4,2,1) along the curve c having parametric equations $x=4t$; $y-2t^2$; $z = t^3$

Expressing everthing in terms of the parameter t, we have

$dx = 4\,dt$; $dy = 4tdt$; $dz = 3t^2\,dt$

$x^2y = (16t^2)(2t^2) = 32\,t^4$

$x = 4t \qquad \therefore \quad dx = 4\,dt$

$xz = (4t)(t^3) = 4\,t^4$

$y = 2\,t^2 \qquad dy = 4t\,dt$

$2yz = (4\,t^2)(\,t^3) = 4\,t^5$

$z = t^3 \qquad \therefore dz = 3t^2\,dt$

$\mathbf{F} = 32\,t^4\,\mathbf{i} + 4\,t^4\,\mathbf{j} - 4\,t^5\,\mathbf{k}$

$dr = 4dt\,\mathbf{i} + 4t\,dt\,\mathbf{j} + 3t^2\,\mathbf{k}$

Then $\int \mathbf{F}.d\mathbf{r} = \int (32t^4\mathbf{i}+4t^4\mathbf{j}-4t^5\mathbf{k}).(4dt\,\mathbf{i} + 4t\,dt\,\mathbf{j} + 3t^2\,dt\,\mathbf{k})$

$$= \int (128t^4+16t^5+12t^7)\,dt$$

Limits: A(0,0,0) $\equiv t = 0$;

B (4, 2, 1) $\equiv t = 1$

$$\int_C \mathbf{F}.d\mathbf{r} = (128t^4 + 16t^5 + 12t^7)dt = \frac{128}{5}\,t^5 + \frac{16}{6}\,t^6 + \frac{12}{8}\,t^8 = \frac{128}{5} + \frac{8}{3} + \frac{3}{2} = 29.76$$

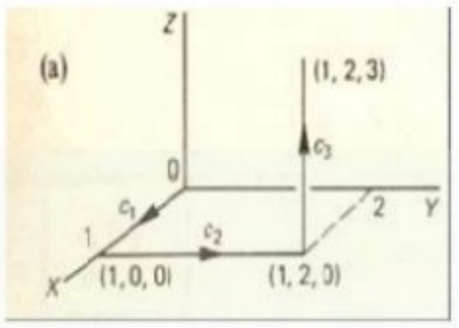
© Copyright Virtual University of Pakistan

Scanned with CamScanner

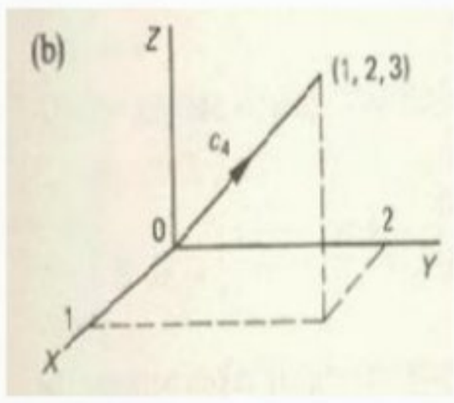
Scanned with CamScanner

Example

If $\mathbf{F(r)} = x^2y\mathbf{i} + 2yz\mathbf{j} + 3z^2x\mathbf{k}$
Evaluate $\int_C \mathbf{F}.d\mathbf{r}$ between A(0,0,0) and B(1,2,3)



B (1, 2, 3)
(a) along the straight line
c_1 from (0, 0, 0) to (1, 0, 0)
then c_2 from (1, 0, 0) to (1, 2, 0)
and c_3 from (1, 2, 0) to (1, 2, 3)
(b) along the straight line c_4 joining (0, 0, 0) to (1, 2, 3).



We first obtain an expression for $\mathbf{F}.d\mathbf{r}$ which is

$\mathbf{F}.d\mathbf{r} = (x^2y\mathbf{i} + 2yz\mathbf{j} + 3z^2x\mathbf{k}).$
$(dx\,\mathbf{i} + dy\mathbf{j} + dz\,\mathbf{k})$

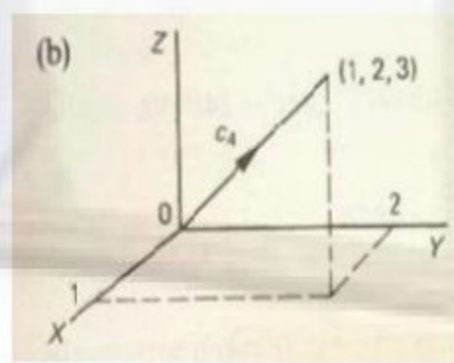
$\mathbf{F}.d\mathbf{r} = x^2y\,dx + 2yz\,dy + 3z^2x\,dz$

$\int \mathbf{F}.d\mathbf{r} = \int x^2y dx + \int 2yz dy + \int 3z^2x dz$

Here the integration is made in three sections, along c_1, c_2 and c_3.

(i) c_1: $y = 0, z = 0, dy = 0, dz = 0$
$\therefore \int_{c_1} \mathbf{F}.d\mathbf{r} = 0+0+0 = 0$

(ii) c_2: The conditions along c_2 are
c_2: $x = 1, z = 0, dx = 0, dz = 0$
$\therefore \int_{c_2} F.dr = 0 + 0 + 0 = 0$



(iii) c_3: $x = 1, y = 2, dx = 0, dy = 0$

$\int_{c_3} F.dr = 0 + 0 + \int_0^3 3z^2 dz = 27$

Summing the three partial results

$\int_{(0,0,0)}^{(1,2,3)} \mathbf{F}.d\mathbf{r} = 0 + 0 + 27 = 27$

$\therefore \int_{c_1+c_2+c_3} \mathbf{F}.d\mathbf{r} = 27$

If t taken as the parameter, the parametric equation of c are $x = t; y = 2t; z = 3t$

$(0, 0, 0) \Rightarrow t = 0$, $(1, 2, 3) \Rightarrow t = 1$ and the limits of t are $t = 0$ and $t = 1$

$\mathbf{F} = 2t^3\mathbf{i} + 12t^2\mathbf{j} + 27t^3\mathbf{k}$

$d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + kdz = dt\,\mathbf{i} + 2dt\,\mathbf{j} + 3dt\,\mathbf{k}$

$\int_{C_4} \mathbf{F}.d\mathbf{r} = \int_0^1 (2t^3\mathbf{i}+12t^3\mathbf{j}+27t^3\mathbf{k}).(\mathbf{i}+2\mathbf{j}+3\mathbf{k})\,dt = \int_0^1 (2t^3+24t^2+81t^3)dt$

$= \int_0^1 (83t^3 + 24t^2)\,dt = \left[83\frac{t^4}{4} + 8t^3\right]_0^1 = \frac{115}{4} = 28.75$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

So the value of the line integral depends on the path taken between the two end points A and B

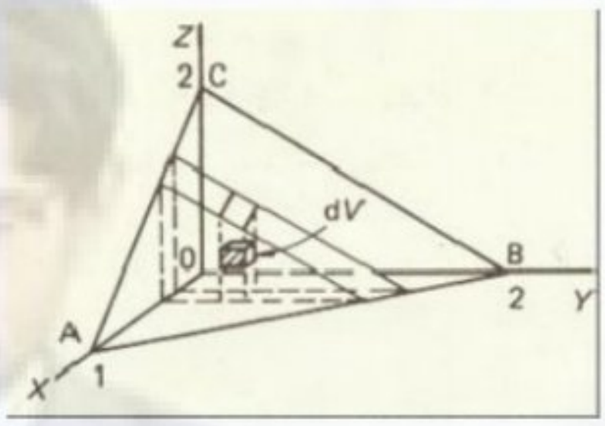
(a) $\int \mathbf{F}.d\mathbf{r}$ via c_1, c_2 and $c_3 = 27$

(b) $\int \mathbf{F}.d\mathbf{r}$ via $c_4 = 28.75$

Example

Evaluate $\int_v F\,dv$ where V is the region bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $2x + y = 2$, and $F = 2z\ \mathbf{i} + y\ \mathbf{k}$. To sketch the surface $2x + y + z = 2$, note that

when $z = 0$, $2x+y=2$ i.e. $y = 2 - 2x$
when $y = 0$, $2x+z=2$ i.e. $z = 2 - 2x$
when $x = 0$, $y+z=2$ i.e. $z = 2 - y$
Inserting these in the planes
$x = 0$, $y = 0$, $z = 0$ will help.
The diagram is therefore.



So $2x + y + z = 2$ cuts the axes at
A(1,0,0); B (0, 2, 0); C (0, 0, 2).
Also $F = 2z\mathbf{i} + y\mathbf{k}$;

$z = 2 - 2x - y = 2(1 - x) - y$

$$\therefore \int_v F dV = \int_0^1 \int_0^{2(1-x)} \int_0^{2(1-x)-y} (2x\mathbf{i}+y\mathbf{k})\,dz\,dy\,dx$$

$$= \int_0^1 \int_0^{2(1-x)} \left[z^2\mathbf{i}+yz\mathbf{k}\right]_{z=0}^{z=2(1-x)-y} dy\,dx$$

$$= \int_0^1 \int_0^{2(1-x)} \{[4(1-x)^2-4(1-x)y+y^2]\mathbf{i}+[2(1-x)y-y^2]k\}\,dy\,dx$$

$$\int_v F dV = \frac{1}{3}(2\mathbf{i}+\mathbf{k})$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

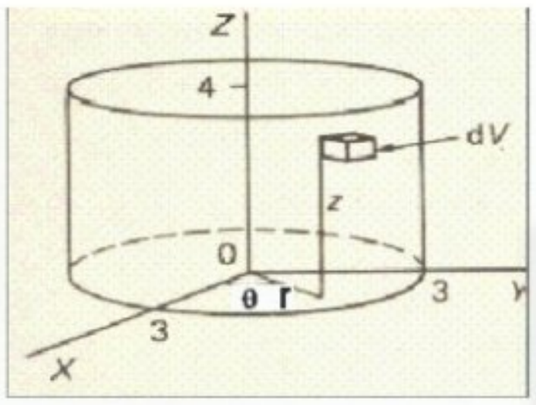
Scanned with CamScanner

Lecture No -37 Examples

Example

Evaluate $\int_v \mathbf{F}dV$ where $\mathbf{F}=2\mathbf{i}+2z\mathbf{j}+y\mathbf{k}$ and V is the region bounded by the planes z = 0, z = 4 and the surface

$x^2+y^2 = 9$.



It will be convenient to use cylindrical polar coordinates (r, θ, z) so the relevant transformations are

$x = r\cos\theta;$ $y = r\sin\theta$ $z = z;$ $dV=rdrd\theta\, dz$

Then $\int_v FdV= \iiint_v (2i+2zj+yk)dxdydz$

Changing into cylindrical polar coordinates with appropriate change of limits this becomes

$$\int_v FdV=\int_{\theta=0}^{2\pi}\int_{r=0}^{3}\int_{z=0}^{4}(2i+2zj+r\sin\theta k)rdzdrd\theta=\int_{\theta=0}^{2\pi}\int_{r=0}^{3}\left[2z\mathbf{i}+z^2\mathbf{j}+r\sin\theta z\mathbf{k}\right]_{z=0}^{4}rdrd\theta$$

$$=\int_0^{2\pi}\int_0^{3}(8\mathbf{i}+16\mathbf{j}+4r\sin\theta\mathbf{k})rdrd\theta= 4\int_0^{2\pi}\int_0^{3}(2r\mathbf{i}+4r\mathbf{j}+r^2\sin\theta\mathbf{k})\,drd\theta=4\int_0^{2\pi}\left[r^2\mathbf{i}+2r^2\mathbf{j}+\frac{r^3}{3}\sin\theta\,\mathbf{k}\right]_0^3 d\theta$$

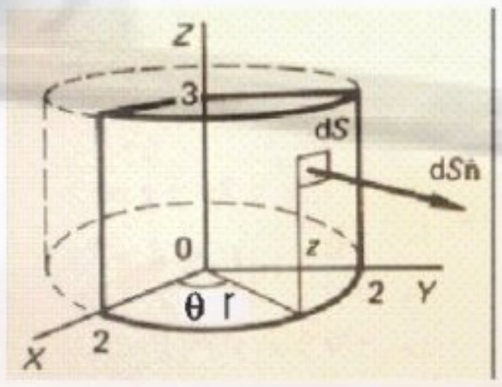
$$=4\int_0^{2\pi}(9\mathbf{i}+18\mathbf{j}+9\sin\theta\mathbf{k})\,d\theta= 36\int_0^{2\pi}(\mathbf{i}+2\mathbf{j}+\sin\theta\mathbf{k})d\theta= 36\left[\theta\mathbf{i}+2\theta\mathbf{j}-\cos\theta\mathbf{k}\right]_0^{2\pi}$$

$$=36\{(2\pi\mathbf{i}+4\pi\mathbf{j}-\mathbf{k})-(-\mathbf{k})\}= 72\pi(\mathbf{i}+2\mathbf{j})$$

Scalar Fields

A scalar field F = xyz exists over the curved surface S defined by $x^2+y^2= 4$ between the planes z = 0 and z = 3 in the first octant.

Evaluate $\int_S \mathbf{F}\,dS$ over this surface.



We have F = xyz S: $x^2+y^2-4 = 0,$ z = 0 to z = 3

$d\mathbf{S} = \hat{\mathbf{n}}\,dS$ where $\hat{\mathbf{n}} = \dfrac{\nabla S}{|\nabla S|}$

Now $\nabla S = \dfrac{\partial S}{\partial x}\mathbf{i}+\dfrac{\partial S}{\partial y}\mathbf{j}+\dfrac{\partial S}{\partial z}\mathbf{k} = 2x\mathbf{i}+2y\mathbf{j}$

$|\nabla S| = \sqrt{4x^2+4y^2} =2\sqrt{x^2+y^2} =2\sqrt{4} = 4$

$\therefore \hat{\mathbf{n}} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{x\mathbf{i}+y\mathbf{j}}{2}$

$\therefore d\mathbf{S} = \hat{\mathbf{n}}dS = \dfrac{x\mathbf{i}+y\mathbf{j}}{2}dS$

$\therefore \int_S FdS=\int_S F\hat{\mathbf{n}}dS= \dfrac{1}{2}\int_S xyz(x\mathbf{i}+y\mathbf{j})dS = \dfrac{1}{2}\int_S (x^2yz\mathbf{i}+xy^2z\mathbf{j})dS$ (1)

We have to evaluate this integral over the prescribed surface.

Changing to cylindrical coordinates with r = 2

$x = 2\cos\theta;$ $y = 2\sin\theta$ $z = z;$ $dS = 2d\theta dz$

$\therefore x^2yz = (4\cos^2\theta)(2\sin\theta)(z)= 8\cos^2\theta\sin\theta z$

$xy^2z = (2\cos\theta)(4\sin^2\theta)(z)=8\cos\theta\sin^2\theta z$

Then result $\int_S FdS = \dfrac{1}{2}\int_S (x^2yz\mathbf{i}+xy^2z\mathbf{j})dS$ becomes.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$\int_S F dS = \frac{8}{2}\int_0^{\pi/2}\int_0^3 (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin 2\theta\mathbf{j})2z dz d\theta = 4\int_0^{\pi/2}\int_0^3 (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin^2\theta\mathbf{j})2z dz\, d\theta$$

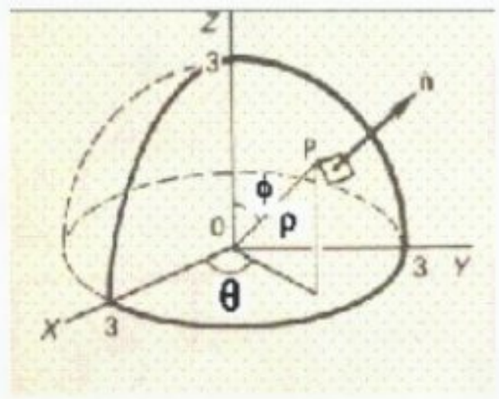
$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin^2\theta\mathbf{j})z^2\Big|_0^3 d\theta = 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin^2\theta\mathbf{j})9 d\theta$$

$$= 36\left[-\frac{\cos^3\theta}{3}\mathbf{i} + \frac{\sin^3\theta}{3}\mathbf{j}\right]_0^{\pi/2} = 12\,(\mathbf{i}+\mathbf{j})$$

Vector Field

A vector field $\mathbf{F}=y\mathbf{i}+2\mathbf{j}+\mathbf{k}$ exists over a surface S defined by $x^2+y^2+z^2=9$ bounded by $x=0, y=0, z=0$ in the first octant,

Evaluate $\int_S \mathbf{F}.dS$ over the surface indicated.



$$dS = \hat{\mathbf{n}}dS; \qquad \hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|}$$

$$S: x^2+y^2+z^2-9=0$$

$$\nabla S = \frac{\partial S}{\partial x}\mathbf{i} + \frac{\partial S}{\partial y}\mathbf{j} + \frac{\partial S}{\partial z}\mathbf{k} = 2x\mathbf{i}+2y\mathbf{j}+2z\mathbf{k}$$

$$\therefore |\nabla S| = \sqrt{4x^2+4y^2+4z^2} = 2\sqrt{x^2+y^2+z^2} = 2\sqrt{9} = 6$$

$$\therefore \hat{\mathbf{n}} = \frac{1}{6}(2x\mathbf{i}+2y\mathbf{j}+2z\mathbf{k}) = \frac{1}{3}(x\mathbf{i}+y\mathbf{j}+z\mathbf{k})$$

$$\int_S \mathbf{F}.dS = \int_S \mathbf{F}.\hat{\mathbf{n}}dS = \int_S (y\mathbf{i}+2\mathbf{j}+\mathbf{k}).\frac{1}{3}(x\mathbf{i}+y\mathbf{j}+z\mathbf{k})dS = \frac{1}{3}\int_S (xy+2y+z)\,dS$$

Before integrating over the surface, we convert to spherical polar coordinates.

$x = 3\sin\phi\cos\theta$; $y = 3\sin\phi\sin\theta$ $z = 3\cos\phi$; $dS = 9\sin\phi d\phi d\theta$

Limits of ϕ and θ are $\phi = 0$ to $\frac{\pi}{2}$; $\theta = 0$ to $\frac{\pi}{2}$

$$\int_S \mathbf{F}.dS = \frac{1}{3}\int_S (xy+2y+z)\,dS$$

$$xy = 3\sin\phi\cos\theta \cdot 3\sin\phi\sin\theta = 9\sin^2\theta\cos\theta$$

$$2y = 2 \cdot 3\sin\phi\sin\theta = 6\sin\phi\sin\theta$$

$$z = 3\cos\phi$$

$$dS = 9\sin\phi\, d\phi\, d\theta$$

Putting these values we get

$$\therefore \int_S \mathbf{F}.dS = \frac{1}{3}\int_0^{\pi/2}\int_0^{\pi/2} (9\sin^2\phi\sin\theta\cos\theta+6\sin\phi\sin\theta+3\cos\phi)9\sin\phi d\phi d\theta$$

$$= 9\int_0^{\pi/2}\int_0^{\pi/2} (3\sin^3\phi\sin\theta\cos\theta+2\sin^2\phi\sin\theta+\sin\phi\cos\phi)d\phi d\theta$$

As we know that

$$\int_0^{\pi/2} \sin^3\phi\, d\phi = \frac{2}{3} \qquad \text{by Wallis Formula}$$

$$\text{also} \int_0^{\pi/2} \sin^2\phi\, d\phi = \frac{1}{2}\frac{\pi}{2}$$

So we get

$$\int_S \mathbf{F}.d\mathbf{S} = 9\int_0^{\pi/2} \left(2\sin\theta\cos\theta+\frac{\pi}{2}\sin\theta+\frac{1}{2}\right)d\theta = 9\left[\sin^2\theta-\frac{\pi}{2}\cos\theta+\frac{\theta}{2}\right]_0^{\pi/2} = 9\left[(1-0+\frac{\pi}{4})-(0-\frac{\pi}{2}+0)\right]$$

© Copyright Virtual University of Pakistan

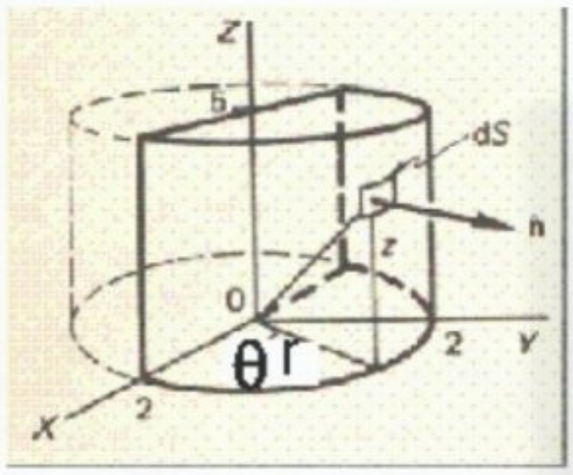
Scanned with CamScanner

Scanned with CamScanner

$$= 9\left[1+\frac{\pi}{4}+\frac{\pi}{2}\right]=9\left(1+\frac{3\pi}{4}\right)$$

Example

Evaluate $\int_S \mathbf{F}.dS$ where $F = 2y\mathbf{j} + z\mathbf{k}$ and S is the surface $x^2 + y^2 = 4$ in the first two octants bounded by the planes $z = 0$, $z = 5$ and $y = 0$.



$$S : x^2 + y^2 - 4 = 0$$

$$\hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial \mathbf{S}}{\partial x}\mathbf{i} + \frac{\partial \mathbf{S}}{\partial y}\mathbf{j} + \frac{\partial \mathbf{S}}{\partial z}\mathbf{k} = 2x\mathbf{i} + 2y\mathbf{i}$$

$$\therefore |\nabla \mathbf{S}| = \sqrt{4x^2 + 4y^2} = 2\sqrt{x^2 + y^2} = 2\sqrt{4} = 4$$

$$\therefore \hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|} = \frac{2x\mathbf{i} + 2y\mathbf{j}}{4} = \frac{1}{2}(x\mathbf{i} + y\mathbf{j})$$

$$\therefore \int_S \mathbf{F}.dS = \int_S \mathbf{F}.\hat{\mathbf{n}}\, dS$$

$$\int_S \mathbf{F}.\hat{\mathbf{n}}\, dS = \frac{1}{2}\int_S (2y^2)\, dS = \int_S y^2\, dS$$

This is clearly a case for using cylindrical polar coordinates.

$$x = 2\cos\theta; \quad y = 2\sin\theta \quad z = z; \quad dS = 2d\theta\, dz$$

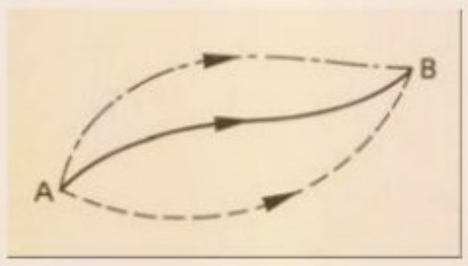
$$\therefore \int_S \mathbf{F}.dS = \int_S y^2 dS = \iint_S 4\sin^2\theta 2 d\theta dz = 8\iint_S \sin^2\theta\, d\theta\, dz$$

Limits: $\theta = 0$ to $\theta = \pi$; $z = 0$ to $z = 5$

$$\int_S \mathbf{F}.dS = 4\int_{z=0}^{5}\int_{\theta=0}^{\pi}(1-\cos 2\theta)\, d\theta dz = 4\int_0^5 \left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} dz = 4\int_0^5 \pi\, dz = 4\pi\left[z\right]_0^5 = 20\pi$$

Conservative Vector Fields

In general, the value of the integral $\int_C \mathbf{F}.d\mathbf{r}$ between two stated points A and B depends on the particular path of integration followed. If, however, the line integral between A and B is independent of the path of integration between the two end points, then the vector field F is said to be conservative.



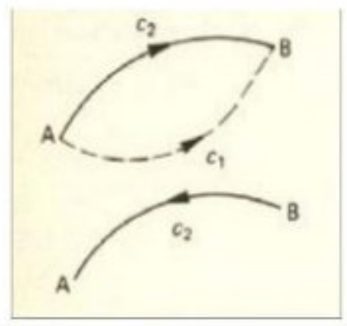
It follows that, for a closed path in a conservative

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

field, $\oint_C \mathbf{F}.d\mathbf{r} = 0$.



For, if the field is conservative,

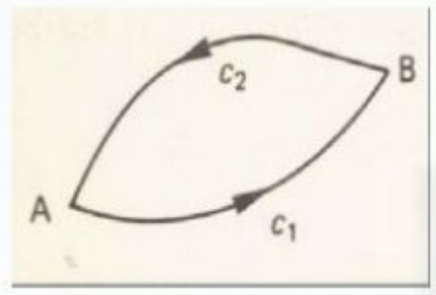
$$\int_{C_1(AB)} \mathbf{F}.d\mathbf{r} = \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$$

But $\int_{c_2(BA)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$

Hence, for the closed path $AB_{c_1} + BA_{c_2}$, $\oint \mathbf{F}.d\mathbf{r}$

$$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} + \int_{c_2(AB)} \mathbf{F}.d\mathbf{r} = \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$$

$$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_1(AB)} \mathbf{F}.d\mathbf{r} = 0$$



© Copyright Virtual University of Pakistan

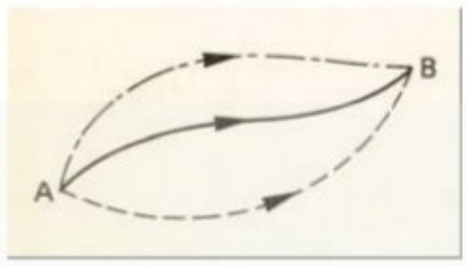
Scanned with CamScanner

Scanned with CamScanner

Lecture No -38 Vector Field

Conservative Vector Fields

In general, the value of the integral $\int_C \mathbf{F}.d\mathbf{r}$ between two stated points A and B depends on the particular path of integration followed. If, however, the line integral between A and B is independent of the path of integration between the two end points, then the vector field F is said to be conservative.



It follows that, for a closed path in a conservative field, $\oint_C \mathbf{F}.d\mathbf{r} = 0$.

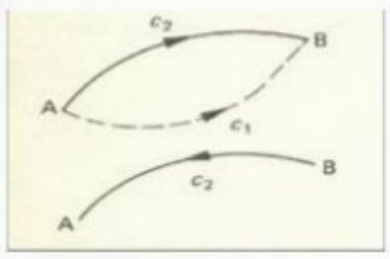
For, if the field is conservative,

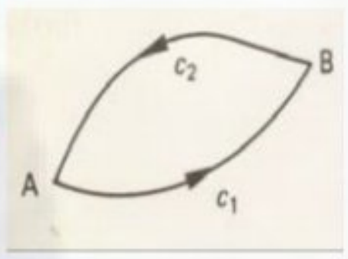
$\int_{C_1(AB)} \mathbf{F}.d\mathbf{r} = \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$ But $\int_{c_2(BA)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$

Hence, for the closed path $AB_{c_1} + BA_{c_2}$, $\oint \mathbf{F}.d\mathbf{r}$

$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} + \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$

$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r} = \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_1(AB)} \mathbf{F}.d\mathbf{r} = 0$

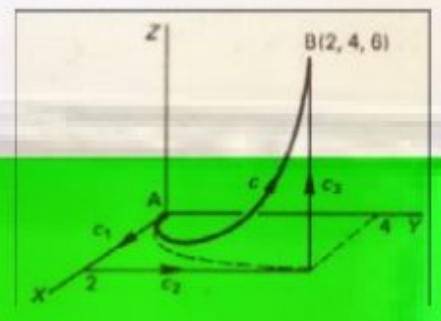




Note that this result holds good only for a closed curve and when the vector field is a conservative field.
Now for some examples

Example

If $\mathbf{F} = 2xyz\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}$, evaluate the line integral $\int \mathbf{F}.d\mathbf{r}$ between A(0,0,0) and B(2,4,6)
(a) along the curve c whose parametric equations are $x = u$, $y = u^2$, $z = 3u$
(b) along the three straight lines c_1:(0,0,0) to (2, 0, 0); c_2 : (2, 0, 0) to (2, 4, 0); c_3: (2, 4, 0) to (2, 4, 6).
Hence determine whether or not F is a conservative field.
First draw the diagram.



(a) $\mathbf{F} = 2xyz\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}$

$x = u;\quad y = 2u;\ z = 3u$

$\therefore \quad dx = du;\quad dy = 2udu;\quad dz = 3du.$

$\mathbf{F}.d\mathbf{r} = (2xyz\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}).(dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k})$

$= 2xyz\,dx + x^2z\,dy + x^2y\,dz$

Using the transformation shown above, we can now express **F**.dr in terms of u.

for $2xyzdx = (2u)(u^2)(3u)du = 6u^4du$

$x^2zdy = (u2)(3u)(2u)du = 6u^4du$

$x^2ydz = (u^2)(u^2)3du = 3u^4du$

$\therefore \quad \mathbf{F}.d\mathbf{r} = 15u^4du$

The limits of integration in u are $u = 0$ to $u = 2$

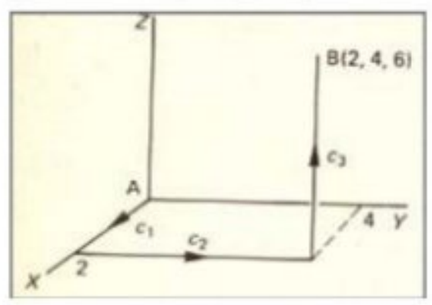
$$\therefore \int_C \mathbf{F}.d\mathbf{r} = \int_0^2 15u^4du = \left[3u^5\right]_0^2 = 96$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(b) The diagram for (b) is as shown. We consider each straigh



$$\int \mathbf{F}.d\mathbf{r} = \int(2xyz\,dx + x^2z\,dy + x^2y\,dz)$$

c_1: (0,0,0) to (2,0,0); $y = 0$, $z = 0$, $dy = 0$,

$$\therefore \quad \int_{C_1} \mathbf{F}.d\mathbf{r} = 0 + 0 + 0 = 0$$

In the same way, we evaluate the line integral along c_2 and c_3.

$$\int_{C_1} \mathbf{F}.d\mathbf{r} = 0;$$

$$\int \mathbf{F}.d\mathbf{r} = \int(2xyz\,dx + x^2z\,dy + x^2y\,dz)$$

c_2: (2,0,0) to (2,4,0); $x = 2$, $z = 0$, $dx = 0$, $dz = 0$

$$\therefore \int_{C_2} \mathbf{F}.d\mathbf{r} = 0+0+0 = 0 \quad \int_{C_2} \mathbf{F}.d\mathbf{r} = 0$$

c_3: (2,4,0) to (2,4,6); $x = 2$, $y = 4$, $dx = 0$, $dy = 0$

$$\therefore \int_{C_3} \mathbf{F}.d\mathbf{r} = 0+0+\int_0^6 16\,dz = \left[16z\right]_0^6 = 96 \quad \therefore \int_{C_3} \mathbf{F}.d\mathbf{r} = 96$$

Collecting the three results together

$$\int_{c_1+c_2+c_3} \mathbf{F}.d\mathbf{r} = 0 + 0 + 96 \quad \therefore \int_{c_1+c_2+c_3} \mathbf{F}.d\mathbf{r} = 96$$

In this particular example, the value of the line integral is independent of the two paths we have used joining the same two end points and indicates that F is a conservative field. It follows that

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix}$$

$$= (x^2 - x^2)\mathbf{i} - (2xy - 2xy)\mathbf{j} + (2xz - 2xz)\mathbf{k} = 0$$

$\therefore$ curl $\mathbf{F} = 0$

So three tests can be applied to determine whether or not a vector field is conservative. They are

(a) $\oint \mathbf{F}.d\mathbf{r} = 0$

(b) curl $\mathbf{F} = 0$

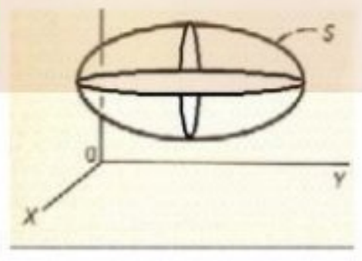
(c) $\mathbf{F} = \text{grad } V$

Any one of these conditions can be applied as is convenient.

Divergence Theorem (Gauss' theorem)

For a closed surface S, enclosing a region V in a vector field F

$$\int_V \text{div } \mathbf{F}\, dV = \int_S \mathbf{F}.d\mathbf{S}$$



In general, this means that the volume integral (tripple integral) on the left-hand side can be expressed as a surface integral (double integral) on the right-hand side.

Example

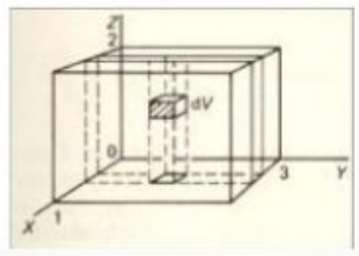
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Verify the divergence theorem for the vector field $\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k}$ taken over the region bonded by the planes $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$.

$dV = dx\, dy\, dz$



we have to show that $\int_V \text{div } \mathbf{F}\, dV = \int_S \mathbf{F}.d\mathbf{S}$

(a) To find $\int_V \text{div } \mathbf{F}\, dV$

$$\text{div } F = \nabla.F = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}\right).(x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k})$$

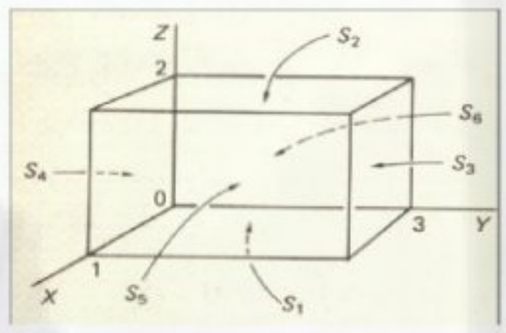
$$= \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(y) = 2x + 0 + 0 = 2x$$

$$\therefore \int_V \text{div } F\, dV = \int_V 2x\, dv = \iiint_V 2x\, dz\, dy\, dx$$

$$\int_V \text{div } \mathbf{F}\, dV = \int_0^1\int_0^3\int_0^2 2x\, dz\, dy\, dx = \int_0^1\int_0^3 \left[2xz\right]_0^2 dy\, dx = \int_0^1 \left[2xz\right]_0^3 dx = \int_0^1 12x\, dx = \left[6x^2\right]_0^1 = 6$$

(b) to find $\int_S F.dS$ i.e. $\int_S \mathbf{F}.\hat{\mathbf{n}}\, dS$

The enclosing surface S consists of six separate plane faces denoted as $S_1, S_2, \ldots S_6$ as shown. We consider each face in turn.



$\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k}$

(i) S_1 (base): $z = 0$; $\hat{\mathbf{n}} = -\mathbf{k}$ (outwards and downwards)

$\therefore \quad \mathbf{F} = x^2\mathbf{i} + y\mathbf{k} \quad dS_1 = dx\, dy$

$$\therefore \int_{S_1} \mathbf{F}.\hat{\mathbf{n}}\, dS = \iint_{S_1} (x2\mathbf{i}+y\mathbf{k}).(-\mathbf{k})\, dy\, dx = \int_0^1\int_0^3 (-y)\, dy\, dx = \int_0^1 \left[-\frac{y^2}{2}\right]_0^3 dx = -\frac{9}{2}$$

(ii) S_2 (top): $z = 2$: $\hat{\mathbf{n}} = \mathbf{k}$ $dS_2 = dx\, dy$

$$\int_{S_2} \mathbf{F}.\hat{\mathbf{n}}\, dS = \iint_{S_2} (x^2\mathbf{i}+z\mathbf{j}+y\mathbf{k}).(\mathbf{k})\, dy\, dx = \int_0^1\int_0^3 y\, dy\, dx = \frac{9}{2}$$

(iii) S_3 (right-hand end): $y = 3$;

$\hat{\mathbf{n}} = \mathbf{j}$ $dS_3 = dx\, dz\, \mathbf{j}$

$\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k}$

$$\therefore \int_{S_3} \mathbf{F}.\hat{\mathbf{n}}\, dS = \iint_{S_3} (x^2\mathbf{i}+z\mathbf{j}+3\mathbf{k}).(\mathbf{j})\, dz\, dx = \int_0^1\int_0^2 z\, dz\, dx = \int_0^1 \left[\frac{y^2}{2}\right]_0^2 dx = \int_0^1 2\, dx = 2$$

(iv) S_4 (left-hand end): $y = 0$, $\hat{\mathbf{n}} = -\mathbf{j}$, $dS_4 = dx\, dz$

$\therefore \int_{S_4} \mathbf{F}.\hat{\mathbf{n}}\, dS = -2$

$$\text{for } \int_{S_4} \mathbf{F}.\hat{\mathbf{n}}\, dS = \iint_{S_4} (x^2\mathbf{i}+z\mathbf{j}+y\mathbf{k}).(-\mathbf{j})\, dz\, dx = \int_0^1\int_0^2 (-z)\, dz\, dx = \int_0^1 \left[-\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 (-2)\, dx = -2$$

Now for the remaining two sides S_5 and S_6. Evaluate these in the same manner, obtaining

$$\int_{S_5} \mathbf{F}.\hat{\mathbf{n}}\, dS = 6; \qquad \int_{S_5} \mathbf{F}.\hat{\mathbf{n}}\, dS = 0$$

Check:

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(v) S_5 (front): $x = 1$; $\hat{\mathbf{n}} = \mathbf{i}$ $dS_5 = dy\,dz$

$\therefore \int_{S_5} \mathbf{F}.\hat{\mathbf{n}}dS = \iint_{S_5} (\mathbf{i}+z\mathbf{j}+y\mathbf{k}).(\mathbf{i})dy\,dz = \iint_{S_5} 1\,dy\,dz = 6$

(vi) S_6 (back): $x = 0$; $\hat{\mathbf{n}} = -\mathbf{i}$ $dS_6 = dy\,dz$

$\therefore \int_{S_6} \mathbf{F}.\hat{\mathbf{n}}dS = \iint_{S_6} (z\mathbf{j}+y\mathbf{k}).(-\mathbf{i})dy\,dz = \iint_{S_6} 0\,dy\,dz = 0$

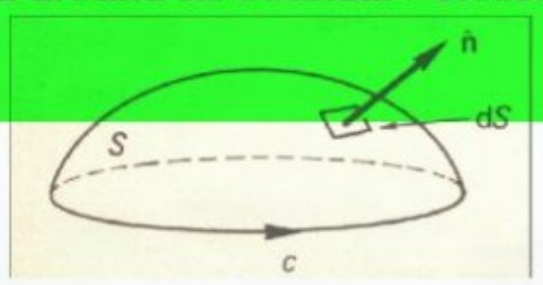
For the whole surface S we therefore have $\int_S \mathbf{F}.dS = -\frac{9}{2} + \frac{9}{2} + 2 - 2 + 6 + 0 = 6$

and from our previous work in section (a) $\int_V \text{div}\,\mathbf{F}\,dV = 6$

We have therefore verified as required that, in this example $\int_V \text{div}\,\mathbf{F}\,dV = \int_S \mathbf{F}.dS$

Stokes Theorem

If F is a vector field existing over an open surface S and around its boundary closed curve C, then $\int_S \text{curl}\,\mathbf{F}.dS = \oint_C \mathbf{F}.d\mathbf{r}$

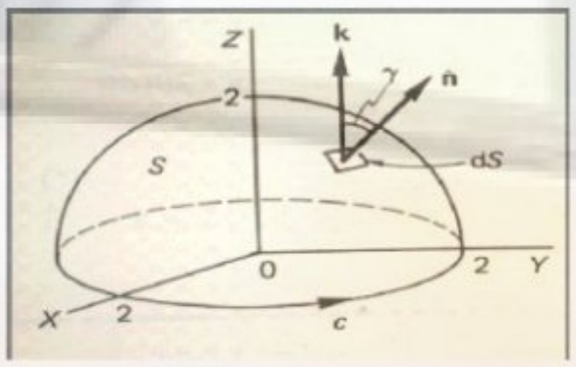


This means that we can express a surface integral in terms of a line integral round the boundary curve. The proof of this theorem is rather lengthy and is to be found in the Appendix. Let us demonstrate its application in the following examples.

Example

A hemisphere S is defined by $x^2 + y^2 + z^2 = 4$ $(z \geq 0)$. A vector field $\mathbf{F} = 2y\mathbf{i} - x\mathbf{j} + xz\mathbf{k}$ exists over the surface and around its boundary c. Verify Stoke's theorem that

$\int_S \text{curl}\,\mathbf{F}.dS = \oint_C \mathbf{F}.d\mathbf{r}$.



$S : x^2 + y^2 + z^2 - 4 = 0$

$\mathbf{F} = 2y\mathbf{i} - x\mathbf{j} + xz\mathbf{k}$ c is the circle $x^2 + y^2 = 4$.

(a) $\oint_C \mathbf{F}.d\mathbf{r} = \int_C (2y\mathbf{i} - x\mathbf{j} + xz\mathbf{k}) \cdot (\mathbf{i}dx + \mathbf{j}dy + \mathbf{k}dz)$

$= \int_C (2ydx - x\,dy + xz\,dz)$

Converting to polar coordintes. $x = 2\cos\theta$; $y = 2\sin\theta$; $z = 0$

$dx = -2\sin\theta\,d\theta$; $dy = 2\cos\theta\,d\theta$; Limits $\theta = 0$ to 2π

$$\oint_C \mathbf{F}.d\mathbf{r} = \int_0^{2/\pi} (4\sin\theta[(-2\sin\theta d\theta] - 2\cos\theta 2\cos\theta d\theta = -4\int_0^{2/\pi} (2\sin^2\theta + \cos^2\theta)\,d\theta$$

$$= -4\int_0^{2/\pi} (1 + \sin^2\theta)\,d\theta = -2\int_0^{2/\pi} (3 - \cos 2\theta)\,d\theta = -2\left[3\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi} = -12\,\pi \quad (1)$$

(b) Now we determine $\int_S \text{curl}\,\mathbf{F}.dS$

$\int \text{curl}\,\mathbf{F}.d\mathbf{S} = \int \text{curl}\,\mathbf{F}.\hat{\mathbf{n}}dS$

$\mathbf{F} = 2y\mathbf{i} - x\mathbf{j} + xz\mathbf{k}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & -x & xz \end{vmatrix} = \mathbf{i}(0-0)-\mathbf{j}(z-0)+\mathbf{k}(-1-2) = -z\mathbf{j} - 3\mathbf{k}$$

$$\text{Now } \hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|} = \frac{2x\mathbf{i}+2y\mathbf{j}+2z\mathbf{k}}{\sqrt{4x^2+4y^2+4z^2}} = \frac{x\mathbf{i}+y\mathbf{j}+z\mathbf{k}}{2}$$

$$\text{Then } \int_S \text{curl } \mathbf{F}.\hat{\mathbf{n}}dS = \int_S (-z\mathbf{k}-3\mathbf{k}).\left(\frac{x\mathbf{i}+y\mathbf{j}+z\mathbf{k}}{2}\right) dS = \frac{1}{2}\int_S (-yz-3z)\, dS$$

$$x = 2\sin\phi\cos\theta;\ y = 2\sin\phi\sin\theta;\ z = 2\cos\phi\ ,dS = 4\sin\phi\, d\phi\, d\theta$$

$$\therefore \int S \text{ curl}\mathbf{F}.\hat{\mathbf{n}}dS = \frac{1}{2}\iint_S (-2\sin\phi\sin\theta\, 2\cos\phi - 6\cos\phi)\, 4\sin\phi\, d\phi\, d\theta$$

$$= -\int_0^{2\pi}\int_0^{\pi/2} (2\sin^2\phi\cos\phi\sin\theta + 3\sin\phi\cos\phi)d\phi d\theta = -4\int_0^{2\pi}\left[\frac{2\sin^3\phi\sin\theta}{3} + \frac{3\sin^2\phi}{2}\right]_0^{\pi/2} d\theta$$

$$= -4\int_0^{2\pi}\left(\frac{2}{3}\sin\theta + \frac{3}{2}\right) d\theta = -12\pi \quad (2)$$

So we have from our two results (1) and (2)

$$\int_S \text{curl } \mathbf{F}.dS = \oint_C \mathbf{F}.d\mathbf{r}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

v.i.p

Lecture No -39 Periodic Functions

Periodic functions

A function f(x) is said to be periodic if its function values repeat at regular intervals of the independent variable. The regular interval between repetitions is the period of the oscillations. $f(x + p) = f(x)$

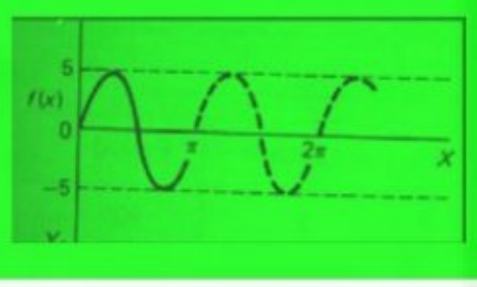
Graphs of y = A sin nx

(a) y = sin x The obvious example of a periodic function is y = sin x, which goes through its complete range of values while x increases from 0° to 360°.

The period is therefore 360° or 2π radians and the amplitude, the maximum displacement from the position of rest.

y = 5sin 2x

The amplitude is 5. The period is 180° and there are thus 2 complete cycles in 360°.



Example

Functions	Amplitude	Period
y = 3 sin 5x	3	72°
y = 2 cos 3x	2	120°
$y = \sin \frac{x}{2}$	1	720°
y = 4 sin 2x	4	180°

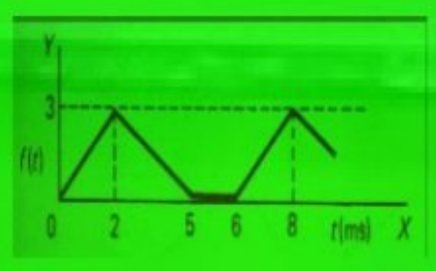
y = A sin nx

Thanking along the same lines, the function y = A sin nx has amplitude = A; period $= \frac{360°}{n} = \frac{2\pi}{n}$; n cycles in 360°.

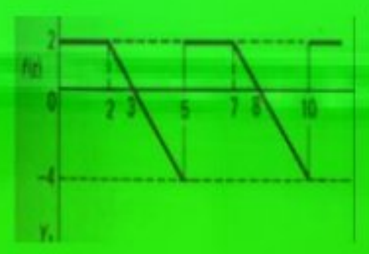
Graphs of y = A cos nx have the same characteristics

Example

period = 8 ms



period = 6 ms



period = 5 cm

Analytical description of a periodic function

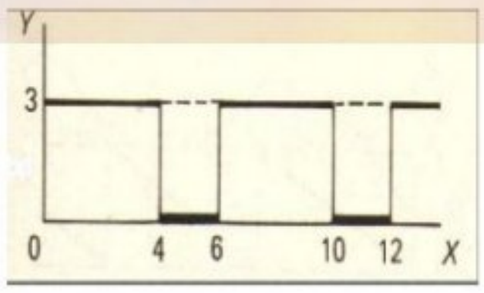
A periodic function can be defined analytically in many cases.

Example

(a) Between x = 0 and x = 4, y = 3,
i.e. $f(x) = 3 \quad 0 < x < 4$

(b) Between x = 4 and x = 6, y = 0,
i.e. $f(x) = 0 \quad 4 < x < 6$.

So we could define the function by
$f(x) = 3 \qquad 0 < x < 4$

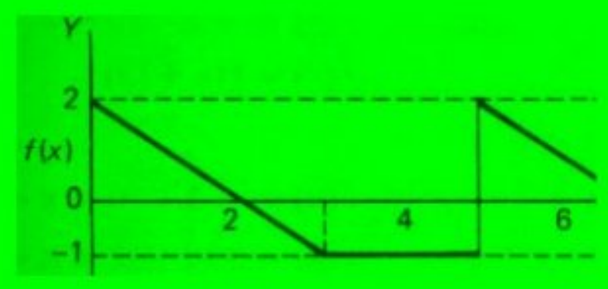


© Copyright Virtual University of Pakistan

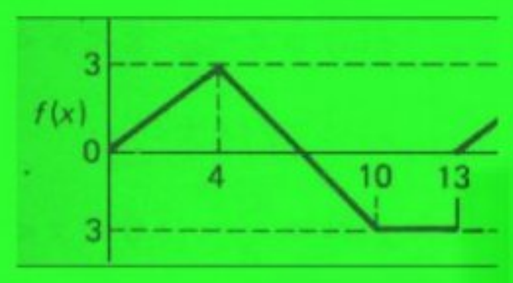
Scanned with CamScanner

Scanned with CamScanner

$f(x) = 0 \qquad 4 < x < 6$
$f(x) = f(x + 6)$
the last line indicating that the function is periodic with period 6 units

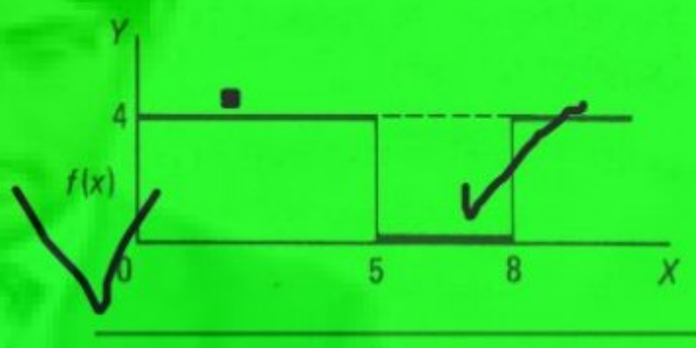


$f(x) = 2 - x \qquad 0 < x < 3$
$f(x) = -1 \qquad 3 < x < 5$
$f(x) = f(x+5)$
$f(x) = \frac{3x}{4} \qquad 0 < x < 4$
$f(x) = 7 - x \qquad 4 < x < 10$
$f(x) = -3 \qquad 10 < x < 13$
$f(x) = f(x + 13)$

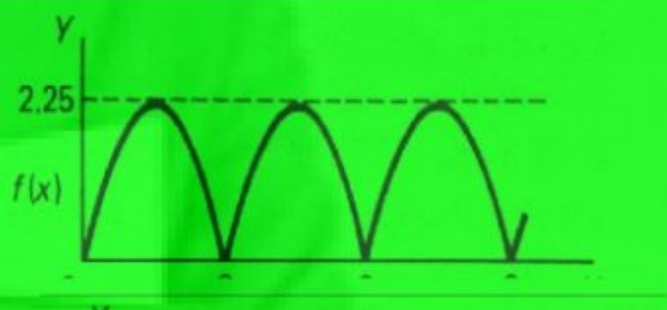


Example
Sketch the graphs of the following inserting relevant values.

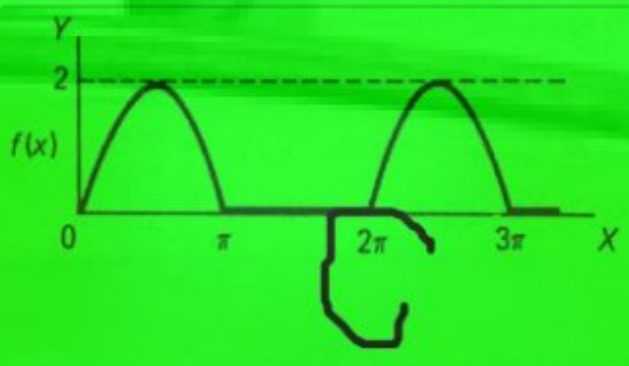
1. $f(x) = 4 \cdot \qquad 0 < x < 5$
   $f(x) = 0 \qquad 5 < x < 8$
   $f(x) = f(x + 8)$



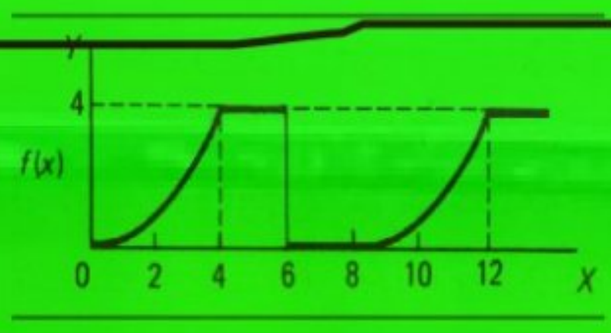
$f(x) = 3x - x^2 \qquad 0 < x < 3$
$f(x) = f(x + 3)$



$f(x) = 2 \sin x \qquad 0 < x < \pi$
$f(x) = 0 \qquad \pi < x < 2\pi$
$f(x) = f(x + 2\pi).$



$f(x) = \frac{x^2}{4} \qquad 0 < x < 4$
$f(x) = 4 \qquad 4 < x < 6$
$f(x) = 0 \qquad 6 < x < 8$
$f(x) = f(x + 8).$



Useful integrals

The following integrals appear frequently in our work on Fourier series, so it will help if we obtain the result in readiness. In each case, m and n are integers other than zero.

(a) $\int_{-\pi}^{\pi} \sin nx\, dx = \left[\frac{-\cos nx}{n}\right]_{-\pi}^{\pi} = \frac{1}{n}\{-\cos n\pi + \cos n\pi\} = 0$

(b) $\int_{-\pi}^{\pi} \cos nx\, dx = \left[\frac{\sin nx}{n}\right]_{-\pi}^{\pi} = \frac{1}{n}\{\sin n\pi + \sin n\pi\} = 0$

(c) $\int_{-\pi}^{\pi} \sin^2 nx\, dx = \frac{1}{2}\int_{-\pi}^{\pi} (1 - \cos 2nx)\, dx = \frac{1}{2}\left[x - \frac{\sin 2nx}{2n}\right]_{-\pi}^{\pi} = \pi \ (n \neq 0)$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(d) $\int_{-\pi}^{\pi} \cos^2 nx\, dx = \frac{1}{2}\int_{-\pi}^{\pi} (1 + \cos 2nx)\, dx = \frac{1}{2}\left[x + \frac{\sin 2nx}{2n}\right]_{-\pi}^{\pi} = \pi \quad (n \neq 0)$

(e) $\int_{-\pi}^{\pi} \sin nx \cos mx\, dx = \frac{1}{2}\int_{-\pi}^{\pi} \{\sin(n+m)x + \sin(n-m)x\}dx = \frac{1}{2}\{0 + 0\} = 0$

from result (a) with $n \neq m$

(f) $\int_{-\pi}^{\pi} \cos nx \cos mx\, dx = \frac{1}{2}\int_{-\pi}^{\pi} \{\cos(n+m)x + \cos(n-m)x\}dx = \frac{1}{2}\{0 + 0\} = 0$

from result (b) with $n \neq m$

Note:

If $n = m$, then

$\int_{-\pi}^{\pi} \cos nx \cos mx\, dx$ becomes $\int_{-\pi}^{\pi} \cos^2 nx\, dx = \pi$ from (d) above.

(g) $\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = -\frac{1}{2}\int_{-\pi}^{\pi} (-2) \sin nx \sin mx dx = -\frac{1}{2}\int_{-\pi}^{\pi} \cos(n+m)x - \cos(n-m)x dx$

$= \frac{1}{2}\int_{-\pi}^{\pi} \{0 - 0) = 0$ from result (b) with $n \neq m$

Note:

If $n = m$, then

$\int_{-\pi}^{\pi} \sin nx \sin mx\, dx$ becomes $\int_{-\pi}^{\pi} \sin^2 nx\, dx = \pi$ from (c) above

Summary of integrals

(a) $\int_{-\pi}^{\pi} \sin nx\, dx = 0$, (b) $\int_{-\pi}^{\pi} \cos nx\, dx = 0$, (c) $\int_{-\pi}^{\pi} \sin^2 nx\, dx = \pi\ (\neq 0)$

(d) $\int_{-\pi}^{\pi} \cos^2 nx\, dx = \pi\ (n \neq 0)$ (e) $\int_{-\pi}^{\pi} \sin nx \cos mx\, dx = 0$

(f) $\int_{-\pi}^{\pi} \cos nx \cos mx\, dx = 0 \quad (n \neq m)$ (g) $\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = 0 \quad (n \neq m)$

$\int_{-\pi}^{\pi} \cos nx \cos mx\, dx = \pi \quad (n = m)$ $\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = \pi \quad (n = m)$

Note

We have evaluated the integrals between $-\pi$ and π, but, provided integration is carried out over a complete periodic interval of 2π, the results are the same. Thus, the limits could just as well be $-\pi$ to π, 0 to 2π, $-\pi/2$ to $3\pi/2$, etc. We can therefore choose the limits to suit the particular problems.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Fourier series

Periodic functions of period 2π

The basis of a Fourier series is to represent a periodic function by a trigonometrical series of the form.

$f(x) = A_0+c_1 \sin(x+\alpha_1) + c_2 \sin(2x+\alpha_2) + c_3 \sin(3x+\alpha_3)+\ldots+c_n \sin(nx+\alpha_n)+ \ldots$

where A_0 is a constant term

$c_1, c_2, c_3 \ldots c_n$ denote the amplitudes of the compound sine terms $\alpha_1, \alpha_2, \alpha_3 \ldots..$ are constant auxiliary angles.

Each sine term, $c_n \sin(nx + \alpha_n)$ can be expanded thus:

$c_n \sin(nx + \alpha_n) = c_n\{\sin nx \cos\alpha_n+\cos n\,x \sin\alpha_n\}= (c_n\sin\alpha_n) \cos nx + (c_n\cos\alpha_n) \sin nx$

$= a_n\cos nx + b_n\sin nx$

the whole series becomes.

$$\mathbf{f(x) = A_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No- 40 Fourier Series

As we know that $A_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$;

which can be written as in the expanded from

$A_0+(a_1\cos x+b_1\sin x)+(a_2\cos 2x+b_2\sin 2x)+ \ldots + (a_n\cos nx+b_n\sin nx)+\ldots\ldots$

$A_0+a_1\cos x+a_2\cos 2x+.+a_n\cos nx+.. .. +b_1\sin x+b_2\sin 2x+\ldots b_n\sin nx+\ldots\ldots$

$f(x) = A_0+a_1\cos x+a_2\cos 2x+.+a_n\cos nx+.. \ldots+b_1\sin x+b_2\sin 2x+\ldots b_n\sin nx+\ldots$

Fourier coefficients

We have defined Fourier series in the form

$f(x) = A_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$; n a positive integer

(a) To find A_0, we integrate f(x) with respect to x from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)dx = \int_{-\pi}^{\pi} A_0\,dx + \sum_{n=1}^{\infty} \left\{ \int_{-\pi}^{\pi} a_n \cos nx\,dx + \int_{-\pi}^{\pi} b_n \sin nx\,dx \right\} = [A_0 x]_{-\pi}^{\pi} + \sum_{n=1}^{\infty} \{0 + 0\} = 2A_0\pi$$

$$2A_0\,\pi = \int_{-\pi}^{\pi} f(x)\,dx$$

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\,dx = 1/2\, a_0\,; \qquad \text{Where } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx$$

(b) To find a_n we multiply f(x) by cos mx and integrate from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)\cos mx\,dx = \int_{-\pi}^{\pi} A_0 \cos mx\,dx + \sum_{n=1}^{\infty} \left\{ \int_{-\pi}^{\pi} a_n \cos nx \cos mx\,dx + \int_{-\pi}^{\pi} b_n \sin nx \cos mx\,dx \right\}$$

$$\int_{-\pi}^{\pi} f(x)\cos mx\,dx = A_0\{0\} + \sum_{n=1}^{\infty} \{a_n(0) + b_n(0)\} = 0 \qquad \text{for } n \neq m$$

$$= 0 + a_n\pi + 0 = a_n\pi \qquad \text{for } n = m$$

$$\therefore\ a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\,dx$$

(c) To find b_n we multiply f(x) by sinmx and integrate from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx = \int_{-\pi}^{\pi} A_0 \sin mx\,dx + \sum_{n=1}^{\infty} \left\{ \int_{-\pi}^{\pi} a_n \cos nx \sin mx\,dx + \int_{-\pi}^{\pi} b_n \sin nx \sin mx\,dx \right\}$$

$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx = A_0\{0\} + \sum_{n=1}^{\infty} \{a_n(0) + b_n(0)\} = 0 \qquad \text{for } n \neq m$$

$$= 0 + 0 + b_n\pi = b_n\pi \qquad \text{for } n = m$$

$$\therefore\ b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\,dx$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Result For Fourier Series

$$f(x)=\frac{1}{2}a_0+\sum_{n=1}^{\infty}\{a_n\cos nx+b_n\sin nx\};$$

(a) $a_0=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = 2\times$mean value of f(x) over a period

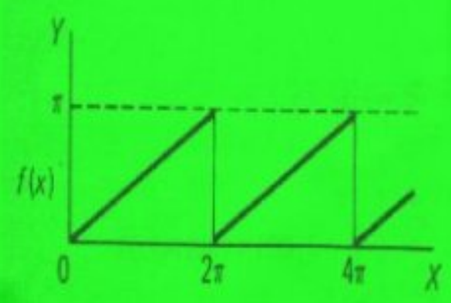
(b) $a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = 2\times$mean value of f(x) cosnx over a period.

(c) $b_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = 2\times$ mean value of f(x) sin nx over a period.

In each case, n = 1, 2, 3, ……

Example

Determine the Fourier series to represent the periodic function shown.



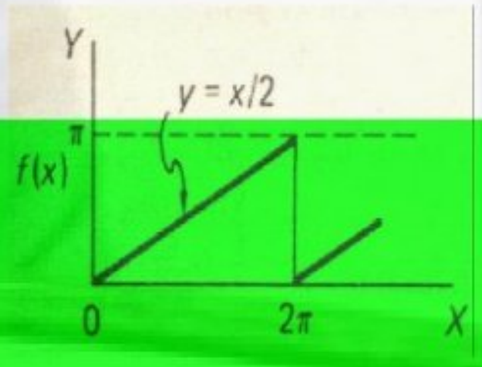
It is more convenient here to take the limits as 0 to 2π.

The function can be defined as

$f(x)=\frac{x}{2}$ $\qquad 0<x<2\pi$

$f(x)=f(x+2\pi)$ $\qquad$ period $=2\pi$.

Now to find the coefficients



(a) $a_0=\frac{1}{\pi}\int_0^{2\pi} f(x)\,dx=\frac{1}{\pi}\int_0^{2\pi}\left(\frac{x}{2}\right)dx=\frac{1}{4\pi}[x^2]_0^{2\pi}=\pi$

$a_0=\pi$

(b) $a_n=\frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx$

$=\frac{1}{\pi}\int_0^{2\pi}\left(\frac{x}{2}\right)\cos nx\,dx$

$$a_n=\frac{1}{2\pi}\int_0^{2\pi} x\cos nx\,dx=\frac{1}{2\pi}\left\{\left[\frac{x\sin nx}{n}\right]_0^{2\pi}-\frac{1}{n}\int_0^{2\pi}\sin nx\,dx\right\}=\frac{1}{2\pi}\left\{(0-0)-\frac{1}{n}(0)\right\}$$

$\therefore a_n=0$

(a) $b_n=\frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$ So we now have

$$b_n=\frac{1}{\pi}\int_0^{2\pi}\frac{x}{2}\sin nx\,dx=\frac{1}{2\pi}\left\{\left[-\frac{x\cos nx}{n}\right]_0^{2\pi}+\frac{1}{n}\int_0^{2\pi}\cos nx\,dx\right\}=-\frac{1}{2\pi n}[2\pi-0]=-\frac{1}{n}$$

$$\mathbf{a_0=\pi;\qquad a_n=0;\qquad b_n=-\frac{1}{n}}$$

Now the general expression for a Fourier series is

$f(x)=\frac{1}{2}a_0+\sum_{n=1}^{\infty}\{a_n\cos nx+b_n\sin nx\}$ Therefore in this case

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \{b_n \sin nx\} = \frac{\pi}{2} + \{-\frac{1}{1}\sin x - \frac{1}{2}\sin 2x - \frac{1}{3}\sin 3x - ...\} \quad \text{since } a_n = 0$$

$$f(x) = \frac{\pi}{2} - \{\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + ...\}$$

Dirichlet Conditions

If the Fourier series is to represent a function f(x), then putting $x = x_1$ will give an infinite series in x_1 and the value of this should converge to the value of $f(x_1)$ as more and more terms of the series are evaluated. For this to happen, the following conditions must be fulfilled.

(a) The function $f(x)$ must be defined and single-valued.

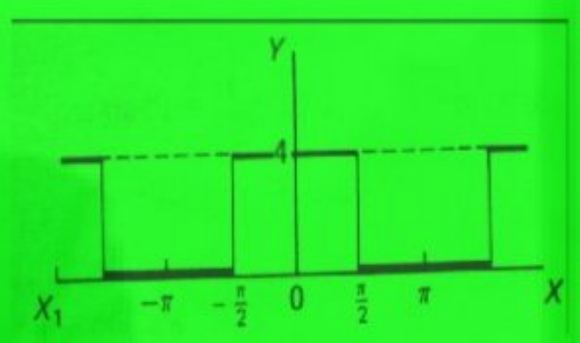
(b) $f(x)$ must be continuous or have a finite number of finite discontinuities within a periodic interval.

(c) $f(x)$ and $f'(x)$ must be piecewise continuous in the periodic interval.

If these Dirichlet conditions are satisfied, the Fourier series converges to $f(x_1)$, if $x = x_1$ is a point of continuity

Example

Find the Fourier series for the function shown.



Consider one cycle between x=0 and x=π.

The function can be defined by

$$f(x) = 0 \qquad -\pi < x < -\frac{\pi}{2}$$

$$f(x) = 4 \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$f(x) = 0 \qquad \frac{\pi}{2} < x < \pi$$

$$f(x) = f(x + 2\pi)$$

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$

The expression for a_0 is $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$ This gives

$$a_0 = \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} 0\,dx + \int_{-\pi/2}^{\pi/2} 4dx + \int_{\pi/2}^{\pi} 0\,dx\right\} = \frac{1}{\pi}[4x]_{-\pi/2}^{\pi/2} = 4$$

$$a_0 = 4$$

$$\text{(b)}\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} (0)\cos nx dx + \int_{-\pi/2}^{\pi/2} 4\cos nx dx + \int_{\pi/2}^{\pi} (0)\cos nx dx\right\} = \frac{1}{\pi}\left\{\int_{-\pi/2}^{\pi/2} 4\cos nx dx\right\}$$

$$= \frac{4}{\pi n}\sin nx\Big|_{-\pi/2}^{\pi/2} = \frac{8}{\pi n}\sin\frac{n\pi}{2}$$

$$a_n = \frac{8}{\pi n}\sin\frac{n\pi}{2}$$

Then considering different integer values of n, we have

If n is even $\qquad a_n = 0$

If n = 1.5, 9.... $\qquad a_n = \frac{8}{n\pi}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

If n = 3, 7, 11, …. $a_n = -\frac{8}{n\pi}$

(c) To find b_n

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2}(0)\sin nx dx + \int_{-\pi/2}^{\pi/2} 4\sin nx dx + \int_{\pi/2}^{\pi}(0)\sin nx dx\right\}$$

$$b_n = \frac{4}{\pi}\int_{-\pi/2}^{\pi/2}\sin nx\,dx = \frac{4}{\pi}\left[\frac{-\cos nx}{n}\right]_{-\pi/2}^{\pi/2} = -\frac{4}{n\pi}\left\{\cos\frac{n\pi}{2} - \cos\left(\frac{-n\pi}{2}\right)\right\} = 0$$

$b_n = 0$

So with $a_0 = 4$: an as stated above; $b_n = 0$;

The Fourier series is

$$f(x) = 2 + \frac{8}{\pi}\left\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{5}\cos 5x - \frac{1}{7}\cos 7x + \ldots..\right\}$$

In this particular example, there are, in fact, no sine terms.

Effect Of Harmonics

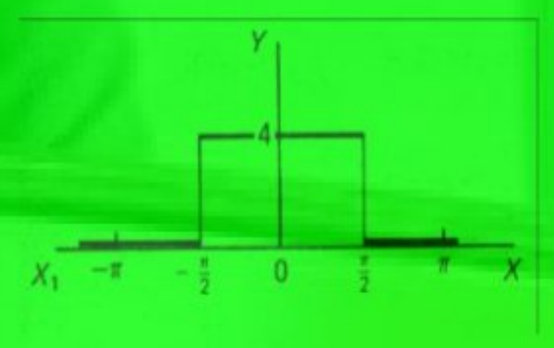
It is interesting to see just how accurately the Fourier series represents the function with which it is associated. The complete representation requires an infinite number of terms, but we can, at least, see the effect of including the first few terms of the series.
Let us consider the waveform shown. We established earlier that the function

$f(x) = 0 \qquad -\pi < x < -\frac{\pi}{2}$

$f(x) = 4 \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$

$f(x) = 0 \qquad \frac{\pi}{2} < x < \pi$

$f(x) = f(x + 2\pi)$

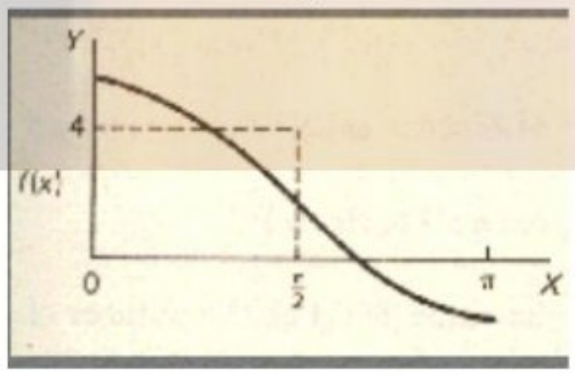


gives the Fourier series

$$f(x) = 2 + \frac{8}{\pi}\left\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x - \frac{1}{7}\cos 7x + \ldots.\right\}$$

If we start with just one cosine term, we can then see the effect o including subsequent harmonics. Let us restrict our attention to just the right-hand half of the symmetrical waveform. Detailed plotting of points gives the following development.

(1) $f(x) = 2 + \frac{8}{\pi}\cos x$

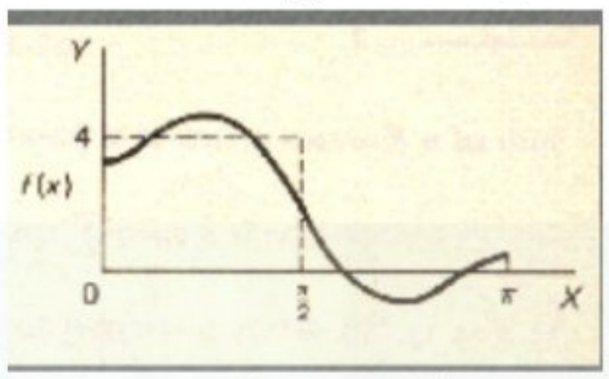


© Copyright Virtual University of Pakistan

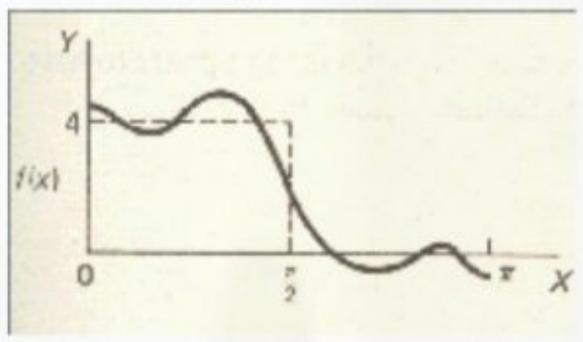
Scanned with CamScanner

Scanned with CamScanner

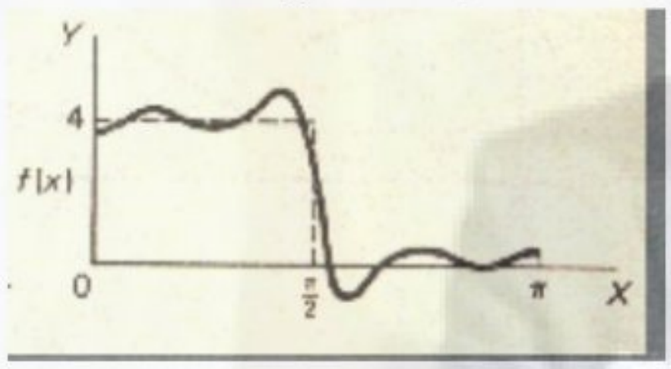
(2) $f(x) = 2 + \frac{8}{\pi} \{\cos x - \frac{1}{3}\cos 3x\}$



(3) $f(x)=2+\frac{8}{\pi} \{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x)$



(4) $f(x) = 2 + \frac{8}{\pi} \{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x - \frac{1}{7}\cos 7x\}$



As the number of terms is increased, the graph gradually approaches the shape of the original square waveform. The ripples increase in number and decrease in amplitude, but a perfectly square waveform is unattainable in practice. For practical purpose, the first few terms normally suffice to give an accuracy of acceptable level.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -41 Examples

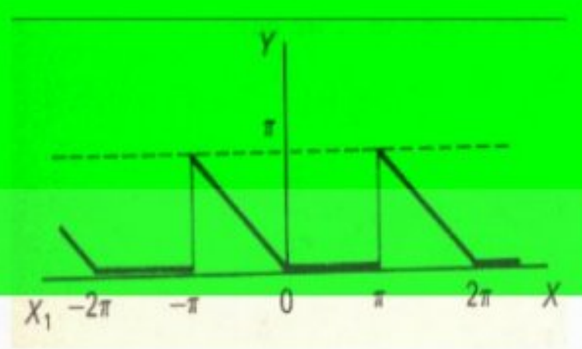
Example

Find the Fourier series for the function defined by

$f(x) = -x \qquad -\pi < x < 0$

$f(x) = 0 \qquad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



The general expressions for a_0, a_n, b_n are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\,dx + \frac{1}{\pi}\int_{0}^{\pi} 0\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\,dx = \frac{1}{\pi}\left[-\frac{x^2}{2}\right]_{-\pi}^{0} = \frac{\pi}{2}$$

(b) To find a_n

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\cos nx\,dx + \frac{1}{\pi}\int_{0}^{\pi} 0\,dx = -\frac{1}{\pi}\int_{-\pi}^{\pi} x\cos nx\,dx$$

$$= -\frac{1}{\pi}\left\{\left[x\frac{\sin nx}{n}\right]_{-\pi}^{0} - \frac{1}{n}\int_{-\pi}^{0}\sin nx\,dx\right\} = -\frac{1}{\pi}\left\{(0-0) - \frac{1}{n}\left[\frac{-\cos nx}{n}\right]_{-\pi}^{0}\right\}$$

$$= -\frac{1}{\pi}\left\{\frac{1}{n}\left[\frac{\cos nx}{n}\right]_{-\pi}^{0}\right\} = -\frac{1}{\pi n^2}\left\{\left[\cos nx\right]_{-\pi}^{0}\right\} = -\frac{1}{\pi n^2}\left[\cos 0 - \cos n\pi\right]$$

$$= -\frac{1}{\pi n^2}\left\{1 - \cos n\pi\right\}$$

But $\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$a_n = -\frac{2}{\pi n^2}$ (n odd) and $a_n = 0$ (n even)

(c) Now to find b_n

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\sin nx\,dx = -\frac{1}{\pi}\int_{-\pi}^{0} x\sin nx\,dx$$

$$= -\frac{1}{\pi}\left\{\left[x\left(\frac{-\cos nx}{n}\right)\right]_{-\pi}^{0} + \frac{1}{n}\int_{-\pi}^{0}\cos nx\,dx\right\} = -\frac{1}{\pi}\left\{\frac{\pi\cos n\pi}{n} + \frac{1}{n}\left[\frac{\sin nx}{n}\right]_{-\pi}^{0}\right\} = -\frac{\cos n\pi}{n}$$

$b_n = -\frac{1}{n}$ (n even) and $b_n = \frac{1}{n}$ (n odd)

So we have

$a_0 = \frac{\pi}{2}$; $a_n = 0$ (n even) and $a_n = -\frac{2}{\pi n^2}$ (n odd)

$b_n = -\frac{1}{n}$ (n even) and $b_n = \frac{1}{n}$ (n odd)

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left(\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \ldots\right) + \left(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \frac{1}{4}\sin 4x + \ldots.\right)$$

It is just a case of substituting n = 1, 2, 3, etc.

In this particular example, we have a constant term and both sine and cosine terms.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Odd And Even Functions

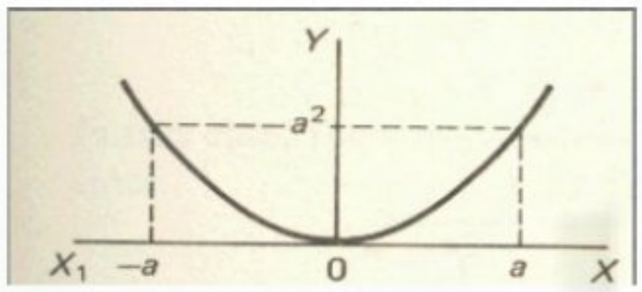
(a) Even functions

A function f(x) is said to be even if $f(-x) = f(x)$ i.e. the function value for a particular negative value of x is the same as that for the corresponding positive value of x. The graph of an even function is therefore symmetrical about the y-axis.

$y = f(x) = x^2$ is an even function since

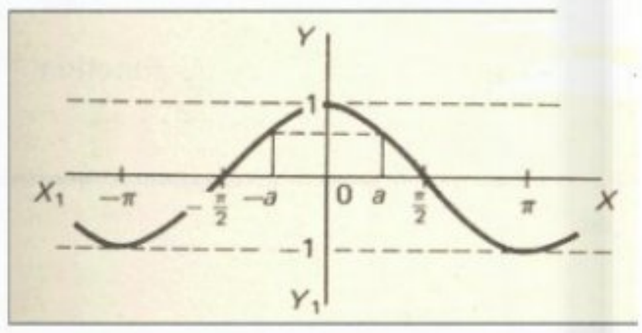
$f(-2) = 4 = f(2)$

$f(-3) = 9 = f(3)$ etc.



$y = f(x) = \cos x$ is an even function since $\cos(-x) = \cos x$

$f(-a) = \cos a = f(a)$



(b) Odd functions

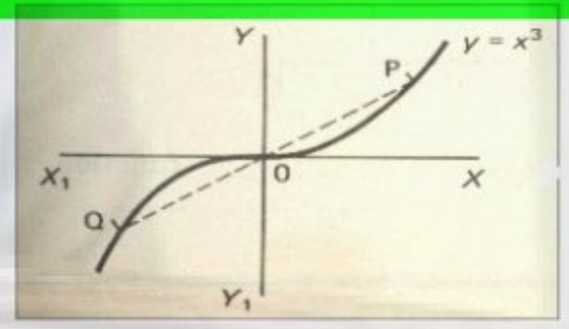
A function f(x) is said to be odd if $f(-x) = -f(x)$

i.e. the function value for a particular negative value of xis numerically equal to that for the corresponding positive value of x but opposite in sign. The graph of an odd function is thus symmetrical about the origin.

$y = f(x) = x^3$ is an odd function since

$f(-2) = -8 = -f(2)$

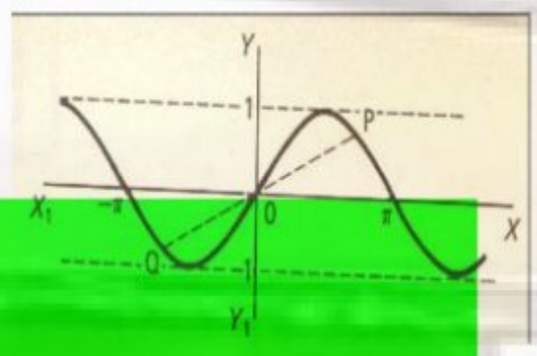
$f(-5) = -125 = -f(5)$ etc.



$y = f(x) = \sin x$ is an odd function

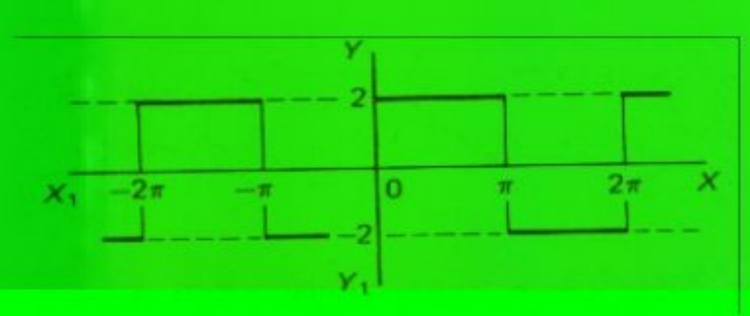
Since $\sin(-x) = -\sin x$

$f(-a) = -f(a)$.

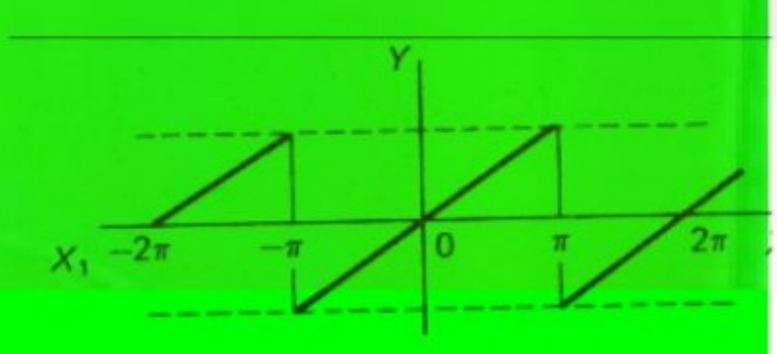


So, for an even function $f(-x) = f(x)$ symmetrical about the y-axis

for an odd function $f(-x) = -f(x)$ symmetrical about the origin.



odd.

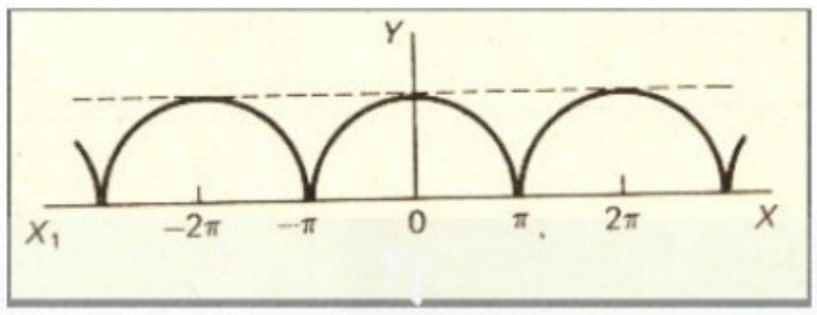


Odd

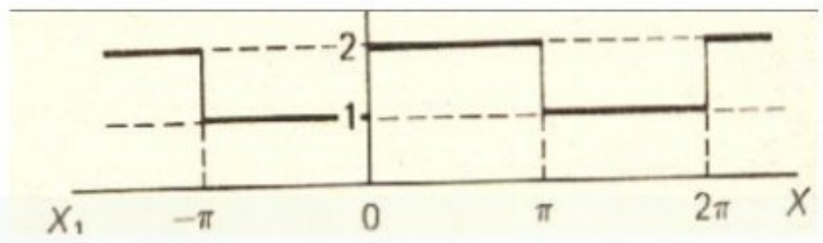
© Copyright Virtual University of Pakistan

Scanned with CamScanner

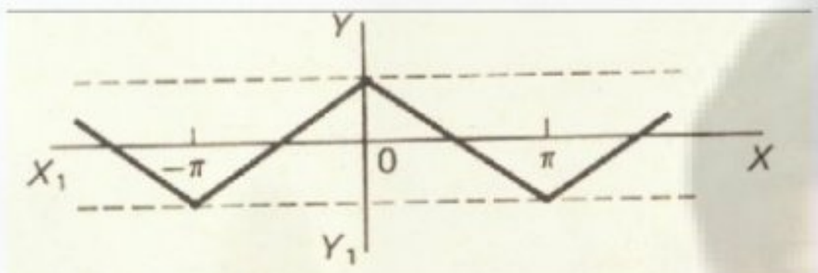
Scanned with CamScanner



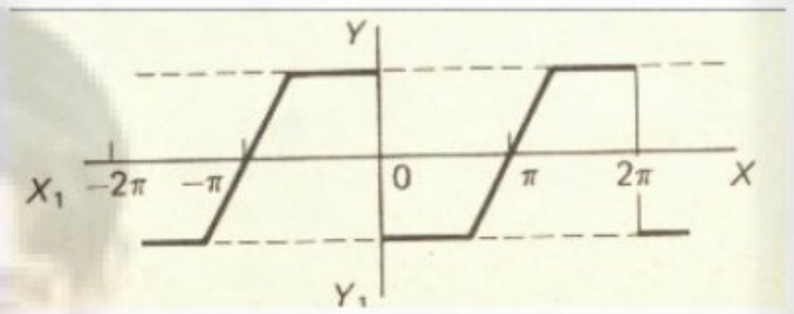
Even



neither



Even



Odd

Products Of Odd And Even Functions

The rules closely resemble the elementary rules of signs.

(even)×(even)=(even) like (+)×(+)=(+) ; (odd)× (odd) = (even) (−)×(−)=(+) ;

(odd)× (even) = (odd) (−)×(+)=(−)

The results can easily be proved.

(a) Two even functions

Let $F(x) = f(x)\, g(x)$ where $f(x)$ and $g(x)$ are even functions.

Then $F(-x) = f(-x)\, g(-x) = f(x)\, g(x)$ since $f(x)$ and $g(x)$ are even.

$\therefore F(-x) = F(x)$

$F(x)$ is even

(b) Two odd functions

Let $F(x) = f(x)\, g(x)$ where $f(x)$ and $g(x)$ are odd functions.

Then $F(-x) = f(-x)\, g(-x) = \{-f(x)\}\, \{-g(x)\}$

since $f(x)$ and $g(x)$ are odd.

$= f(x)\, g(x) = F(x)$

$\therefore F(-x) = F(x)$

$F(x)$ is even

Finally

(c) One odd and one even function

Let $F(x) = f(x)\, g(x)$ where $f(x)$ is odd and $g(x)$ even.

Then $F(-x) = f(-x)\, g(-x) = -f(x)\, g(x) = -F(x)$

$\therefore F(-x) = -F(x)$

$F(x)$ is odd

So if f(x) and g(x) are both even, then f(x) g(x) is even and if f(x) and g(x) are both odd, then f(x) g(x) is even but if either f(x) or g(x) is even and the other odd. Then f(x) g(x) is odd.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

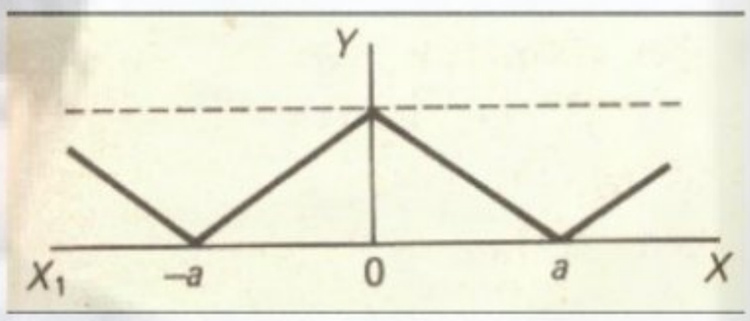
Example
State whether each of the following products is odd, even, or neither.

1. $x^2 \sin 2x$	odd	(E) (O) = (O)
2. $x^3 \cos x$	odd	(O) (E) = (O)
3. $\cos 2x \cos 3x$	even	(E) (E) = (E)
4. $x \sin nx$	even	(O) (O) = (E)
5. $3 \sin x \cos 4x$	odd	(O) (E) = (O)
6. $(2x + 3) \sin 4x$	neither	(N) (O) = (N)
7. $\sin^2 x \cos 3x$	even	(E) (E) = (E)
8. $x^3 e^x$	neither	(O) (N) = (N)
9. $(x^4 + 4) \sin 2x$	odd	(E) (O) = (O)

Two useful facts emerge from odd and even functions.

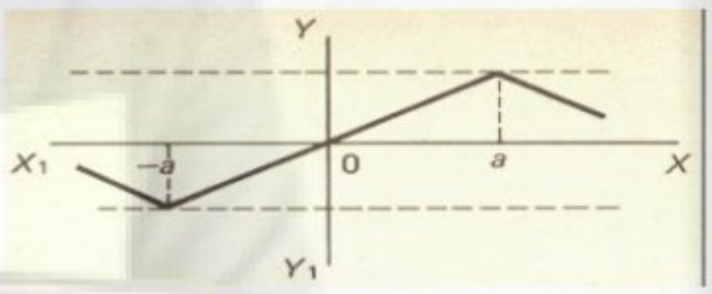
(a) For an even function

$$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)\,dx$$



(b) For an odd function

$$\int_{-a}^{a} f(x)\,dx = 0$$



THEOREM 1

If f(x) is defined over the interval $-\pi < x < \pi$ and f(x) is even, then the Fourier series for f(x) contains cosine terms only. Included in this is a_0 which maybe regarded as $a_n \cos nx$ with $n = 0$.

Proof:

(a) $$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)dx \qquad \therefore \quad a_0 = \frac{2}{\pi}\int_{0}^{\pi} f(x)\,dx$$

(b) $$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx.$$

But f(x) cos nx is the product of two even functions and therefore itself even.

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx. \qquad \therefore\ a_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$$

(c) $$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

Since f(x) sin nx is the product of an even function and an odd function, it is itself odd.

$$\therefore\ b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = 0.$$

$$\therefore\ b_n = 0$$

Therefore, there are no sine terms in the Fourier series for f(x).

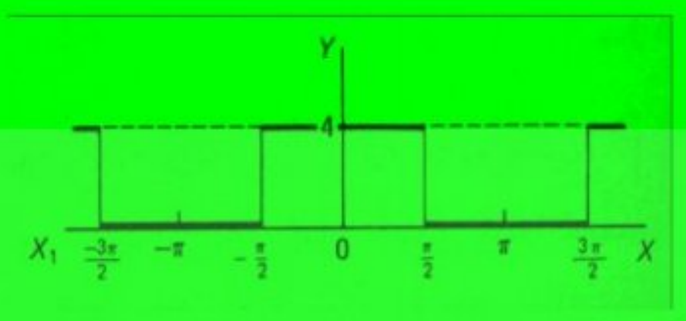
© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example

The waveform shown is symmetrical about the y-axis. The function is therefore even and there will be no sine terms in the series.



$$\therefore \quad f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$(a)\ a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\,dx. = \frac{2}{\pi}\int_{0}^{\pi/2} 4\,dx$$

$$= \frac{2}{\pi}[4x]_0^{\pi/2} = 4$$

$$(b)\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi/2} 4\cos nx\,dx = \frac{8}{\pi}\left[\frac{\sin nx}{n}\right]_0^{\pi/2}$$

$$= \frac{8}{\pi n}\sin\frac{n\pi}{2}$$

But $\sin\frac{n\pi}{2} = 0$ for n even

$= 1$ for n 1, 5, 9, ...

$= -1$ for n = 3, 7, 11, ...

$a_n = 0$ (n even);

$a_n = \frac{8}{\pi n}$ (n = 1, 5, 9, ...);

$a_n = -\frac{8}{\pi n}$ (n = 3, 7, 11 .....)

(c) $b_n = 0$, since f(x) is an even function. Therefore, the required series is

$$f(x) = 2 + \frac{\pi}{8}\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x - \frac{1}{7}\cos 7x + \ldots.\}$$

Theorem 2:

If f(x) is an odd function defined over the interval $-\pi < x < \pi$, then the Fourier series for f(x) contains sine terms only.

Proof:

Since f(x) is an odd function

$$\int_{-\pi}^{0} f(x)\,dx = -\int_{0}^{\pi} f(x)\,dx.$$

$$(a)\ a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx.$$

But f(x) is odd

$$\therefore \quad a_0 = 0$$

$$(b)\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

Remembering that f(x) is odd and cosnx is even, the product f(x) cosnx is odd.

$$\therefore\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} (\text{odd function})\,dx$$

$$\therefore \quad a_n = 0$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(b) $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx$

and since f(x) and sin nx are each odd, the product f(x) sin nx is even.

Then $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}$ (even function) $dx = \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx\, dx$

$\therefore\ b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx\, dx$

So,

If f(x) is odd function then $a_0 = 0;\ a_n = 0;$

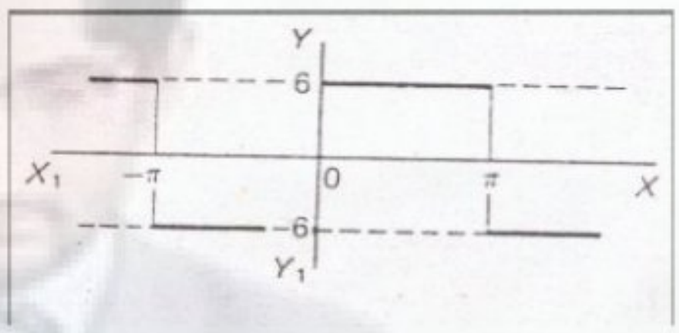
$b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx\, dx$ i.e. the Fourier series contains sine terms only.

Example

$f(x) = -6 \qquad -\pi < x < 0$

$f(x) = 6 \qquad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



We can see that this is an odd function;

$a_0 = 0$ and $a_n = 0$

$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx.$

f(x) sin nx is a product of two odd functions and is therefore even.

$\therefore\ b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx\, dx$

$b_n = \frac{2}{\pi}\int_{0}^{\pi} 6 \sin nx\, dx = \frac{12}{\pi}\left[\frac{-\cos nx}{n}\right]_0^{\pi} = \frac{12}{\pi}\left[\frac{\cos nx}{n}\right]_{\pi}^{0} = \frac{12}{\pi}\ [\cos 0 - \cos n\pi]$

$= \frac{12}{\pi n}\ (1 - \cos n\pi).$

$b_n = 0$ (n even) $\qquad b_n = \frac{24}{\pi n}$ (n odd)

So the series is $f(x) = \frac{24}{\pi}\{\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \ldots.\}$

Of course,

if f(x) is neither an odd nor an even function, then we must obtain expressions for a_0, a_n and b_n in full.

© Copyright Virtual University of Pakistan

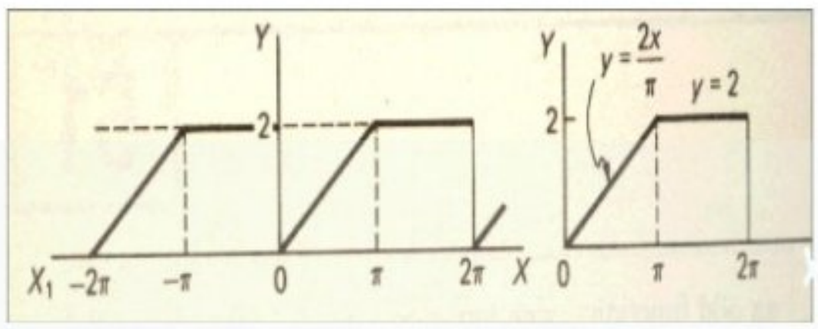
Scanned with CamScanner

Scanned with CamScanner

Lecture No -42 Examples

Example

Determine the Fourier series for the function shown.



This is neither odd nor even.

Therefore we must find a_0, a_n and b_n.

$$f(x) = \frac{1}{2}\, a_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$

$$f(x) = \frac{2}{\pi} x \qquad 0 < x < \pi$$

$$= 2, \qquad \pi < x < 2\pi$$

$$(a)\ a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)dx = \frac{1}{\pi}\left\{\int_0^{\pi} \frac{2}{\pi} x\, dx + \int_{\pi}^{2\pi} 2\, dx\right\} = \frac{1}{\pi}\left\{\left[\frac{x^2}{\pi}\right]_0^{\pi} + \left[2x\right]_{\pi}^{2\pi}\right\} = \frac{1}{\pi}\ \{\pi + 4\pi - 2\pi\}$$

$$\therefore\ a_0 = 3$$

$$(b)\ a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\, dx = \frac{1}{\pi}\left\{\int_0^{\pi}\left(\frac{2}{\pi}x\right)\cos nx\, dx + \int_{\pi}^{2\pi} 2\cos nx\, dx\right\}$$

$$= \frac{1}{\pi}\left\{\frac{2}{\pi}\int_0^{\pi} x\cos nx\, dx + 2\int_{\pi}^{2\pi} \cos nx dx\right\} = \frac{2}{\pi}\left\{\frac{1}{\pi}\int_0^{\pi} x\cos nx\, dx + 2\int_{\pi}^{2\pi} \cos nx dx\right\}$$

$$= \frac{2}{\pi}\left[\left\{\frac{1}{\pi}\left[\frac{x\sin nx}{n}\right]_0^{\pi} - \frac{1}{\pi n}\int_0^{2\pi} \sin nx dx\right\} + \int_{\pi}^{2\pi} \cos nx dx\right] = \frac{2}{\pi}\left\{0 - \frac{1}{n\pi}\left[-\frac{\cos nx}{n}\right]_0^{\pi} + \left[\frac{\sin nx}{n}\right]_{\pi}^{2\pi}\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{n\pi}\left[\frac{\cos nx}{n}\right]_0^{\pi}\right\} = \frac{2}{\pi^2 n^2}\ \{\cos n\pi - \cos 0\} = \frac{2}{\pi^2 n^2}\ \{\cos n\pi - 1\}$$

$a_n = 0$ (n even); $a_n = \frac{-4}{\pi^2 n^2}$ (n odd)

(c) To find b_n, we proceed in the same general manner

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\, dx = \frac{1}{\pi}\left\{\int_0^{\pi}\left(\frac{2}{\pi}x\right)\sin nx\, dx + \int_{\pi}^{2\pi} 2\sin nx\, dx\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{\pi}\left[\frac{-x\cos nx}{n}\right]_0^{\pi} + \frac{1}{\pi n}\int_0^{\pi} \cos nx dx + \int_{\pi}^{2\pi} \sin nx dx\right\} = \frac{2}{\pi}\left\{\frac{1}{\pi n}(-\pi\cos n\pi) + \frac{1}{\pi n}\left[\frac{\sin nx}{n}\right]_0^{\pi} + \left[\frac{-\cos nx}{n}\right]_{\pi}^{2\pi}\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{1}{n}\cos n\pi + (0-0) - \frac{1}{n}(\cos 2\pi n - \cos n\pi)\right\} = \frac{2}{\pi}\left\{-\frac{1}{n}\cos n\pi + (0-0) - \frac{1}{n}\cos 2\pi n + \frac{1}{n}\cos n\pi\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{1}{n}\cos 2n\pi\right\} = -\frac{2}{\pi n}\ \cos 2n\pi$$

But $\cos 2n\pi = 1$.

$$\therefore\ b_n = -\frac{2}{\pi n}$$

So the required series is

$$f(x) = \frac{3}{2} - \frac{4}{\pi^2}\left\{\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \ldots\right\} - \frac{2}{\pi}\left\{\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + \frac{1}{4}\sin 4x \ldots\right\}$$

Sum of a Fourier series at a point of discontinuity

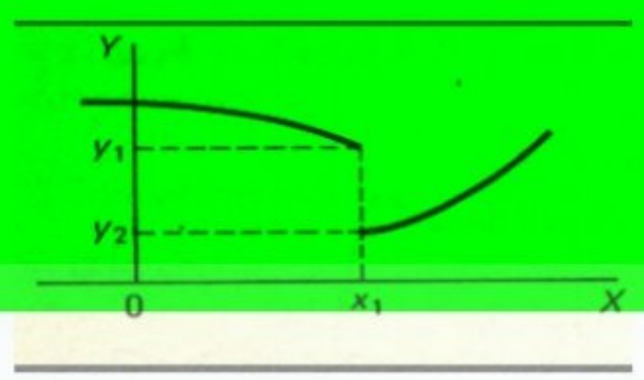
© Copyright Virtual University of Pakistan

Scanned with CamScanner

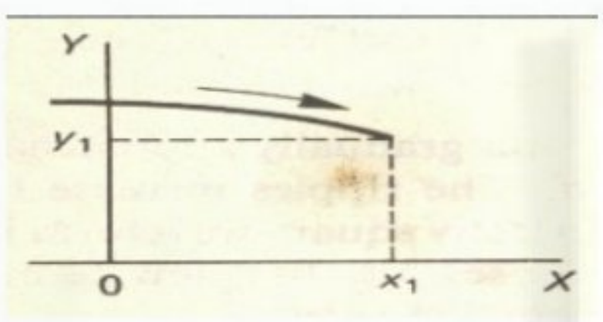
Scanned with CamScanner

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$

At $x = x_1$, the series converges to the value $f(x_1)$ as the number of terms including increases to infinity.

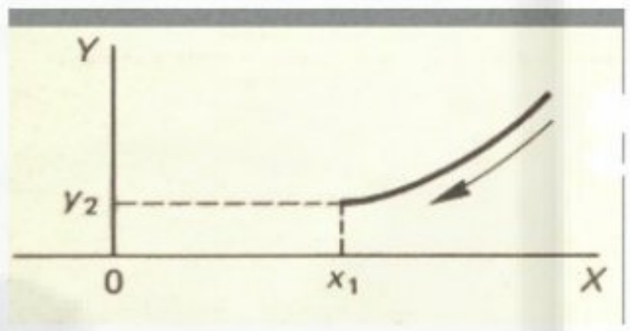


A particular point of interest occurs at a point of finite discontinuity or 'jump' of the function $y = f(x)$.
At $x = x_1$, the function appears to have two distinct values, y_1 and y_2.



If we approach $x = x_1$ from below that value, the limiting value of $f(x)$ is y_1.

If we approach $x = x_1$ from above that value, the limiting value of $f(x)$ is y_2.



To distinguish between these two values we write
$y_1 = f(x_1 - 0)$ denoting immediately before $x = x_1$
$y_2 = f(x_1 + 0)$
denoting immediately after $x = x_1$.
In fact, if we substitute $x = x_1$ in the Fourier series for $f(x)$, it can be shown that the series converges to the value

$\frac{1}{2} \{f(x_1 - 0) + f(x_1 + 0)\} = \frac{1}{2} (y_1 + y_2)$, the average of y_1 and y_2.

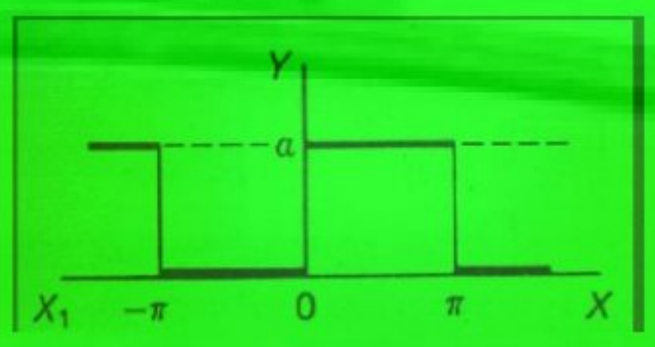
Example

Consider the function

$f(x) = 0 \qquad -\pi < x < 0$

$f(x) = a \qquad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



$$(a)\ a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi} \int_{0}^{\pi} a\,dx$$

$$= \frac{1}{\pi} [ax]_0^{\pi} = a$$

$$\therefore \quad a_0 = a$$

$$(b)\ a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\,dx = \frac{1}{\pi} \int_{0}^{\pi} a \cos nx\,dx = \frac{a}{\pi} \left[\frac{\sin nx}{n}\right]_0^{\pi} = 0$$

$$\therefore \quad a_n = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\,dx = \frac{1}{\pi} \int_{0}^{\pi} a \sin nx\,dx 0 = \frac{a}{\pi} \left[\frac{-\cos nx}{n}\right]_0^{\pi} = \frac{a}{n\pi} (1 - \cos n\pi)$$

But $\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$$b_n = 0 \ \text{(n even)} \quad \text{and} \quad b_n = \frac{2a}{n\pi} \ \text{(n odd)}$$

© Copyright Virtual University of Pakistan

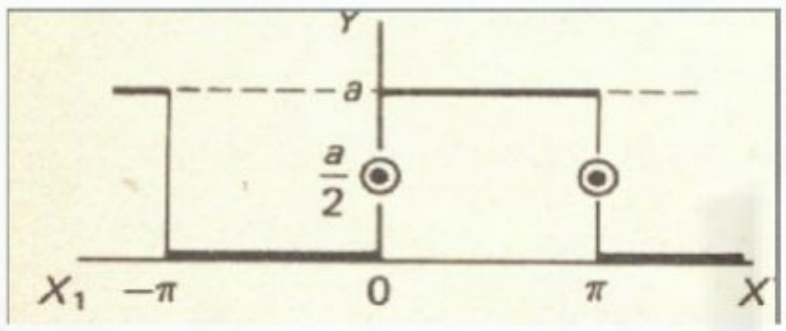
Scanned with CamScanner

Scanned with CamScanner

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} b_n \sin nx$$

$$f(x) = \frac{a}{2} + \frac{2a}{\pi} \{\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + ...\}$$

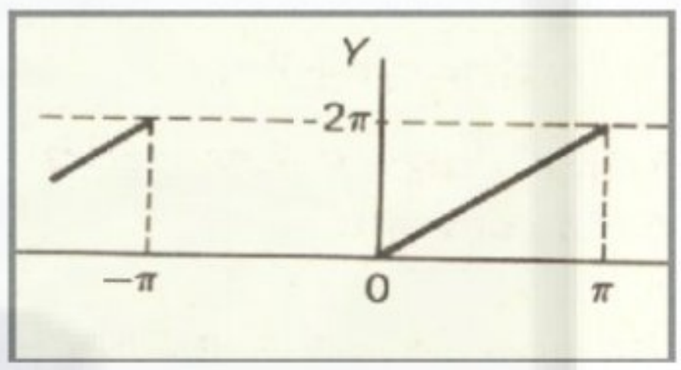
A finite discontinuity, or `jump', occurs at x = 0. If we substitute x = 0 in the series obtained, all the sine terms vanish and we get f(x) = a/2, which is, in fact, the average of the two function values at x = 0.



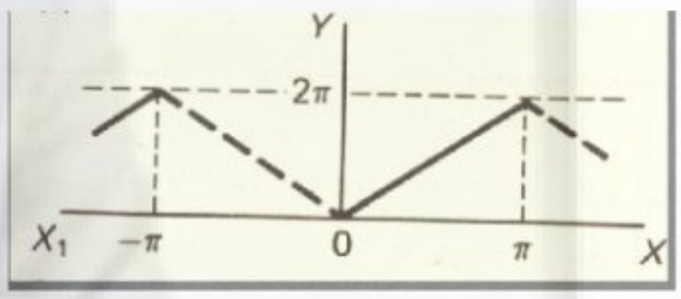
Note also that at x = π, another finite discontinuity occurs and substituting x = π in the series gives the same result. Because of this ambiguity, the function is said to be `undefined' at x = 0, x = π, etc.

Half-Range Series

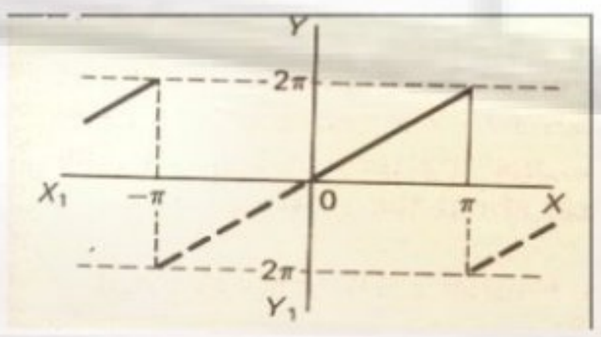
Sometime a function of period 2π is defined over the range 0 to π, instead of the normal – π to π, or 0 to 2π. We then have a choice of how to proceed. For example, if we are told between x = 0 and x = π, f(x) = 2x, then, since the period is 2π, we have no evidence of how the function behave between x = – π and x = 0.



If the waveform were as shown in (a), the function would be an even function, symmetrical about the y-axis and the series would have only cosine terms (including possibly a_0).



On the other hand, if the waveform were as shown in (b), the function would be odd, being symmetrical about the origin and the series would have only sine terms.



Example

A function f(x) is defined by

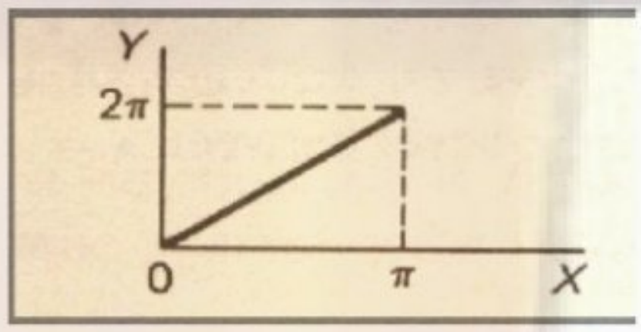
$f(x) = 2x \qquad 0 < x < \pi$

$f(x) = f(x + 2\pi)$

Obtain a half-range cosine series to represents the function.

To obtain a cosine series, i.e. a series with no sine terms, we need an even function.

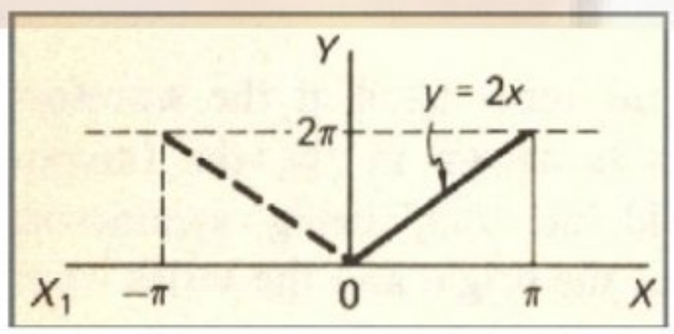
Therefore, we assume the waveform between x = – π and x = 0 to be as shown, making the total graph symmetrical about the y-axis.



Now we can find expressions for the Fourier coefficients as usual.

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx = \frac{2}{\pi}\int_0^{\pi} 2x dx = \frac{2}{\pi}\left[x^2\right]_0^{\pi} = 2\pi$$

$\therefore \quad a_0 = 2\pi$



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$$

$$a_n = \frac{2}{\pi}\int_{0}^{\pi} 2x\cos nx\,dx = \frac{4}{\pi}\int_{0}^{\pi} x\cos nx\,dx = \frac{4}{\pi}\left\{\left[\frac{x\sin nx}{n}\right]_0^{\pi} - \frac{1}{n}\int_{0}^{\pi}\sin nx\,dx\right\}$$

$$= \frac{4}{\pi}\left\{(0-0) - \frac{1}{n}\left[\frac{-\cos nx}{n}\right]_0^{\pi}\right\} = \left(\frac{4}{\pi n^2}\right)(\cos n\pi - 1)$$

$\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$\therefore\ a_n = 0$ (n even) and $a_n = -\frac{8}{\pi n^2}$ (n odd)

All that now remains is b_n which is zero, since f(x) is an even function, i.e. $b_n = 0$

So $a_0 = 2\pi$, $a_n = 0$ (n even) and $a_n = -\frac{8}{\pi n^2}$ (n odd),

$b_n = 0$. Therefore

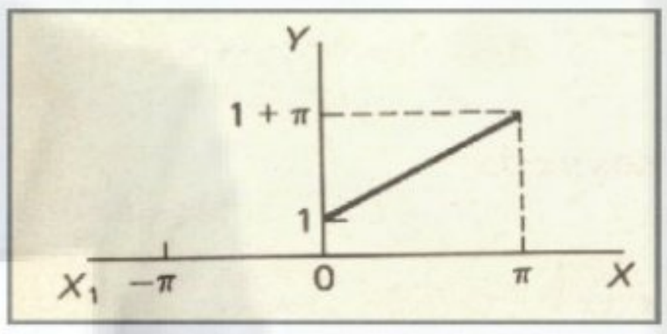
$$f(x) = \pi - \frac{8}{\pi}\left\{\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \ldots.\right\}$$

Example

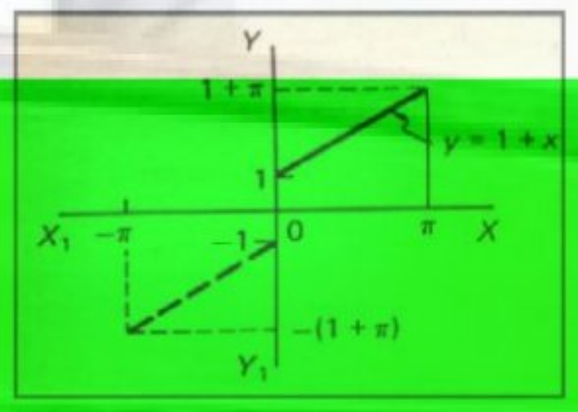
Determine a half-range sine series to represent the function f(x) defined by

$f(x) = 1 + x \quad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



We choose the waveform between $x = -\pi$ and $x = 0$ so that the graph is symmetrical about the origin. The function is then an odd function and the series will contain only sine terms.



$\therefore\ a_0 = 0$ and $a_n = 0$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\sin nx\,dx$$

$$b_n = \frac{2}{\pi}\int_{0}^{\pi} (1+x)\sin nx\,dx = \frac{2}{\pi}\left\{\left[(1+x)\frac{-\cos nx}{n}\right]_0^{\pi} + \frac{1}{n}\int_{0}^{\pi}\cos nx\,dx\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{1+\pi}{n}\cos n\pi + \frac{1}{n} + \frac{1}{n}\left[\frac{\sin nx}{n}\right]_0^{\pi}\right\} = \frac{2}{\pi}\left\{\frac{1}{n} - \frac{1+\pi}{n}\cos n\pi\right\} = \frac{2}{\pi n}\{1 - (1+\pi)\cos n\pi\}$$

$\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$\therefore\ b_n = -\frac{2}{n}$ (n even)

$\therefore\ = \frac{4+2\pi}{\pi n}$ (n odd)

Substituting in the general expression $f(x) = \sum_{x=1}^{\infty} b_n \sin nx$ we have

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$f(x)=\frac{4+2\pi}{\pi}\{\sin x+\frac{1}{3}\sin 3x+\frac{1}{5}\sin 5x+...\}-2\{\frac{1}{2}\sin 2x+\frac{1}{4}\sin 4x+\frac{1}{6}\sin 6x+....)$$

and the required series obtained

$$f(x)=\left(\frac{4}{\pi}+2\right)\{\sin x+\frac{1}{3}\sin 3x+\frac{1}{5}\sin 5x+..\}-2\{\frac{1}{2}\sin 2x+\frac{1}{4}\sin 4x+\frac{1}{6}\sin 6x+.....\}$$

So knowledge of odd and even functions and of half-range series saves a deal of unnecessary work on occasions.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No.-43 Functions With Periods Other Than 2π

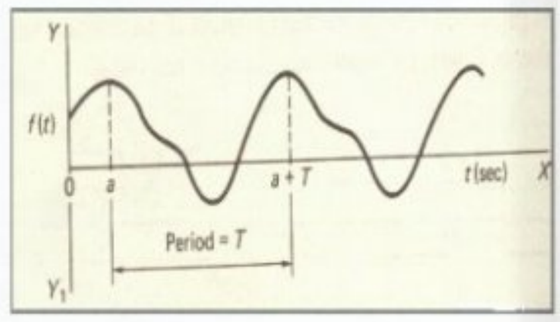
So far, we have considered functions f(x) with period 2π. In practice, we often encounter functions defined over periodic intervals other than 2π, e.g. from 0 to T, $-\frac{T}{2}$ to $\frac{T}{2}$ etc.

Functions With Period T

If y = f(x) is defined in the range $-\frac{T}{2}$ to $\frac{T}{2}$, i.e. has a period T, we can convert this to an interval of 2π by changing the units of the independent variable.

In many practical cases involving physical oscillations, the independent variable is time (t) and the periodic interval is normally denoted by T, i.e.

$$f(t) = f(t + T)$$



Each cycle is therefore completed in T seconds and the frequency f hertz (oscillations per second) of the periodic function is therefore given by $f = \frac{1}{T}$. If the angular velocity, ω radians per seconds, is defined by $\omega = 2\pi f$, then

$$\omega = \frac{2\pi}{T} \text{ and } T = \frac{2\pi}{\omega}$$

The angle, x radians, at any time t is therefore $x = \omega t$ and the Fourier series to represent the function can be expressed **as**

$$\mathbf{f(t) = \frac{1}{2} a_0 + \sum_{x=1}^{\infty} \{a_n Cos\ n\omega t + b_n Sin\ n\omega t\}}$$

which can also be written in the form

$$\mathbf{f(t) = \frac{1}{2} A_0 + \sum_{x=1}^{\infty} B_n \sin(n\omega t + \phi_n)} \qquad n = 1, 2, 3, ...$$

Fourier Coefficients

With the new variable, the Fourier coefficients become:

$$f(t) = \frac{1}{2} a_0 + \sum_{x=1}^{\infty} \{a_n \cos n\omega t + b_n \sin n\omega t\}$$

$$a_0 = \frac{2}{T}\int_0^T f(t)\,dt = \frac{\omega}{\pi}\int_0^{2\pi/\omega} f(t)\,dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega t\,dt = \frac{\omega}{\pi}\int_0^{2\pi/\omega} f(t)\cos n\omega t\,dt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t\,dt = \frac{\omega}{\pi}\int_0^{2\pi/\omega} f(t)\sin n\omega t\,dt$$

We can see that there is very little difference between these expressions and those that have gone before. The limits can, of course, be 0 to T, $-\frac{T}{2}$ to $\frac{T}{2}$, $-\frac{\pi}{\omega}$ to $\frac{\pi}{\omega}$, 0 to $\frac{2\pi}{\omega}$ etc. as is convenient, so long as they cover a complete period.

Example

Determine the Fourier series for a periodic function defined by

© Copyright Virtual University of Pakistan

Scanned with CamScanner

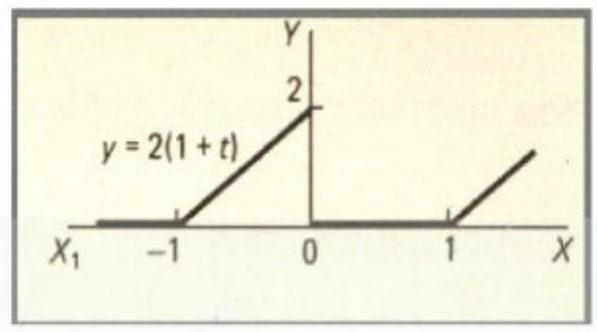
Scanned with CamScanner

$f(t) = 2(1+t) \qquad -1 < t < 0$

$f(t) = 0 \qquad 0 < t < 1$

$f(t) = f(t+2)$

The first step is to sketch the wave which is.



$$f(t) = \frac{1}{2}a_0 + \sum_{x=1}^{\infty} \{a_n \cos n\omega t + b_n \sin n\omega t\})$$

$T = 2$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\,dt = \frac{2}{2}\int_{-1}^{1} f(t)\,dt = \int_{-1}^{0} 2(1+t)\,dt + \int_{0}^{1} (0)\,dt = \left[2t + t^2\right]_{1}^{0} = 1$$

$\therefore \quad a_0 = 1$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos n\omega t\,dt = \frac{2}{2}\int_{-1}^{1} f(t)\cos n\omega t\,dt = \int_{-1}^{0} 2(1+t)\cos n\omega t\,dt$$

$$a_n = 2\left\{\left[(1+t)\frac{\sin n\omega t}{n\omega}\right]_{-1}^{0} - \frac{1}{n\omega}\int_{-1}^{0} \sin n\omega t\,dt\right\} = 2\left\{(0-0) - \frac{1}{n\omega}\left[-\frac{\cos n\omega t}{n\omega}\right]_{-1}^{0}\right\}$$

$$= \frac{2}{n^2\omega^2}(1 - \cos n\omega)$$

Now $T = \frac{2\pi}{\omega}$

$\therefore \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi \quad \therefore a_n = \frac{2}{n^2\omega^2}(1 - \cos n\pi)$

$\therefore \quad a_n = 0$ (n even)

$= \frac{4}{n^2\omega^2}$ (n odd)

Now for b_n

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\sin n\omega t\,dt$$

$$b_n = \frac{2}{2}\int_{-1}^{0} 2(1+t)\sin n\omega t\,dt = 2\left\{\left[(1+t)\frac{-\cos n\omega t}{n\omega}\right]_{-1}^{0} + \frac{1}{n\omega}\int_{-1}^{0}\cos n\omega t\,dt\right\}$$

$$= 2\left\{-\frac{1}{n\omega} + \frac{1}{n\omega}\left[\frac{\sin n\omega t}{n\omega}\right]_{-1}^{0}\right\} = 2\left\{-\frac{1}{n\omega} + \frac{1}{n^2\omega^2}(\sin n\omega)\right\}$$

As before $\omega = \pi$

$\therefore \quad b_n = -\frac{2}{n\omega}$

So the first few terms of the series give

$$f(t) = \frac{1}{2} + \frac{4}{\omega^2}\left\{\cos \omega t + \frac{1}{9}\cos 3\omega t + \frac{1}{25}\cos 5\omega t + ..\right\} - \frac{2}{\omega}\left\{\sin \omega t + \frac{1}{2}\sin 2\omega t + \frac{1}{3}\sin 3\omega t + \frac{1}{4}\sin 4\omega t..\right\}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

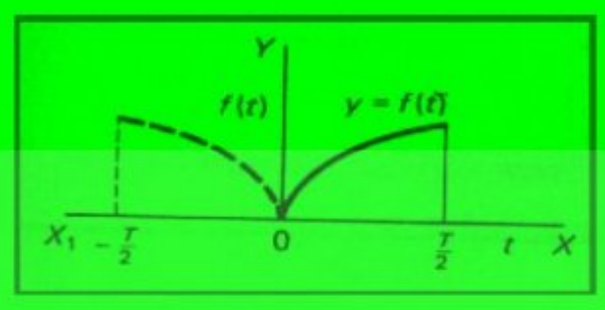
Half-Range Series

The theory behind the half-range sine and cosine series still applies with the new variable.

(a) Even function

Half-range cosine series

$y = f(t) \qquad 0 < t < \frac{T}{2}$



$f(t) = f(t + T)$

symmetrical about the y-axis.

With an even function, we know that $b_n = 0$

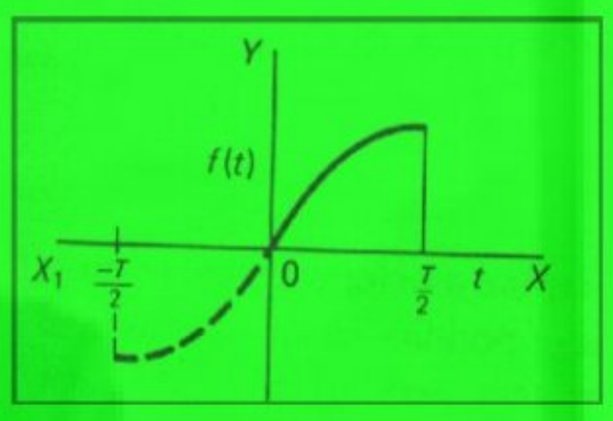
$$\therefore \quad f(t) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t$$

where $a_0 = \frac{4}{T}\int_0^{T/2} f(t)\,dt$ and $a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega t\,dt$

(b) Odd function

Half-range sine series

$y = f(t) \qquad 0 < t < \frac{T}{2}$



$f(t) = f(t + T)$

symmetrical about the origin.

$\therefore a_0 = 0$ and $a_n = 0$

$$f(t) = \sum_{x=1}^{\infty} b_n \sin n\omega t;$$

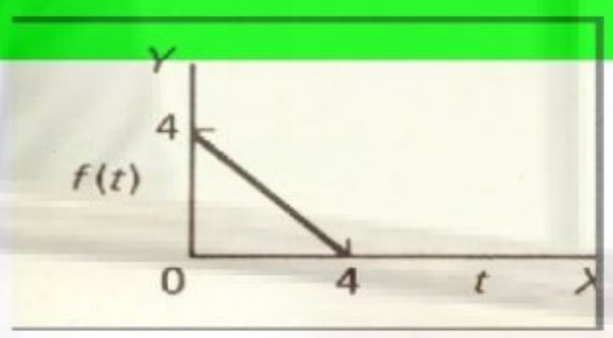
$$b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega t\,dt$$

Example

A function f(t) is defined by

$f(t) = 4 - t, \qquad 0 < t < 4.$

We have to form a half-range cosine series to represent the function in this interval.



First we form an even function, i.e. symmetrical about the y-axis.

$$a_0 = \frac{4}{T}\int_0^{T/2} f_1(t)\,dt = \frac{4}{8}\int_0^{4}(4 - t)\,dt = \frac{1}{2}\int_0^{4}(4 - t)\,dt$$

$$= \frac{1}{2}\left|4t - \frac{t^2}{2}\right|_0^4 = \frac{1}{2}\left[4(4) - \frac{(4)^2}{2}\right] = \frac{1}{2}[16 - 8] = \frac{1}{2}(8) = 4$$

$$a_n = \frac{4}{T}\int_0^{T/2} f_1(t)\cos n\omega t\,dt = \frac{4}{8}\int_0^{4}(4 - t)\cos n\omega t\,dt$$

Simple integration by parts gives

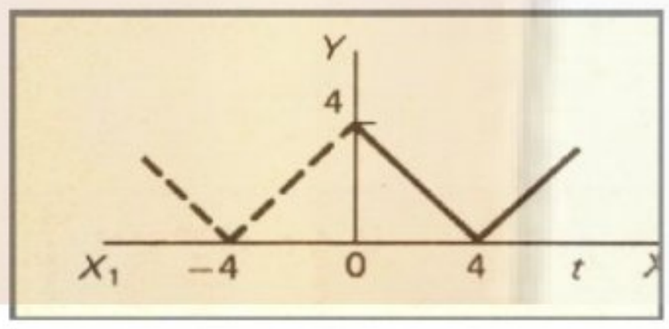
$$a_n = \frac{1}{2}\left\{-\frac{2\sin 4n\omega}{n\omega} - \frac{1}{n^2\omega^2}(\cos 4n\omega - 1)\right\}$$

But $\omega = \frac{2\pi}{T} = \frac{2\pi}{8} = \frac{\pi}{4}$

$$\therefore a_n = \frac{1}{2}\left\{-\frac{2\sin n\pi}{n\omega} - \frac{1}{n^2\omega^2}(\cos n\pi - 1)\right\}$$

$n = 1, 2, 3, \ldots\ldots$

$\sin n\pi = 0;$



© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$\cos n\pi = 1$ (n even); $\cos n\pi = -1$ (n odd)

$\therefore\ a_n = 0$ (n even)

$a_n = \frac{1}{n^2\omega^2}$ (n odd)

$\therefore\ f(t)=2+\frac{1}{\omega^2}\left\{\cos \omega t+\frac{1}{9}\cos 3\omega t+\frac{1}{25}\cos 5\omega t+\ldots\right\}$ where $\omega = \frac{\pi}{4}$.

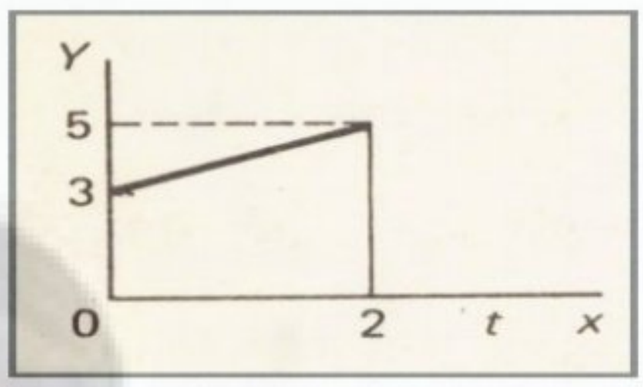
Example

A function f(t) is defined by

$f(t) = 3 + t \qquad 0 < t < 2$

$f(t) = f(t + 4)$

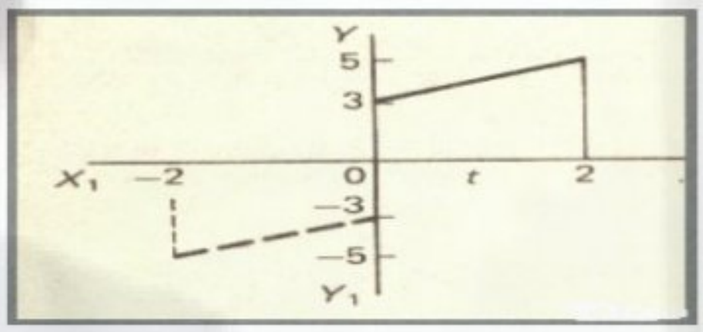
Obtain the half-range sine series for the function in this range.



Sine series required. Therefore, we form an odd function, symmetrical about the origin.

$a_0 = 0; \qquad a_n = 0; \qquad T = 4$

$f(t) = \sum_{n=1}^{\infty} b_n \sin n\omega t$



$\therefore\ b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega t\, dt = \int_0^2 (3 + t)\sin n\omega t\, dt$

$\int_0^2 (3 + t)\sin n\omega t\, dt = \left|\frac{(3+t)\cos n\omega t}{-n\omega}\right|_0^2 - \int_0^2 \frac{\cos n\omega t}{-n\omega}\, dt$

$= \frac{(3 + 2)\cos n\omega 2}{-n\omega} - \frac{3}{-n\omega} + \frac{1}{n\omega}\left|\frac{\sin n\omega t}{n\omega}\right|_0^2 = \frac{3}{n\omega} - \frac{5}{n\omega}\cos 2n\omega + \frac{1}{n^2\omega^2}\left[\frac{\sin 2n\omega}{n\omega} - 0\right]$

But $T = \frac{2\pi}{\omega} \qquad \therefore \qquad \omega = \frac{2\pi}{T} = \frac{\pi}{2}$

$= \frac{3}{n\omega} - \frac{5}{n\omega}\cos n\pi + \frac{1}{n^2\omega^2}\left[\frac{\sin n\pi}{n\omega}\right]$

$b_n = \frac{1}{n\omega}(3 - 5\cos 2n\omega) + \frac{1}{n^2\omega^2}(\sin 2n\omega)$

$\therefore\ b_n = \frac{1}{n\omega}(3 - 5\cos n\pi) + \frac{1}{n^2\omega^2}(\sin n\pi) = -\frac{2}{n\omega}$ (n even)

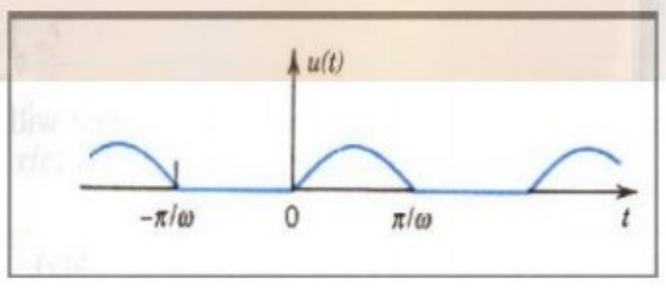
$= \frac{8}{n\omega}$ (n odd)

$\therefore\ f(t)=\frac{2}{\omega}\left\{4\sin\omega t - \frac{1}{2}\sin 2\omega t+\frac{4}{3}\sin 3\omega t - \frac{1}{4}\sin 4\omega t..\right\}$

Half-Wave Rectifier

A sinusoidal voltage E sin ωt, where t is time, is passed through a half-wave rectifier that clips the negative portion of the wave

Find the Fourier series of the resulting periodic functions.



$u(t) = 0 \qquad$ if $\qquad -T/2 < t < 0$

$= E \sin \omega t \qquad 0 < t < T/2$

here $T = \frac{2\pi}{\omega}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} u(t)\,dt = \frac{2}{T}\int_{-T/2}^{0} 0\,dt + \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\,dt = \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\,dt = \frac{\omega}{\pi}\int_{0}^{\pi/\omega} E\sin\omega t\,dt$$

$$= \frac{\omega}{\pi} E\left|\frac{-\cos\omega t}{\omega}\right|_{0}^{\pi/\omega} = \frac{2E}{\pi}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} u(t)\cos n\omega t\,dt = \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\cos n\omega t\,dt = \frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} 2\sin\omega t\cos n\omega t\,dt$$

$$= \frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\sin(1+n)\omega t + \sin(1-n)\omega t]\,dt$$

If n = 1 then integral on the right is zero
and if n = 2, 3, …… then we obtain.

$$a_n = \frac{\omega E}{2\pi}\left[-\frac{\cos(1+n)\omega t}{(1+n)\omega} - \frac{\cos(1-n)\omega t}{(1-n)\omega}\right]_{0}^{\pi/\omega} = \frac{\omega E}{2\pi}\left[\frac{-\cos(1+n)\pi + 1}{(1+n)\omega} + \frac{-\cos(1-n)\pi + 1}{(1-n)\omega}\right]$$

$$= \frac{\omega E}{2\pi\omega}\left[\frac{-\cos(1+n)\pi + 1}{(1+n)} + \frac{-\cos(1-n)\pi + 1}{(1-n)}\right]$$

if n is odd then $a_n = 0$

if n is even then $a_n = \frac{E}{2\pi}\left(\frac{2}{1+n} + \frac{2}{1-n}\right) = \frac{E}{2\pi}\left[\frac{2-2n+2+2n}{(1+n)(1-n)}\right] = \frac{2E}{(1-n)(1+n)\pi}$

$$= \frac{2E}{(1-n^2)\pi}$$

For b_n we have

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} u(t)\sin n\omega t\,dt = \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\sin n\omega t\,dt = -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} -2\sin\omega t\sin n\omega t\,dt$$

$$= -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\cos(1+n)\omega t - \cos(1-n)\omega t]\,dt$$

If n = 1

$$b_n = -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\cos 2\omega t - 1]\,dt = -\frac{\omega E}{2\pi}\left|\frac{\sin 2\omega t}{2\omega} - t\right|_{0}^{\pi/\omega} = -\frac{\omega E}{2\pi}(-\pi/\omega) = E/2$$

if $n \neq 1$

$$b_n = -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\cos(1+n)\omega t - \cos(1-n)\omega t]\,dt = -\frac{\omega E}{2\pi}\left[\frac{\sin(1+n)\omega t}{(1+n)\omega} - \frac{\sin(1-n)\omega t}{(1-n)\omega}\right]_{0}^{\pi/\omega}$$

$$= -\frac{\omega E}{2\pi}\left[\frac{\sin(1+n)\pi}{(1+n)\omega} - \frac{\sin(1-n)\pi}{(1-n)\omega}\right] = 0 \qquad \text{for } n = 2, 3, 4, \ldots\ldots$$

$$u(t) = \frac{1}{2}a_0 + \sum_{n=2}^{\infty} a_n \cos n\omega t$$

$$u(t) = \frac{E}{\pi} + \frac{E}{2}\sin\omega t - \frac{2E}{\pi}\left(\frac{1}{1.3}\cos 2\omega t + \frac{1}{3.5}\cos 4\omega t + \ldots\right)$$

Lecture No -44 Laplace Transforms

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$\mathcal{L}\{e^{3t}\} = \frac{1}{s-3}; \mathcal{L}\{t^3\} = \frac{3!}{s^4}; \mathcal{L}\{2\sin 3t + \cos 3t\} = \frac{s+6}{s^2+9}$$

And $L^{-1}\{\frac{1}{s-3}\} = e^{3t}$; $L^{-1}\{\frac{3!}{s^4}\} = t^3$; $L^{-1}\{\frac{s+6}{s^2+9}\} = 2\sin 3t + \cos 3t$

We show that $L\{t^3\} = \frac{3!}{s^4}$ For this consider the integral

$$L(t^3) = \int_0^\infty t^3 e^{-st} dt = [t^3 \frac{e^{-st}}{-s}]_0^\infty - \int_0^\infty 3t^2 e^{-st} \frac{1}{-s} dt = \frac{-t^3}{se^{st}} \Big|_0^\infty + \frac{3}{s}\int_0^\infty t^2 e^{-st} dt$$

$$= 0 + \frac{3}{s}\{t^2 \frac{e^{-st}}{-s} \Big|_0^\infty + \frac{2}{s}\int_0^\infty te^{-st} dt\} = \frac{3}{s}\frac{2}{s}\int_0^s te^{-st} dt = \frac{3.2}{s^2}\{t \frac{e^{-st}}{-s} \Big|_0^\infty + \frac{1}{s}\int_0^\infty e^{-st} dt\}$$

$$= \frac{3.2}{s^3}\int_0^\infty e^{-st} dt = \frac{3.2}{s^3}\frac{-1}{s}\int_0^\infty e^{-st}(-s)dt = \frac{-3.2.1}{s^4} e^{-st} \Big|_0^\infty = \frac{-3.2.1}{s^4}(0-1) = \frac{3.2.1}{s^4} = \frac{3!}{s^4}$$

So that $L\{t^3\} = \frac{3!}{s^4}$

Laplace Transform

The Laplace Transform of a function F(t) is denoted by L{F(t)} and is defined as the integral of F(t) e^{-st} between the limits t=0 and t = ∞

$$L\{F(t)\} = \int_0^\infty e^{-st} F(t) dt.$$

In all cases, the constant parameter s is assumed to be positive and large enough to ensure that the product F(t) e^{-st} converges to zero as t → ∞ , what ever the function F(t).

In determining the transform of any function , you will appreciate that the limits are substituted for t, so that the result will be a function of s.

Laplace Transform of F(t) = a (constant).

That is

$$L(a) = \int_0^\infty ae^{-st} dt = a\int_0^\infty e^{-st} dt$$

$$= a\frac{e^{-st}}{-s} \Big|_0^\infty = \frac{-a}{s}\{0-1\} = \frac{a}{s}$$

Example

Find the laplace transform of the form e^{at} that F(t)= e^{at} where a is a constant.

$$L(e^{at}) = \int_0^\infty e^{at} e^{-st} dt = \int_0^\infty e^{-(s-a)t} dt = [\frac{e^{-(s-a)t}}{-(s-a)}]_0^\infty$$

$$= -\frac{1}{s-a}\left[\frac{1}{e^{(s-a)t}}\right], \quad s > a$$

$$= -\frac{1}{s-a}[0-1] = \frac{1}{s-a}$$

$$L\{e^{at}\} = \frac{1}{s-a}$$

So we already have two standard transforms

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$L\{a\}=\frac{a}{s}$; $L\{e^{at}\}=\frac{1}{s-a}$;

$L\{4\}=\frac{4}{s}$; $L\{e^{4t}\}=\frac{1}{s-4}$;

$L\{-5\}=\frac{-5}{s}$; $L\{e^{-2t}\}=\frac{1}{s+2}$;

Laplace transform is always a function of s.

Complex Numbers Power of i

Every time a factor i^4 occurs , it can be replace by the factor 1, so that the power of i is reduced to one of the four results above.

$i^9=(i^4)^2 i=(1)^2 i=1.\ i=i$

$i^{20}=(i^4)^5=(1)^5=1$

$i^{30}=(i^4)^7 i^2=(1)^7(-1)=1(-1)=-1$

$i^{15}=(i^4)^3 i^3=1(-i)=-i$

Complex Numbers

z = 3+ 5 i , is called a complex number where 3 is real part and 5 is imaginary part of the complex number.

In general z = a+ b i , is called a complex number where a is real part and b is imaginary part of the complex number. So,

Complex Number = (Real Part)+ *i* (Imaginary Part)

Conjugate complex numbers

For a complex number a + *i* b , the complex number a - *i* b is called the conjugate of a + *i* b. Conjugate complex numbers are identical except the signs in the middle for the brackets.

- (4 + 5 i) and (4 - 5 i) are conjugate complex numbers
- (6 + 2 i) and (2 + 6 i) are not conjugate complex numbers
- (5 - 3 i) and (-5 +3 i) are not conjugate complex numbers

Remember

The product of complex number by its conjugate is always entirely real.

$(3+4i)(3-4i)=9+16=25$

$(a+bi)(a-bi)=a^2+b^2$

Euler Formaula

As we know that the series expansion of e^x ,cos x and sin x are given as

$e^x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\ldots\ldots\ldots\ldots$

$\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\ldots\ldots\ldots\ldots$

$\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\ldots\ldots\ldots\ldots$

Replace x by (it) , we get

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$e^{(it)} = 1 + (it) + \frac{(it)^2}{2!} + \frac{(it)^3}{3!} + \frac{(it)^4}{4!} + \ldots\ldots\ldots\ldots$$

$$e^{(it)} = 1 + (it) - \frac{(t)^2}{2!} - \frac{i(t)^3}{3!} + \frac{(t)^4}{4!} + \frac{i(t)^5}{5!} - \frac{(t)^6}{6!} - \frac{i(t)^7}{7!} + \ldots\ldots\ldots\ldots$$

$$e^{(it)} = [1 - \frac{(t)^2}{2!} + \frac{(t)^4}{4!} - \frac{(t)^6}{6!} + \ldots\ldots\ldots\ldots] + i[t - \frac{(t)^3}{3!} + \frac{(t)^5}{5!} - \frac{(t)^7}{7!} + \ldots\ldots\ldots\ldots]$$

where

$$\cos t = 1 - \frac{(t)^2}{2!} + \frac{(t)^4}{4!} - \frac{(t)^6}{6!} + \ldots\ldots\ldots\ldots$$

$$\sin t = t - \frac{(t)^3}{3!} + \frac{(t)^5}{5!} - \frac{(t)^7}{7!} + \ldots\ldots\ldots\ldots$$

$$e^{it} = \cos t + i \sin t$$

$R(e^{it}) = \cos t$ and $I(e^{it}) = \sin t$

The Laplace transform of F(t) = sin at

$$L(\text{Sin at}) = L(I(e^{iat})) = I\int_0^\infty e^{iat} e^{-st} dt = I\int_0^\infty e^{-(s-ia)t} dt = I\left\{\left[\frac{e^{-(s-ia)t}}{-(s-ia)}\right]_0^\infty\right\}$$

$$= I\left\{\frac{1}{-(s-ia)}(0-1)\right\} = I\left\{\frac{1}{(s-ia)}\right\}$$

$$L\{\sin at\} = I\{\frac{s+ia}{s^2+a^2}\} = I[\frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}]$$

$$\therefore L\{\sin at\} = \frac{a}{s^2+a^2}$$

We can use the same method to determined to determine L{cos at }.

Since cos at is the real part of e^{iat}, written as $R(e^{iat})$

$$L\{\cos at\} = R\{\frac{s+ia}{s^2+a^2}\} = R[\frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}]$$

$$L\{\cos at\} = \frac{s}{s^2+a^2}$$

The Transform of F(t) = t^n where n is a positive integer.

By the definition $L(t^n) = \int_0^\infty t^n e^{-st} dt$ integrating by parts

$$L(t^n) = \int_0^\infty t^n e^{-st} dt = \left[t^n \frac{e^{-st}}{-s}\right]_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1} e^{-st} dt = -\frac{1}{s}\left[t^n e^{-st}\right]_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1} e^{-st} dt$$

$$\therefore \left[t^n \frac{e^{-st}}{-s}\right]_0^\infty = 0 - 0 = 0 \qquad \therefore L\{t^n\} = \frac{n}{s}\int_0^\infty t^{n-1} e^{-st} dt \text{ ----------(1)}$$

you will notice that $\int_0^\infty t^{n-1} e^{-st} dt$ is identical to $\int_0^\infty t^n e^{-st} dt$ except that n is replaced by (n-1)

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

If $if\ I_n = \int_0^\infty t^n e^{-st} dt, then\ I_{n-1} = \int_0^\infty t^{n-1} e^{-st} dt$ and the result (1) becomes

$$I_n = \frac{n}{s} I_{n-1}$$

This is reduction formula and , if we now replace n by (n-1) we get

$$I_{n-1} = \frac{n-1}{s} I_{n-2}$$

If we replace n by (n-1) again in this last result , we have

$$I_{n-2} = \frac{n-2}{s} I_{n-3}$$

$$So\ \ I_n = \int_0^\infty t^n e^{-st} dt = \frac{n}{s} I_{n-1} = \frac{n}{s}.\frac{n-1}{s} I_{n-2} = \frac{n}{s}.\frac{n-1}{s}.\frac{n-2}{s} I_{n-3} = \frac{n}{s}.\frac{n-1}{s}.\frac{n-2}{s}.\frac{n-3}{s} I_{n-4}$$

So finally, we have

$$I_n = \frac{n}{s}.\frac{n-1}{s}.\frac{n-2}{s}.\frac{n-3}{s}.\frac{n-4}{s} \ldots\ldots\ldots\ldots \frac{2}{s}.\frac{1}{s} I_0$$

But

$$I_0 = L\{t^0\} = L\{1\} = \frac{1}{s}$$

$$I_n = \frac{n}{s}\frac{n-1}{s}\frac{n-2}{s}\frac{n-3}{s}\frac{n-4}{s} \ldots\ldots\ldots\ldots \frac{2}{s}\frac{1}{s} I_0 = \frac{n!}{s^{n+1}}$$

$$L\{t^n\} = \frac{n!}{s^{n+1}}$$

$$L\{t\} = \frac{1}{s^2}; L\{t^2\} = \frac{2!}{s^3}; L\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4}$$

Laplace Transform of F(t) = sinh at and F(t) = cosh at.

Starting from the exponential definitions of sinh at and cosh at

$$i.e.\quad \sinh at = \frac{1}{2}(e^{at} - e^{-at})\ and\ \cosh at = \frac{1}{2}(e^{at} + e^{-at})$$

We proceed as follow

$a)\quad F(t) = \sinh at$

$$\text{Sinh at} = \int_0^\infty \sinh at\ e^{-st} dt = \frac{1}{2}\int_0^\infty (e^{at} - e^{-at}) e^{-st} dt = \frac{1}{2}\int_0^\infty (e^{at}e^{-st} - e^{-at}e^{-st}) dt = \frac{1}{2}\int_0^\infty (e^{-(s-a)t} - e^{-(s+a)t}) dt$$

$$= \frac{1}{2}\left\{\left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty - \left[\frac{e^{-(s+a)t}}{-(s+a)}\right]_0^\infty\right\} = \frac{1}{2}\left\{\frac{1}{-(s-a)}(0-1) - \frac{1}{-(s+a)}(0-1)\right\} = \frac{1}{2}\left\{\frac{1}{(s-a)} - \frac{1}{(s+a)}\right\}$$

$$= \frac{1}{2}\left\{\frac{s+a-s+a}{(s-a)(s+a)}\right\} = \frac{1}{2}\left\{\frac{2a}{s^2-a^2}\right\} = \left\{\frac{a}{s^2-a^2}\right\}$$

$b)\quad F(t) = \cosh at$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

$$L(\cosh at) = L(\frac{1}{2}(e^{at} + e^{-at})) = \frac{1}{2}\int_0^\infty (e^{at} + e^{at})e^{-st}dt = \frac{1}{2}\left[\int_0^\infty e^{-(s-a)t}dt + \int_0^\infty e^{-(s+a)t}dt\right]$$

$$= \frac{1}{2}\left\{\left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty + \left[\frac{e^{-(s+a)t}}{-(s+a)}\right]_0^\infty\right\} = \frac{1}{2}\left\{\frac{1}{-(s-a)}(0-1) + \frac{1}{-(s+a)}(0-1)\right\} = \frac{1}{2}\left\{\frac{1}{(s-a)} + \frac{1}{(s+a)}\right\}$$

$$= \frac{1}{2}\left\{\frac{s+a+s-a}{(s-a)(s+a)}\right\} = \frac{1}{2}\left\{\frac{2s}{s^2-a^2}\right\} = \left\{\frac{s}{s^2-a^2}\right\}$$

$$L(\cosh at) = \left\{\frac{s}{s^2-a^2}\right\}$$

Several Standard Results

$L\{a\} = \frac{a}{s}$; $L\{e^{at}\} = \frac{1}{s-a}$; $L\{t^n\} = \frac{n!}{s^{n+1}}$

$L\{\sin at\} = \frac{a}{s^2+a^2}$; $L\{\cos at\} = \frac{s}{s^2+a^2}$

$L\{\sinh at\} = \frac{a}{s^2-a^2}$; $L\{\cosh at\} = \frac{s}{s^2-a^2}$

We can , of course, combine these transforms by adding or subtracting as necessary , but they must not be multiplied together to form the transform of a product.

Example

$a)\ L\{2\sin 3t + \cos 3t\} = 2L(\sin 3t) + L(\cos 3t) = 2.\frac{3}{s^2+9} + \frac{s}{s^2+9} = \frac{s+6}{s^2+9}$

$b)\ L\{4e^{2t} + 3\cosh 4t\} = 4L(e^{2t}) + L(3\cosh 4t) = 4.\frac{1}{s-2} + 3.\frac{s}{s^2-16} = \frac{4}{s-2} + \frac{3s}{s^2-16}$

$$= \frac{7s^2-6s-64}{(s-2)(s^2-16)}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Lecture No -45 Theorems

Theorem 1
The First Shift Theorem

The first Shift theorem states that if $\mathbf{L\{F(t)\}= f(s)}$ then $\mathbf{L\{e^{(-at)}\ F(t)\}= f(s+a)}$
The transform $\mathbf{L\{e^{(-at)}\ F(t)\}}$ is thus the same as $\mathbf{L\{F(t)\}}$ with s everywhere in the result replaced by (s+a)

Example

$$L\{\sin 2t\}= \frac{2}{s^2+4} \quad \textbf{then} \quad L\{e^{-3t}\ \sin 2t\}= \frac{2}{(s+3)^2+4}= \frac{2}{s^2+6s+13}$$

Example

$L\{t^2\}=\frac{2}{s^3}$; $L\{t^2e^{4t}\}$ is the same with s replaced by (s-4)

So $L\{t^2e^{4t}\}=\frac{2}{(s-4)^3}$

Theorem 2
Multiplying by t

If $L\{F(t)\}= f(s)$ **then** $L\{t(F(t)\}=-\frac{d}{ds}\{f(s)\}$

Example $L\{\sin 2t\}= \frac{2}{s^2+4}$

And $L\{t \sin 2t\} = -\frac{d}{ds}\left[\frac{2}{(s^2+4)}\right]=\frac{4s}{(s^2+4)^2}$

Example $L\{t\cos 3t\}=-\frac{d}{ds}\left(\frac{s}{s^2-9}\right)=-\frac{(s^2-9)-s(2s)}{(s^2-9)^2}=-\frac{s^2-9-2s^2}{(s^2-9)^2}=\frac{s^2+9}{(s^2-9)^2}$

We could, if necessary, take this a stage further and find

$L\{t\cos 3t\}=-\frac{d}{ds}\left(\frac{s^2+9}{(s^2-9)^2}\right)=\frac{2s(s^2+27)}{(s^2-9)^3}$

Theorem Obviously extends the range of function that we can deal with. So, in general

If $L\{F(t)\}= f(s)$ **then** $L\{t^n(F(t)\}=(-1)^n\frac{d^n}{ds^n}\{f(s)\}$

Theorem
Dividing by t

If $L\{F(t)\}= f(s)$ **then** $L\{\frac{F(t)}{t}\}=\int_s^\infty f(s)ds$

Example Determine $L\{\frac{\sin at}{t}\}$

$As\ L\{\sin at\}=\frac{a}{s^2+a^2}$

$\therefore L\{\frac{\sin at}{t}\}=\int_s^\infty \frac{a}{s^2+a^2}ds=\left[\tan^{-1}(\frac{s}{a})\right]_s^\infty=\frac{\pi}{2}-\tan^{-1}(\frac{s}{a})=\tan^{-1}(\frac{a}{s})$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example Determine $L\{\frac{1-\cos 2t}{t}\}$

As $L\{1-\cos 2t\}=\frac{1}{s}-\frac{s}{s^2+4}$

Then by Theorem 3,

$$L\{\frac{1-\cos 2t}{t}\}=\int_s^\infty(\frac{1}{s}-\frac{s}{s^2+4})ds=\left[\ln s-\frac{1}{2}\ln(s^2+4)\right]_s^\infty=\left[\frac{1}{2}.2\ln s-\frac{1}{2}\ln(s^2+4)\right]_s^\infty$$

$$=\left[\frac{1}{2}\ln s^2-\frac{1}{2}\ln(s^2+4)\right]_s^\infty=\frac{1}{2}\left[\ln s-\ln(s^2+4)\right]_s^\infty=\frac{1}{2}\left[\ln\frac{s^2}{(s^2+4)}\right]_s^\infty$$

When $s\to\infty$ *then* $\ln\frac{s^2}{(s^2+4)}\to\ln 1=0$

$$L\{\frac{1-\cos 2t}{t}\}=\int_s^\infty(\frac{1}{s}-\frac{s}{s^2+4})ds=-\frac{1}{2}\left[\ln\frac{s^2}{(s^2+4)}\right]=\ln\left[\frac{s^2}{(s^2+4)}\right]^{-\frac{1}{2}}=\ln\sqrt{\frac{s^2+4}{s^2}}$$

Standard Forms

F(t)	$\mathcal{L}\{F(t)\}=f(s)$
a	$\frac{a}{s}$
e^{at}	$\frac{1}{s-a}$
sin at	$\frac{a}{s^2+a^2}$
cos at	$\frac{a}{s^2+a^2}$
sinh at	$\frac{a}{s^2-a^2}$
cosh at	$\frac{s}{s^2-a^2}$
t^n	$\frac{n!}{s^{n+1}}$

(n a positive integer)

Theorem 1 The First Shift Theorem

If $L\{F(t)\}=f(s)$ then $L\{e^{(-at)}F(t)\}=f(s+a)$

Theorem 2 Multiplying by t

If $L\{F(t)\}=f(s)$ **then** $L\{t(F(t)\}=-\frac{d}{ds}\{f(s)\}$

Theorem 3 Dividing by t

If $L\{F(t)\}=f(s)$ **then** $L\{\frac{F(t)}{t}\}=\int_s^\infty f(s)ds$

Provided $\lim_{t\to 0}\left\{\frac{F(t)}{t}\right\}$ *exists.*

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Inverse Transforms

Here we have the reverse process i.e. given a Laplace transform, we have to find the function of t to which it belongs. For example, we know that $\frac{a}{s^2+a^2}$ is the Laplace Transform of **sin at** , so we can now write $L^{-1}\left[\frac{a}{s^2+a^2}\right]=\sin at$, the symbol L^{-1} indicating the inverse transform and not a reciprocal.

(a) $L^{-1}\left\{\frac{1}{s-2}\right\}=e^{-2t}$

(b) $L^{-1}\left\{\frac{s}{s^2+25}\right\}=\cos 5t$

(c) $L^{-1}\left\{\frac{4}{s}\right\}=4$

(d) $L^{-1}\left\{\frac{12}{s^2-9}\right\}=4\sinh 3t$

But what about $L^{-1}\left\{\frac{3s}{s^2-s-6}\right\}$, it happens that we can write $\frac{3s}{s^2-s-6}$ as the sum of two simpler functions $\frac{1}{s+2}+\frac{1}{s-3}$ which , of course , makes all the difference , since we can now proceed. $L^{-1}\left\{\frac{3s}{s^2-s-6}\right\}=L^{-1}\left\{\frac{1}{s+2}\right\}+L^{-1}\left\{\frac{1}{s-3}\right\}=e^{-2t}+2e^{3t}$

Rules of Partial Fractions

1. The numerator must be of lower degree than denominator. If it is not , then we first divide out.
2. Factorise the denominator into its prime factors. These determine the shapes of the partial fraction.
3. A linear factor (s+a) gives a partial fraction $\frac{A}{s+a}$ is a constant to be determined.
4. A repeated factor $(s+a)^2$ gives $\frac{A}{s+a}+\frac{B}{(s+a)^2}$
5. Similarly $(s+a)^3$ gives $\frac{A}{s+a}+\frac{B}{(s+a)^2}+\frac{C}{(s+a)^3}$
6. A quadratic Factor (s^2+ps+q) gives $\frac{Ps+Q}{s^2+ps+q}$
7. Repeated quadratic Factor $(s^2+ps+q)^2$ gives $\frac{Ps+Q}{s^2+ps+q}+\frac{Rs+T}{(s^2+ps+q)^2}$

So $\frac{s-19}{(s-2)(s+5)}$ has partial fraction of the form $\frac{A}{(s-2)}+\frac{B}{(s+5)}$ and $\frac{3s^2-4s+11}{(s+3)(s-2)^2}$ has partial fraction $\frac{A}{(s+3)}+\frac{B}{(s-2)}+\frac{C}{(s-2)^2}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Example

To determine $L^{-1}\left\{\frac{5s+1}{s^2-s-12}\right\}$

a) First we check that the numerator is of lower degree than the denominator. In fact this is so.

b) Factories the denominator

$$\frac{5s+1}{s^2-s-12}=\frac{5s+1}{(s-4)(s+3)}=\frac{A}{(s-4)}+\frac{B}{(s+3)}$$

We therefore have an identity

$$\frac{5s+1}{s^2-s-12}=\frac{A}{(s-4)}+\frac{B}{(s+3)}$$

which is true for any value of s we care to substitute
If we multiply through by the denominator (s^2-s-12) we have

$5s + 1 \equiv A(s + 3) + B(s - 4)$

We now substitute covenant values for s

i) Let $(s -4) = 0$ that is $s = 4$ therefore $21 = A(7) + B(0) \Rightarrow A = 3$
ii) Let $(s + 3) = 0$ that is $s = -3$ therefore $B = 2$

So $\frac{5s+1}{s^2-s-12}=\frac{3}{(s-4)}+\frac{2}{(s+3)}$

$$L^{-1}\left\{\frac{5s+1}{s^2-s-12}\right\}=3e^{4t}+2e^{-3t}$$

Example Determined $L^{-1}\left\{\frac{9s-8}{s^2-2s}\right\}$

$$L\{F(t)\}=\frac{9s-8}{s^2-2s}$$

a) Numerator of first degree ; denominator of second degree.

b) $\frac{9s-8}{s^2-2s}=\frac{A}{s}+\frac{B}{s-2}$

c) Multiply by $s(s-2)$

$\therefore 9s - 8 \equiv A(s - 2) + B(s)$

d) Put $s = 0$

$\therefore -8 \equiv A(-2) + B(0) \quad \therefore A = 4$

e) Put $s - 2$, i.e. $s = 2$

$\therefore 10 = A(0) + B(0) \quad \therefore B = 5$

$$\therefore F(t) = L^{-1}\left\{\frac{4}{s}+\frac{5}{s-2}\right\} = 4 + 5e^{2t}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Table of inverse transforms

Standard transforms

f(s)	F(t)
$\frac{a}{s}$	a
$\frac{1}{s+a}$	e^{-at}
$\frac{n!}{s^{n+1}}$	t^n (n a positive integer)
$\frac{1}{s^n}$	$\frac{t^{n-1}}{(n-1)!}$ (n a positive integer)
$\frac{a}{s^2+a^2}$	sin at
$\frac{s}{s^2+a^2}$	Cos at
$\frac{a}{s^2-a^2}$	Sinh at
$\frac{s}{s^2-a^2}$	Cos ax

Transforms Of Derivatives

Let $F'(t)$ denote the first derivative of $F(t)$ with respect to t , $F''(t)$ denote the second derivative of $F(t)$ with respect to t, etc.

Then $L\{F'(t)\}=\int_0^\infty e^{-st}F'(t)dt$ by definition ,

Integrating By Parts

$$L\{F'(t)\}=\left[e^{-st}F(t)\right]_0^\infty-\int_0^\infty F(t)(-se^{-st})dt$$

when $t\to 0, e^{-st}F(t)\to 0$

$$L\{F'(t)\}=-F(0)+s\int_0^\infty e^{-st}F(t)dt$$

$$L\{F'(t)\}=-F(0)+sL\{F(t)\}$$

$$L\{F''(t)\}=-F'(0)+sL\{F'(t)\}=-F'(0)+s\left[-F(0)+sL\{F(t)\}\right]$$

$$L\{F''(t)\}=s^2L\{F(t)\}-sF(0)+F'(0)$$

$$L\{F'''(t)\}=s^3L\{F(t)\}-s^2F(0)-sF'(0)-F''(0)$$

$$L\{F^{iv}(t)\}=s^4L\{F(t)\}-s^3F(0)-s^2F'(0)-sF''(0)-F'''(0)$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

Differential Equation And Its Solution

$\frac{dx}{dt} - 2x = 4$ ----------(1)

Its Solution is $x = -2 + 3e^{2t}$, To verify it we find $\frac{dx}{dt}$

$\frac{dx}{dt} = 6e^{2t}$ then

$$\frac{dx}{dt} - 2x = 6e^{2t} - 2(-2 + 3e^{2t})$$

$$= 6e^{2t} + 4 - 6e^{2t} = 4$$

So equation (1) is satisfied. Hence $x = -2 + 3e^{2t}$ is solution of $\frac{dx}{dt} - 2x = 4$

Example Solve the differential equation $\frac{dx}{dt} - 2x = 4$ given that at $t = 0$, $x = 1$

Taking Laplace transform as

$$L\left[\frac{d}{dt}(x(t)) - 2L(x(t))\right] = L(4) \Rightarrow L\left[\frac{d}{dt}(x(t))\right] - 2L(x(t)) = L(4) \Rightarrow sL(x(t)) - x(0) - 2L(x(t)) = \frac{4}{s}$$

$$(s-2)L(x(t)) - x(0) = \frac{4}{s} \Rightarrow (s-2)L(x(t)) - 1 = \frac{4}{s} \Rightarrow (s-2)L(x(t)) = \frac{4}{s} + 1 \Rightarrow (s-2)L(x(t)) = \frac{4+s}{s}$$

$$\Rightarrow L(x(t)) = \frac{4+s}{s(s-2)} \Rightarrow x(t) = L^{-1}\left[\frac{4+s}{s(s-2)}\right]$$ ----------(1)

First we do the partial fraction of $\frac{4+s}{s(s-2)}$

$$\frac{4+s}{s(s-2)} = \frac{A}{s} + \frac{B}{(s-2)}$$

$\Rightarrow 4 + s = A(s-2) + B(s)$ ----------(2)

Put $s = 0$ in equation (2) ; $4 = -2A$; $A = -2$

Put $s = 2$ in equation (2) ; $6 = B(2)$; $B = 3$

So $\frac{4+s}{s(s-2)} = \frac{-2}{s} + \frac{3}{s-2}$

Equation # (1) becomes

$$x(t) = L^{-1}\left[\frac{4+s}{s(s-2)}\right] = L^{-1}\left[\frac{-2}{s}\right] + L^{-1}\left[\frac{3}{s-2}\right] = -2 + 3e^{2t}$$

Solution of differential equation by laplace transforms

To solve a differential equation by Laplace transforms, we go through Laplace transforms, we go through four distinct stages.

(a) Re- write the equation in term of Laplace transforms.
(b) Insert the given initial conditions.
(c) Rearrange the equation algebraically to give the transform of the solution.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner

(d) Determine the inverse transform to obtain the particular solution.

Solve the equation

$\frac{d^2x}{dt^2} - 3\frac{d\ x}{dt} + 2x = 2e^{3t}$ given that at t = 0 , x = 5 and $\frac{dx}{dt} = 7$

$x''(t) - 3x'(t) + 2x(t) = 2e^{3t}$

Given $x(0) = 5$, $x'(0) = 7$

$L\left(x''(t)\right) - 3L(x'(t)) + 2L(x(t)) = 2L(e^{3t})$

$s^2L\{x(t)\} - s\,x(0) - x'(0) - 3\{s\ L(x(t)) - x(0)\} + 2L\,x(t) = \frac{2}{s-3}$

We rewrite the equation in term of its transforms.

$L\left[\frac{d^2x}{dt^2}\right] - 3L\left[\frac{d\ x}{dt}\right] + 2L[x] = 2L\left[e^{3t}\right]$

$\left[[s^2L(x(t)) - s\,x(0) - x'(0)] - 3[s\,L(x(t)) - x(0)] + 2L(e^{3t})\right]$

At t = 0 , x = 5 , $\frac{dx}{dt} = 7$

So $x(0) = 5$, $x'(0) = 7$

$s^2L(x(t)) - s(5) - 7 - 3\{sL(x(t)) + 3(5)\} + 2L(x(t)) = \frac{2}{s-3}$

$s^2L(x(t)) - 3sL(x(t)) + 2L(x(t)) = \frac{2}{s-3} - 8 + 5s$

$(s^2 - 3s + 2)L(x(t)) = \frac{2 - 8s + 24 + 5s^2 - 15s}{s-3}$

$L(x(t)) = \frac{2 - 8s + 24 + 5s^2 - 15s}{(s-1)(s-2)(s-3)} = \frac{5s^2 - 23s + 24}{(s-1)(s-2)(s-3)}$

Making Partial fraction of R.H.S , We have

$L(x(t)) = \frac{A}{(s-1)} + \frac{B}{(s-2)} + \frac{C}{(s-3)}$

After solving these we get A= 3 , B = 2 and C = 0

So $L(x(t)) = \frac{3}{(s-1)} + \frac{2}{(s-2)} + \frac{0}{(s-3)}$

$L(x(t)) = \frac{3}{(s-1)} + \frac{2}{(s-2)}$

$x(t) = L^{-1}\{\frac{3}{(s-1)}\} + L^{-1}\{\frac{2}{(s-2)}\} = 3e^{t} + 2e^{2t}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Scanned with CamScanner